Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	BioReperia AB
Short name / acronym used in projects	BR
Participant Identification Code (PIC), if available	*917010744*
Website(s)	www.bioreperia.com
Country	Sweden
Organisation type	☐ Academic ☒ Non-academic ☐ Hospital / clinic ☒ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Wahlbecksgatan 25, 582 13 Linköping
Department(s), institute(s), research group(s) or unit(s) that would be involved	Same as above
Additional institutional resources relevant to collaboration	Pre-clinical: Independent laboratories including complete zebrafish lab, cell culture lab and visualization facility (fluorescence microscopy). Own biobank including sample collections from ongoing clinical studies. Organizational: Fully implemented quality management system (QMS) according to criteria for producers and users of clinical diagnostics (ISO-13485 and -15189 respectively). Commercial: CE-marked and regulatory approved ZTX®-test for supporting treatment planning of cancer patients.

Main contact person for this profile	Emmelie Persson, CEO

Main contact email	Emmelie.persson@bioreperia.com
Other contact person(s), if relevant	Anna Erkstam, COO, anna.erkstam@bioreperia.com Gabriela Vazquez-Rodriguez, Project leader, gabriela.vazquez.rodriguez@bioreperia.com

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	☒ Retina
☐ Inherited retinal dystrophies (IRDs)	☐ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
☒ Advanced therapies / regenerative medicine	☒ Gene therapy / cell therapy / tissue engineering
☒ Inflammation / biomarkers / neuroprotection	☒ Drug delivery / formulations / nanomedicine
☒ Drug discovery / screening / repurposing	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☒ Health technology assessment	☒ AI-assisted diagnosis / software / digital tools
☒ Clinical trials / translational validation	☐ Other (please specify):

Free Keywords for internal searchable database (please provide up to 10)	Zebrafish; drug development; diagnostics; regulatory; laboratory quality management; ocular cancer
Geographies of interest for exchanges (optional)	

Overview of your research / innovation focus
BioReperia is a contract research organization and clinical diagnostics company active primarily within oncology. BioReperias CE-marked zebrafish tumor xenograft (ZTX) platform is used by academics and pharma companies for early, intermediate and late stage, pre-clinical drug development as well as within or in support of clinical drug trials. The platform is also used for precision medicine to identify drugs that are active on the patients cancer and thus to individualize treatment. The company was founded in 2015 and currently has four employees. *BioReperia is mainly doing research within clinical validation of the ZTX platform. We are, however, also interested in projects related to continued product development (e.g. developing automation and new ZTX capabilities), and ZTX-associated technologies such as drug toxicity, pharmacokinetics, imaging, and cancer pathophysiology (inflammation, angiogenesis, etc) using zebrafish larvae.* *BioReperia is also interested in collaborating within regulatory issues (mainly coupled to MDR and IVDR), IP-management within diagnostic companies, quality assurance including inter-laboratory validation, strategic business models for CROs and diagnostic companies, and efficient management of small companies with limited human resources but large production requirements.*

Why your team could be relevant to a translational eye-health and vision consortium
We can offer expertise and resources mainly within ocular cancer models. We have, however, experience in studying drug effects on hyaloid vessel development in zebrafish larvae, and ocular toxicities of drugs including retinal degeneration or inhibited development. *Zebrafish are becoming increasingly popular and relevant for cancer research and personalized medicine. Zebrafish tumor models consistently demonstrate high accuracy for predicting the efficacy of oncological therapies in patients. They are therefore recognized as a high potential novel test for personalized treatment planning in cancer patients, especially for patients with rare cancers (such as*

ocular cancers) where large, population-based studies of drug efficacy are lacking and there is a large unmet need for more effective treatments.

Zebrafish larvae are also increasingly used for development of drugs or understanding disease mechanisms within non-malignant diseases. Many pathophysiological processes that are important during tumorigenesis and cancer progression, e.g. inflammation, angiogenesis, disturbed metabolism, etc, are also relevant for non-malignant ocular diseases. Zebrafish lend themselves to studies where these processes can be investigated in unmatched spatiotemporal detail in vivo, as the transparent, living larvae continue their development (or disease development) under the fluorescence microscope, allowing sub-cellular visualization of pathophysiological processes in real time.

Within translational research, academic discoveries (basic science) are applied to different contexts either within commercialization (e.g. a drug development project) or clinical translation. BioReperia has extensive experience in these processes and can support collaborators who are interested in knowing more about how to take the next translational steps along the journey to clinical use and societal impact.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

- *Larval zebrafish disease models*
- *Zebrafish tumor models*
- *Cell culture*
- *Immunohistochemistry*
- *Fluorescence microscopy*
- *Commercialization of technologies/methodologies, drug candidates, or diagnostic tests*
- *Clinical translation of pre-clinical research including clinical trials*

Infrastructure, equipment, facilities or datasets relevant to consortium work

- *Zebrafish lab*
- *Cell culture labs*
- *Microscope facilities including two top-tier fluorescence dissection microscopes*
- *Image analysis software for automated quantification of structures in fluorescent images*

Training capacity and knowledge-transfer potential

- *Experienced in training secondees on both short and long secondments*
- *Secondees will gain knowledge in zebrafish models, cell-based (in vitro) models, or associated techniques (microinjection, cell culture, visualization)*
- *Secondees will gain knowledge in standardized laboratory operations that ensure reproducibility and quality (clinical lab procedures)*
- *Secondees will be in a translational environment working closely with pharma customers as well as clinical oncological surgeons, medical oncologists and pathologists*
- *Secondees will gain knowledge of business start-up and scale-up, including an opportunity to discuss business plans, business models, funding/investment strategies, and practical/concrete activities that increase the chance of commercial success*

What your team would hope to gain from participating

- *Broadening our network which may lead to a large customer base – in particular a base for using ZTX models for treatment planning of patients with ocular cancers*
- *Knowledge of the management and pathophysiology of ocular cancers*
- *More experience in ocular cancer disease models, drug delivery principles, mode-of-action studies, and other aspects related to ocular cancer drug development*
- *Access to a larger recruitment base for highly-skilled, future employees*

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Esri Hazael Juares	Lab Engineer	BioReperia	Technical	C	Esri.juares@bioreperia.com
Zaheer Ali	Lab Engineer	BioReperia	Technical	C	Zaheer.ali@bioreperia.com
Franciele Dietrich Zagonel	Customer Developer	BioReperia	Adminstrative	C	Franciele.dietrich@bioreperia.com

Main disciplines represented in your team	Oncology, drug development, zebrafish models, business development, investment, clinical trials
Experience in EU / international / multi-partner collaborative projects	Participation in three prior EU-staff-exchange programs, two Eurostars consortia (one as coordinator) and in multiple product or process development projects founded by the Swedish Innovation Council. Have executed >50 projects with international drug companies/academic groups from all continents targeting a broad range of mechanisms, drug types, and stages in the drug development process.
Experience with clinical, translational or patient-facing research (if relevant)	☒ Yes ☐ No ☐ Not applicable | Short note: Mainly within cancer

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

Ding Y, van der Kolk KJ, van der Ent W, Scotto de Mase M, Kowald S, Huizing J, Teuton JV, Mishra G, Kempers M, Almter R, Kunz S, Munier L, Koschmann C, Waszak S, Di Donato V, Dyballa S, ten Dijke P, Esguerra C, Jensen LD and de Sonneville J, "Automated microinjection for zebrafish xenograft models", npj Biomedical Innovations, 2025, 2(1), 13

Clarke NC, McCabe E, Jensen LD, Creaven BS, Costello DA, "Salicylaldehyde benzoylhydrazone protects against ferroptosis in models of neurotoxicity and behavioural dysfunction, in vitro and in vivo", J. Mol. Neurosci., 75(2), 77

Lindahl G, Fjellander S, Selvaraj K, Vildeval M, Ali Z, Almter R, Erkstam A, Vazquez Rodriguez G, Abrahamssom A, Kersley AR, Fahlgren A, Kjölhede P, Linder S, Dabrosin C and Jensen LD, "Zebrafish tumor xenograft models: A novel prognostic approach to epithelial ovarian cancer", NPJ Precision Oncology, 8(1):53

Sundaramurthi H, Tonelotto V, Wynne K, O'Connell F, O'Reilly E, Costa-Garcia M, Kovacshazi C, Kittel A, Marcone S, Blanco A, Pallinger E, Hambalko S, Riulats Rodriguez JM, Ferdinandy P, O'Sullivan J, Matallanas D, Jensen LD, Giricz Z and Kennedy BN, "Ergolide mediates anti-cancer effects on metastatic uveal melanoma cells and modulates their cellular and extracellular vesicle proteosomes", Open Research Europe, 3:88

Ali Z, Vildevall M, Vazquez-Rodriguez G, Tandiono D, Vamvakaris I, Evangelou G, Lolas G, Syrigos KN, Villanueva A, Wick M, Omar S, Erkstam A, Schueler J, Fahlgren A and Jensen LD, "Zebrafish patient-derived xenograft models predict lymph node involvement and treatment outcome in non-small cell lung cancer", J. Exp. Clin. Cancer Res., 41(1):58.

Up to 5 relevant previous projects, grants, networks or collaborations

EVEREST: MSCA-SE project wherein BioReperia is collaborating with 20+ EU partners to advance research in EV biology including applications within cancer research and novel therapies (international)

CRYSTAL3: MSCA-RISE project wherein BioReperia is collaborating with ~15 EU partners to advance research in cysteine leukotriene signaling and its involvement in various diseases (international)

3D-NEONET: MSCA-RISE project wherein BioReperia is collaborating with ~15 EU partners to advance research in eye disease, cancer and cardiovascular disease (international)

ROBO-FISH: A Eurostars project in which Dr. Jensen worked with LSM to develop new injection and visualization robots for creation and analysis of larval zebrafish tumor xenograft models. (international)

MECHANO-CHIP: A Eurostars project in which Dr. Jensen was collaborating with Dutch partners to develop microfluidic systems for cancer research (international)

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)
Know-how related to the ZTX platform is mostly protected through NDAs and restricted access to critical protocols. *Numerous clinical protocols for diagnostic validation of ZTX®-Predict have been co-developed with clinical collaborators in different malignancies.* *Software for quantification of structures found in fluorescence images of zebrafish larvae have been developed in house and is not available outside of BioReperia* *BioReperia have co-developed two robots for automated tumor implantation and imaging of tumor-implanted zebrafish larvae with the company LifeScienceMethods in the Netherlands.*

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	☒ Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	☐ Only Researchers ☒ Technical Staff/Innovation ☒ Managerial / project support ☒ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only ☒ Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	Planning need to be done well ahead of arrival due to limited HR-resources and capacity to supervise and manage secondments on site.

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
Zebrafish cancer models *Translational research profile and experience working with drug developers and in clinical studies/contexts* *Extensive experience in commercialization of research outcomes including start-up and scale-up* *Extensive experience in quality control of laboratory processes including clinical diagnostics*

Capabilities or expertise you would especially like to access through the consortium
Clinical expertise and samples from patients with ocular cancer *Cross-platform comparisons with ocular cancer organoids, spheroids, tumoroids, mouse xenografts/orthoxenografts, and patient treatment outcomes.* *Expertise in development of drugs targeting ocular cancers*

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	☒ Yes ☐ No
☒ We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
Examples: PIC pending approval, institutional approval, legal status, hosting rules, IP constraints, data access, ethics, clinical permissions, staffing continuity, resource availability. NA

9. Additional collaborators you would like us to consider

If there are any additional organisations, teams or contacts - current collaborators, previous collaborators or desired future collaborators - that you think could be relevant for this MSCA-SE consortium, please list them below. The coordination team may contact them to explore their potential interest in joining the consortium-building process.

Please provide the following details (as many entries as needed).

Type of organisation: ☐ **Academic** ☐ **Clinical** ☒ **SME/industry** ☐ **NGO/foundation** ☐ **Public body** ☐ **Other**

Organisation / team name: Any Pharma company, preferably medium or big. Any clinical partner interested in developing new treatment planning methodologies.

Country:

Website:

Existing link with your team: ☐ **Current collaborator** ☐ **Previous collaborator** ☐ **Desired collaborator** ☐ **Other**

Suggested contact person/email (if known):

Why could they be relevant for this MSCA-SE consortium?

Any additional comments or sensitivities before contacting them?

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

☒ **Understood**

Name: Lasse Jensen
Date completed: July 6th 2026

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)

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1. Organisation and administrative profile

Organisation legal name	Blafar Ltd
Short name / acronym used in projects	BLAFAR
Participant Identification Code (PIC), if available	906676364
Website(s)	www.blafar.com
Country	Ireland
Organisation type	☐ Academic X Non-academic ☐ Hospital / clinic X SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Building 10, Cherrywood Business Park, Cherrywood, Loughlinstown Co. Dublin, D18 T3Y1 Ireland
Department(s), institute(s), research group(s) or unit(s) that would be involved	NA
Additional institutional resources relevant to collaboration	• Biomaterials synthesis and formulation laboratories • Injectable hydrogel design and characterisation platform • Functional biopolymer modification technologies • Quality management system supporting translational R&D (ISO 13485) • Access to advanced analytical facilities through University College Dublin, including polymer characterisation, rheology and materials analysis • Established collaborations with academic, clinical and industrial partners supporting translational biomaterials research

Main contact person for this profile	Dr Hongyun Tai, CEO and Managing Director
Main contact email	Hongyun.tai@blafar.com
Other contact person(s), if relevant	Prof Wenxin Wang, Founder & Chief Scientific Advisor

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	☐ Retina
☐ Inherited retinal dystrophies (IRDs)	☐ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
x Advanced therapies / regenerative medicine	x Gene therapy / cell therapy / tissue engineering
☐ Inflammation / biomarkers / neuroprotection	x Drug delivery / formulations / nanomedicine
☐ Drug discovery / screening / repurposing	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	☐ Other (please specify):

Free Keywords for internal searchable database (please provide up to 10)	Injectable hydrogels, Biomaterials engineering, Controlled drug delivery, Regenerative medicine, Tissue engineering, Biomaterial-enabled therapeutic delivery, Functional biopolymers, Ophthalmic biomaterials, Translational biomaterials, Advanced therapeutic medicinal products (ATMPs)
Geographies of interest for exchanges (optional)	

Overview of your research / innovation focus

Please describe the main focus of the team / unit that could participate. Keep it concise but specific (recommended: 150-250 words).

Blafar Ltd is an Irish research-intensive biotechnology SME developing advanced biomaterial platforms for regenerative medicine, advanced therapeutic medicinal products (ATMPs) and next-generation medical technologies. The participating team focuses on the design, synthesis and translation of functional biopolymers, injectable hydrogel systems and biomaterial-enabled therapeutic delivery technologies that address unmet clinical needs.

Our research integrates polymer chemistry, biomaterials engineering and translational product development to create biocompatible materials for controlled drug delivery, cell and gene therapy, tissue engineering and regenerative medicine. Particular expertise has been developed in chemically modified hyaluronic acid-based biomaterials and injectable hydrogel platforms with tunable mechanical properties, degradation profiles and therapeutic release kinetics.

The team actively collaborates with academic researchers, clinicians and industrial partners to accelerate the translation of biomaterial innovations from laboratory research towards clinical and commercial application. Current application areas include ophthalmology, musculoskeletal repair, wound healing and advanced drug delivery, while maintaining a platform technology approach that enables adaptation to a broad range of therapeutic indications.

Through participation in European collaborative research projects, Blafar aims to combine scientific excellence with industrial innovation, contributing expertise in biomaterial design, translational development and interdisciplinary

collaboration to support the development of next-generation healthcare technologies.

Why your team could be relevant to a translational eye-health and vision consortium

Please explain the specific added value your organisation / team could bring to a consortium centred on visual health, translational ophthalmology and related enabling technologies. ***Please think broadly: the contribution does not need to be directly ophthalmology-related. Interdisciplinarity is strongly encouraged, and expertise from apparently distant fields may open unexpected routes to innovation, new paradigms, and creative solutions to complex challenges that currently limit progress from research to patient benefit****.*

Blafar brings complementary expertise in advanced biomaterials, injectable hydrogel systems and translational product development that can enable innovative solutions for visual health beyond conventional ophthalmology research. Rather than focusing on a single disease area, our platform technologies are designed to support the controlled delivery of therapeutic molecules, cells and other advanced therapeutics, creating opportunities for interdisciplinary collaboration across regenerative medicine, drug delivery and tissue engineering.

Our team can contribute expertise in the design of functional biopolymers, biomaterial-assisted therapeutic delivery, formulation development and translation towards clinically relevant products. These capabilities are applicable to a wide range of ophthalmic challenges, including sustained intraocular drug delivery, regenerative approaches for ocular tissues, biomaterial-based cell and gene therapy delivery, and next-generation medical devices.

As an innovation-driven SME with experience in European collaborative projects, Blafar also contributes an industrial and translational perspective that complements academic and clinical research. We aim to help bridge the gap between laboratory discovery and practical healthcare solutions through technology development, regulatory awareness, quality-by-design principles and commercialisation planning.

By combining expertise in biomaterials engineering with close collaboration across academia, healthcare and industry, Blafar seeks to contribute enabling technologies that can accelerate innovation, foster interdisciplinary knowledge exchange and improve the translation of research into patient benefit.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

Please provide bullet points covering your strongest capabilities, methods, models, analytical approaches, translational assets or innovation strengths.

Technology Platforms

- Advanced biomaterials platform – Design and synthesis of functional natural and synthetic biomaterials for regenerative medicine and advanced therapeutic applications.
- Injectable hydrogel platform – Development of tunable hydrogel systems for sustained delivery of therapeutic molecules, cells and extracellular vesicles.
- Drug delivery and formulation technologies – Biomaterial-enabled delivery systems for small molecules, biologics and advanced therapeutics, with applications including ophthalmology.

Technical Expertise

- Functionalisation and engineering of hyaluronic acid and other biopolymers.
- Polymer synthesis, hydrogel fabrication and formulation optimisation.
- Physicochemical and rheological characterisation of biomaterials and hydrogel systems.
- Design of biomaterials with tunable mechanical properties, degradation profiles and controlled release characteristics.

Translation and Innovation

- Translation of biomaterial technologies from laboratory research towards clinically relevant products.
- Prototype development, design optimisation and manufacturability assessment.
- Experience in quality management, technology transfer, intellectual property and industrial collaboration.
- Development of biomaterial technologies for medical devices, regenerative medicine and advanced therapeutic medicinal products (ATMPs).

Knowledge Exchange and Collaboration

- Interdisciplinary collaboration between academia, clinicians and industry.
- Training in biomaterial design, hydrogel technologies and translational product development.
- Experience participating in European collaborative research and staff exchange projects.
- Support for innovation planning, commercialisation pathways and industry-oriented research translation.

Infrastructure, equipment, facilities or datasets relevant to consortium work

Examples: specialised laboratories, clean rooms, imaging platforms, -omics, biobanks, clinical cohorts, animal models, software tools, rehabilitation platforms, regulatory support or prototyping capacities.

- **Biomaterials Research Laboratory** for polymer synthesis, biomaterial functionalisation, hydrogel fabrication and formulation development.
- **Hydrogel and biomaterial characterisation capabilities**, including physicochemical analysis, rheological evaluation and formulation optimisation.
- **Prototype development capability** for injectable biomaterials, controlled therapeutic delivery systems and translational biomaterial technologies.
- **Quality management system (ISO 13485)** supporting research, technology development, documentation and innovation management for translational biomaterials.
- **Access to advanced analytical facilities through University College Dublin (UCD)**, including polymer characterisation, materials analysis and specialised instrumentation.
- **Access to academic and clinical collaboration networks** supporting in vitro evaluation, preclinical studies and application-specific validation of biomaterial technologies.
- **International industrial collaboration network** supporting technology translation, product development and future manufacturing scale-up.

Training capacity and knowledge-transfer potential

Please indicate what inbound staff could realistically learn in your organisation (scientific, technical, clinical, translational, regulatory, entrepreneurial or managerial skills).

Blafar provides a unique cross-sector training environment where researchers gain first-hand experience of the innovation pathway—from biomaterial design and laboratory development to quality-managed translational research and industrial collaboration.

Scientific and Technical Skills

- Design and functionalisation of natural and synthetic biomaterials.
- Injectable hydrogel formulation, optimisation and characterisation.
- Biomaterial engineering for controlled therapeutic delivery and regenerative medicine.
- Polymer synthesis, biomaterial modification and formulation development.

Translational Research and Innovation

- Translating biomaterial technologies from laboratory research towards clinically relevant products.
- Prototype development, design optimisation and technology validation.
- Integration of scientific research with industrial product development.

Quality Management and Regulatory Awareness

- Practical exposure to quality management systems supporting biomaterials research and innovation.
- Experience in design documentation, risk-based thinking and quality-by-design approaches for translational biomaterial development.
- Introduction to regulatory considerations and product development pathways for biomaterial-based medical technologies.

Industry–Academia Collaboration

- Working within multidisciplinary teams involving academic researchers, clinicians and industrial partners.
- Experience of collaborative R&D, European collaborative projects and cross-sector knowledge exchange.

What your team would hope to gain from participating
Examples: access to methods, models, technologies, patient-oriented validation, translational pathways, industrial insight, clinical exposure, data science, regulatory know-how, joint publications or longer-term strategic collaboration. ***Again, please think broadly and beyond ophthalmology.***
Blafar views participation in the consortium as an opportunity to strengthen long-term collaboration with leading academic, clinical and industrial partners in translational ophthalmology and regenerative medicine. We are particularly interested in expanding our understanding of ophthalmic disease mechanisms, preclinical models, clinical needs and translational pathways that will help guide the future development of our biomaterial platform technologies. The consortium would also provide valuable opportunities to identify new biomedical applications for Blafar's biomaterial platform technologies, validate biomaterial-enabled therapeutic delivery systems in clinically relevant settings, establish new interdisciplinary collaborations and explore emerging applications in ocular regeneration, advanced therapies and precision medicine. Through staff exchanges, we hope to foster long-term scientific partnerships, co-develop innovative research ideas, generate high-impact publications and prepare future collaborative European research proposals. We also value the opportunity for our researchers to gain exposure to complementary methodologies, clinical perspectives and translational research environments that will enhance both scientific excellence and industrial innovation.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Dr Hongyun Tai	CEO				Hongyun.tai@blafar.com
Prof Wenxin Wang	CSO				Wenxin.wang@ucd.ie
Dr Rijiang Song	CTO				Rijiang.song@blafar.com
TBC	Product Development Scientist				
TBC	Product Development Scientist				

Main disciplines represented in your team	Biomaterials Science; Polymer Chemistry; Biomedical Engineering; Drug Delivery; Regenerative Medicine; Tissue Engineering; Materials Science; Quality Management; Technology Translation; Innovation Management.
Experience in EU / international / multi-partner collaborative projects	Blafar has extensive experience participating in European and international collaborative research projects involving academic institutions, SMEs, clinicians and industry partners. The company has contributed to multiple EU-funded collaborative programmes, including Marie Skłodowska-Curie Staff Exchanges, Horizon Europe and Enterprise Ireland-supported international projects, with expertise spanning biomaterials research, translational innovation, staff exchange, technology transfer and cross-sector collaboration.
Experience with clinical, translational or patient-facing research (if relevant)	x Yes ☐ No ☐ Not applicable | Short note: collaboration with clinical partners.

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements
Please prioritise achievements most relevant to your possible contribution. Please include DOI, URL or identifier.
1. Yu, H. et al.; Wang, W. 'MSC-derived exosomes injectable hyaluronic acid hydrogel for enhanced chronic wound healing'. JOURNAL OF CONTROLLED RELEASE, 2025, DOI: 10.1016/j.jconrel.2025.113985.

2. Chen, L. et al.; Wang, W. 'Biphasic release of betamethasone from an injectable HA hydrogel implant for alleviating lumbar disc herniation induced sciatica'. ACTA BIOMATERIALIA, 2024, 176, 173-189, DOI: 10.1016/j.actbio.2024.01.016.

3. Xu, Q. et al.; Wang, W. 'Versatile Hyperbranched Poly(β-hydrazide ester) Macromers as Injectable Antioxidative Hydrogels'. ACS Applied Materials & Interfaces, 2018, 10(46), 39494-39504. DOI: 10.1021/acsami.8b15006.

4. Wang, X. et al.; Wang, W. 'An injectable Chitosan-based self-healable hydrogel system as an antibacterial wound dressing'. Materials, 2021, 14(20), 5956. DOI: 10.3390/ma14205956.

5. He, Z. et al.; Wang, W. 'Reactive Oxidative Species (ROS): Utilizing Injectable Antioxidative Hydrogels and ROS-Producing Therapies to Manage the Double-Edged Sword'. Journal of Materials Chemistry B, 2021, 9, 6326-6346. DOI: 10.1039/D1TB00728A.

Up to 5 relevant previous projects, grants, networks or collaborations

Please provide detailed information (Title, Acronym, funding agency, GA-number, dates). Please indicate your role (coordinator / beneficiary / partner / service provider / collaborator). Indicate if international.

Note: GA numbers are not provided in the supplied CV/PDF for the entries below. Proposal number/call identifiers are included where available.

1. NET4MAT - Building knowledge and tools for the sustainable microbial fighting through sensing and responsive polysaccharide-based materials. Funding/programme: HORIZON-MSCA-2022-SE-01 / HORIZON-TMA-MSCA-SE. Proposal number: 101131205. Dates: 2024-2027. Role: Blafar Limited partner; Prof Wenxin Wang listed as a leading Blafar researcher. International: yes, 13-organisation consortium.

2. Drug Discovery and Delivery NEtwork for ONcology and Eye Therapeutics (3D NEONET). Funding/programme: EU Horizon 2020, H2020-MSCA-RISE-2016. Dates: 2017-2020. Role: Co-PI; €80,500 allocated to Prof Wang's sub-project from €945,000 total award. International: yes.

3. Novel Cell Migration Assay Based on Microtissue Technology and Tissue-Specific Matrices (MATRIXASSAY). Funding/programme: EU Horizon 2020, H2020-MSCA-RISE-2014. Dates: 2015-2019. Role: Co-PI; €54,000 allocated to Prof Wang's sub-project from €949,500 total award. International: yes.

4. Drug-Free Antibacterial Hybrid Biopolymers for Medical Applications (HyMedPoly). Funding/programme: EU Horizon 2020, H2020-MSCA-ITN-2014. Dates: 2014-2018. Role: Co-PI; €119,413 allocated to Prof Wang's sub-project from €3,918,055 total award. International: yes.

5. Novel combination of biopolymers and manufacturing technologies for production of a peripheral nerve implant containing an internal aligned channels array (NEURIMP). Funding/programme: EU FP7, FP7-NMP-2013-SME-7. Dates: 2013-2017. Role: Co-PI; €379,640 allocated to Prof Wang's sub-project from €3,304,705 total award. International: yes.

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)

Please indicate only what can be shared at this stage.

Know-how

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	x Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	x Only Researchers xTechnical Staff/Innovation x Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only x Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	Potentially, some of our staff need to apply for visa.

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice

Please summarise the 2-3 strengths that should be first visible in an internal searchable profile.

Blafar is an Irish translational biomaterials innovation SME specialising in advanced biomaterial platform technologies for regenerative medicine, advanced therapeutic medicinal products (ATMPs), therapeutic delivery and medical devices. Through close collaboration with academia, clinicians and industry, Blafar translates innovative biomaterials into clinically relevant healthcare solutions.

Capabilities or expertise you would especially like to access through the consortium
Please indicate the types of teams, methods, infrastructures or translational pathways you would like to connect with in the consortium. Blafar would particularly welcome collaboration with teams offering complementary expertise in ophthalmic biology, disease modelling, preclinical evaluation and clinical translation. We are interested in connecting with researchers and clinicians who can help validate biomaterial platform technologies in clinically relevant models and identify unmet clinical needs that can drive future technology development. We are especially interested in collaborating with partners working in: • Ophthalmic disease models and preclinical validation. • Ocular drug delivery, regenerative medicine and advanced therapeutic development. • Cell and gene therapy, extracellular vesicle (EV) therapeutics and tissue engineering. • Clinical translation, regulatory science and patient-oriented evaluation. • International industrial–academic collaborations that support technology translation and future collaborative research.

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	x Yes ☐ No
x We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
NA

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

x Understood

Name: Hongyun Tai
Date completed: 05/07/2026

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)

Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	Centre of Biotechnology of Sfax
Short name / acronym used in projects	CBS
Participant Identification Code (PIC), if available	988821978
Website(s)	https://cbs.rnrt.tn
Country	Tunisia
Organisation type	X Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company x Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Sid Mansour Road km06 BP1177, Sfax, 3018, Tunisia
Department(s), institute(s), research group(s) or unit(s) that would be involved	Laboratory of Molecular and Cellular Screening Processes (LPCMC)
Additional institutional resources relevant to collaboration	Unit of provision of analytical services of CBS (UPSA) (https://cbs.rnrt.tn/services-analytiques-upsa) Unit of technology transfer of CBS (https://cbs.rnrt.tn/unite-specialisee-de-transfert-de-technologie) Unit of Valorization of Research Results (UVRR) of CBS (https://cbs.rnrt.tn/unite-specialisee-de-valorisation-des-resultats-de-la-recherche-uvrr) Unit of QSE of CBS (https://cbs.rnrt.tn/unite-specialisee-de-qualite-securite-environnement-et-accreditation)

Main contact person for this profile	Mohamed Sami Aifa, Professor, Head of a research lab, LPCMC / CBS
Main contact email	Sami.aifa@cbs.rnrt.tn
Other contact person(s), if relevant	Monia Mlaouah, secretary of LPCMC, monia.mlaouah.cbs@gmail.com

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	☐ Retina
☐ Inherited retinal dystrophies (IRDs)	☐ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
☐ Advanced therapies / regenerative medicine	☐ Gene therapy / cell therapy / tissue engineering
☐ Inflammation / biomarkers / neuroprotection	☐ Drug delivery / formulations / nanomedicine
x Drug discovery / screening / repurposing	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	X Other (please specify): signal transduction and Genetic Engineering

Free Keywords for internal searchable database (please provide up to 10)	Drug screening, drug repurposing, cell culture, recombinant proteins, Signal transduction; Docking and molecular dynamics
Geographies of interest for exchanges (optional)	Spain; Sweden; Ireland; Germany; Italy; France

Overview of your research / innovation focus
LPCMC, as a laboratory within the Centre of Biotechnology of Sfax, was created in 2015. *Our research focus on the following general and specific objectives:* *• Implementation of multidisciplinary approaches to address health challenges based on expertise in Genomics, Proteomics, Cell Signaling and Bioinformatics* *• Train technical staff in molecular and cellular technologies and the use of statistical and bioinformatics tools* *• Screen and characterize chemical libraries (natural or synthetic) for various properties (antiproliferative, antibiofilm)* *• Develop and apply simple and high-throughput DNA analysis technologies for molecular diagnosis and identification of genes responsible for pathologies* *• Analyze, study and appraise by statistical and computer processing of biological data*

Why your team could be relevant to a translational eye-health and vision consortium
The Laboratory of Molecular and Cellular Screening Processes (LPCMC) of the Centre of Biotechnology of Sfax (CBS, Tunisia) is a Tunisian laboratory whose core activity is the development and application of molecular and cellular screening processes for the discovery and evaluation of therapeutic molecules, with a strong focus on drug screening and drug repurposing.

In relationship to the present consortium topic, the LPCMC focuses on screening, repurposing, and improvement of therapeutic molecules, production and isolation of molecules with therapeutic potential and development of *in vitro*, *ex vivo*, and *in vivo* screening processes. We adopt a targeted molecular screening through enzymatic or biochemical tests on specific therapeutic targets (e.g. kinases, cholinesterase, COMT).

We work on cellular phenotypic screening via the culture of cancer, infectious, or other cellular lineages to detect cytotoxic, anti-inflammatory, antiviral, or other effects. We conduct cell treatment and the measurement of viability, proliferation, apoptosis, or other phenotypic parameters.

In addition, the LPCMC has established links with university hospitals and private laboratories in Sfax and Tunis to provide standardized, reliable, and reproducible molecular diagnostic solutions.

The laboratory addresses repurposing as a strategy to identify new therapeutic indications for drugs already on the market and take advantage of the known safety profile of these drugs to accelerate development and reduce costs and risks compared to classical drug discovery.

We work on the target difficult-to-treat diseases or under-treated pathologies such as cancers, infectious diseases, rare diseases, etc. and follow an integrative biology approach to link molecular, cellular, and clinical data. In cancer we screen molecules and extracts on cell lines (e.g., breast, lung, melanoma, etc.) and evaluate the selective cytotoxicity and antitumor efficacy. In infectious disease, the LPCMC has worked on the detection of SARS-CoV-2 mutations and monitoring of viral variability (research project federated with other laboratories in Tunisia) and the development of antiviral screening tests and monitoring of viral replication.

We work, also, on some metabolic and neurodegenerative diseases within the broader framework of research in integrative biology and personalized medicine.

All our approaches are valuable for any disease investigation and could applied in the present consortium centred on visual health.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

The LPCMC has **integrated expertise in molecular and cellular screening**, **high-throughput screening platforms**, and the **capacity to link molecular, cellular, and clinical data** to accelerate drug repurposing and the discovery of new therapies.

Core Expertise of LPCMC

- Integrative biology approach combining molecular, cellular, and clinical data
- Development of in silico, in vitro, ex vivo, and in vivo screening processes to evaluate therapeutic activity of compounds.
- Screening, repurposing, and optimization of therapeutic molecules, including drugs already on the market.
- Identification of new therapeutic indications for existing drugs, targeting difficult-to-treat diseases (cancers, infectious diseases, rare diseases).
- Isolation and purification of therapeutic-potential compounds (often from natural sources: plants, microorganisms, marine sources).
- Collaboration with university hospitals and private laboratories to develop standardized, reliable, and reproducible molecular diagnostic solutions.

Main Methods and Techniques

- Enzymatic and biochemical assays on specific therapeutic targets (kinases, cholinesterase, COMT).
- Characterization of inhibitory or activating activity of compounds on molecular targets.
- Use of cancer and non-cancer cell lines to evaluate Selective cytotoxicity, Cell proliferation, Apoptosis, Anti-inflammatory and antagonism effects
- Measurement of phenotypic parameters like viability, cell morphology...
- DNA-based analyses (PCR, mutation detection, viral variability tracking).
- Tracking of SARS-CoV-2 mutations and viral variability in Tunisia.
- Use of animal models for preclinical validation.
- Ex vivo tests on tissues or primary cells.

Platforms and Technical Capabilities

- Integration of high-throughput screening technologies enabling processing of large numbers of samples.

- Maintained cell culture for repeatable screening tests.
- Access to classical cancer cell lines (breast, lung, melanoma, leukemia, etc.) to evaluate anticancer activity.
- Standardized molecular diagnostic services for public and private partners.

Therapeutic and Pathological Target Areas

- Screening of molecules and natural extracts on cancer cell lines.
- Evaluation of selective cytotoxicity and antitumor efficacy.
- Examples of cell lines: breast cancer, lung cancer, colorectal cancer, HEK293.
- Development of antibiofilm screening assays.
- Tracking of viral mutations (SARS-CoV-2).
- Research within the framework of personalized medicine and integrative biology.

Valorization and Collaborations

- Development of patents and products for pharmaceutical, paramedical, and agro-food industries.
- Service provision: screening, molecular diagnostics, technical expertise.
- International and regional collaborations within research consortia and valorization projects.
- Links with universities, hospitals, and private laboratories in Sfax and Tunis.

Potential LPCMC Contributions to a Drug Screening & Repurposing Project

- Phenotypic screening: Tests on cancer or infectious cell lines to identify active compounds.
- Molecular screening: Enzymatic assays on therapeutic targets (kinases, proteases, etc.).
- Drug repurposing: Identification of new indications for existing drugs.
- Isolation of bioactive compounds: Extraction and purification of molecules from natural sources.
- Preclinical validation: In vitro, ex vivo, and in vivo tests to validate therapeutic efficacy.
- Molecular diagnostics: Mutation detection tests and viral variability tracking.
- Test standardization: Development of reproducible protocols for high-throughput screening.
- Clinical collaboration: Links with hospitals and laboratories for access to clinical samples

.

Infrastructure, equipment, facilities or datasets relevant to consortium work

- *Cell Culture Lab*: equipped with laminar flow hoods, incubators, ovens, water baths, stereo, light and fluorescence microscopes, cell counters, transfectors, centrifuges, liquid nitrogen tanks, -80°C and -20°C freezers, among others.
- *Microbiological Lab*: equipped with laminar flow hoods, incubators, ovens, water baths, pHmeters , plate reader
- *Molecular Biology Lab:* equipped with PCR and RT-PCR equipment, laminar flow hood, centrifuges, gel electrophoresis system of nucleic acids and proteins, Western blot system.
- *Zebra fish equipment*

- *Shared Labs*: equipped with autoclaves, laminar flow hoods, ovens, precision balances, pipettes, ice machines, sterilization equipment, electroporator , centrifuges, and, among others
- Unit of provision of analytical services of CBS (UPSA) (https://cbs.rnrt.tn/services-analytiques-upsa): equipped with chromatographic machines, GC-MS, MALDI-LC-MS, Sanger sequencing, New generation sequencing, imaging facilities, flow cytometry among others.
- Unit of Valorization of Research Results (UVRR) of CBS (https://cbs.rnrt.tn/unite-specialisee-de-valorisation-des-resultats-de-la-recherche-uvrr): equipped with fermenters, large scale facilities of biomolecule production,
- Hall of technology : equipped with Supercritical CO2 machine, 300l fermenter among others

Examples: specialised laboratories, clean rooms, imaging platforms, -omics, biobanks, clinical cohorts, animal models, software tools, rehabilitation platforms, regulatory support or prototyping capacities.

Training capacity and knowledge-transfer potential

Visitors can benefit from technical skills in in silico screening of drugs, in vitro screening through cell culture methods.
Downstream analysis of cytotoxicity including apoptosis/ necrosis evaluation, signal transduction interference evaluation through protein analysis (SDS-PAGE, Western blot with appropriate antibodies).
Anti-kinase, anti-COMT, anti-cholinesterase activity screening.
QRT-PCR analysis of inflammatory genes
Moreover, the visitors can learn the processes of fermentation in order to produce bioactive molecules.
Screening of antibiofilm activity and quorum sensing interference evaluation.
Genomic DNA extraction, RNA extraction, Sanger sequencing, NGS sequencing, Bioinformatics analysis of sequences.
Training on HPLC, GC-MS and LPCMS analysis are available through the UPSA unit (CBS).

What your team would hope to gain from participating

The LPCMC would hope to gain from participating in a staff exchange consortium a Capacity Building in Research and Innovation (R&I) through:

- Training in interdisciplinary approaches (combining molecular biology, cell biology, bioinformatics, and clinical data).
- Acquisition of advanced screening technologies and platforms (e.g., high-throughput screening, high-content screening, microfluidics) that are not available in Tunisia.
- Learning new methods in drug discovery, drug repurposing, and personalized medicine.
- Transfer of knowledge from partner organizations in Europe and beyond back to Tunisia.
- Introduction of best practices in research management, quality control, and standardization of screening protocols.
- The laboratory can increase its research and innovation capacities by gaining experience in cutting-edge methodologies and technologies.
- Establishment of long-term collaborations with prestigious European and international research institutions and companies.
- Development of sustainable collaborative projects between academic and non-academic sectors.
- Expansion of networks for researchers and staff, increasing the laboratory's visibility on the international stage.
- Opportunities for future joint projects, co-publications, and co-supervision of PhD students.
- Seconded staff members (researchers, technical staff, managers) gain new knowledge, skills, and transferable competencies.
- Improved employability and career prospects for researchers, from PhD candidates to postdoctoral researchers.
- Access to mentoring and career guidance from experienced researchers in partner organizations.
- Staff return to LPCMC after secondment to pass on knowledge and foster collaboration within the laboratory.
- Access to advanced research infrastructures (omics facilities, bioinformatics platforms) that may not be available in Tunisia.
- Opportunities for high-impact publications based on collaborative research.
- Participation in cutting-edge research in drug screening, repurposing, and personalized medicine.
- Contribution to Excellence in Research that boosts jobs, growth, and investment.
- Access to drug libraries from European partners
- Understanding of regulatory pathways for drug repurposing and approval in Europe
- Methodological innovation development of new screening assays and protocols tailored to local disease contexts

access to methods, models, technologies, patient-oriented validation, translational pathways, industrial insight, clinical exposure, data science, regulatory know-how, joint publications or longer-term strategic collaboration. ***Again, please think broadly and beyond ophthalmology.***

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Sami Aifa	Professor	LPCMC	Researcher	Experienced	Sami.aifa@cbs.rnrt.tn

Sami Mnif	Associate professor	LPCMC	Researcher	Experienced	Sami.mnif@cbs.rnrt.tn
Najla Kharrat	Associate professor	LPCMC	Researcher	Experienced	Najla.kharrat@cbs.rnrt.tn
Bochra Hakim	Engineer	LPCMC	Researcher	Experienced	bochrahakim@yahoo.fr
Monia Mlaouah	Lab Secretary	LPCMC	Administrative	Experienced	monia.mlaouah.cbs@gmail.com
Fares Elghali	Posdoctorant	LPCMC	Researcher	Experienced	fares7ghali@gmail.com
Kawther Bedchich	PhD student	LPCMC	Researcher	Early-Stage	bedchich.kawther18@gmail.com
Salim Kraiem	PhD student	LPCMC	Researcher	Early-Stage	salimkraiem190@gmail.com
Hajer Jlidi	PhD student	LPCMC	Researcher	Early-Stage	hajerdlo98@gmail.com

Main disciplines represented in your team	Cell biology, immunology, genetics, biomaterials, data science, genetic engineering, AI, signal transduction, microbiology.
Experience in EU / international / multi-partner collaborative projects	I was a postdoctorant in Linköping University, Sweden (2000-2003)
Experience with clinical, translational or patient-facing research (if relevant)	☐ Yes x No ☐ Not applicable | Short note:

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

1.Faten Abdelli, Karim Jellali, Estefanía Anguita, María González-Muñoz, Eduardo Villalobo, Ivan Madroñal, Juan Alcalde, Mamdouh Ben Ali, Jihene Elloumi-Mseddi, Ikram Jemel, Francesc Tebar, Carlos Enrich, ***Sami Aifa****, Antonio Villalobo. The role of the calmodulin-binding and calmodulin-like domains of the epidermal growth factor receptor in tyrosine kinase activation. J Cell Physiol. 2021 Jul;236(7):4997-5011. doi: 10.1002/jcp.30205.*

2. Dorra Abdelmalek; Fahmi Smaoui; Fakher Frikha; Riadh ben Marzoug; Dhouha Msalbi; Amal Souissi; ***Mohamed Sami Aifa*** *(2023). Computational identification of new TKI as potential noncovalent reversible EGFR$^{L858R/T790M}$ inhibitors: VHTS, molecular docking, DFT study and molecular dynamic simulation. J Biomol Struct Dyn. 2023 Jun 22:1-18. doi: 10.1080/07391102.2023.2223663*

3. Msalbi, D., Jellali, F., Elloumi-Mseddi, J. Hakim B, Sahli E and ***Aifa S****. Toxicity evaluation of synthetic glucocorticoids against breast cancer cell lines MDA-MB-231, MCF-7 and human embryonic kidney HEK293. Med Oncol* ***40****, 309 (2023). https://doi.org/10.1007/s12032-023-02189-1*

4. Fares Elghali; Dhouha Msalbi; Fakher Frikha; Mona Alonazi; Emna Sahli; Bochra Hakim; Sami Mnif; Abir Ben Bacha; ***Sami Aifa*** *(2024) Evaluation of the Combinatory Anticancer Effect of Chemotherapeutic Compounds and Prodigiosin against HCT-116, LoVo, and A549 Cell lines. ACS Omega. DOI: 10.1021/acsomega.4c04760*

5. Fakher Frikha, ***Sami Aifa*** *(2025). Evaluation of N-palmitoylethanolamine (PEA) binding to nuclear receptors through docking and molecular dynamics studies, Computers in Biology and Medicine, Volume 184, 2025, 109421, https://doi.org/10.1016/j.compbiomed.2024.109421.*

Up to 5 relevant previous projects, grants, networks or collaborations

International Projects as PI

- AP/040803/11: Collaborative Project Tunisia-Spain with Pr Antonio VILLALOBO (Instituto de Investigaciones Biomedicas-Madrid) for 2012: « Importance de l'interaction calmoduline/EGFR dans l'activité tyrosine kinase du récepteur ».
- A/019018/08: Collaborative project Tunisia –Spain with Pr Antonio VILLALOBO (Instituto de Investigaciones Biomedicas-Madrid) for 2009: "Nuevos mecanismos de regulacion del receptor del factor de creciemiento epidermico en tumores" A/019018/08.
- A/5444/06 and A/8197/07: Collaborative Project Tunisia-Spain with Pr Antonio VILLALOBO (Instituto de Investigaciones Biomedicas- Madrid) for 2007 and 2008: "Estudio de la interaction CaM-BD/CaM-LD del receptor del factor de crecimientp epidérmico (EGFR) en tumores".
- COOPA20053: Project I-coop+ (spain) with Prof. Antonio Villalobo 01/01/2015-31/12/2016: "The CaM-BD/CaM-LD of ErbB2 as therapeutic targets in breast cáncer.
National project as a partner
- PRC2024-D3P06 : « Les dystrophies musculaires en Tunisie : apport de l'étude génétique, épigénétique et fonctionnelle pour un diagnostic précis et précoce en vue d'une thérapie ciblée ». For 2025-2027.

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)

National Patents

*1-Mehdi EL ARBI, Ali RHOUMA, Pascal PIGEON, Siden TOP, Ahmed REBAI, **Sami AIFA**, Gérard JAOUEN (2011). Les ferrocenyl diaryl butenes : des composes organométalliques non phytotoxiques et d'intérêt phytopharmaceutique contre les phytopathogènes de l'olivier et des autres fruitiers. TN 2011/0077*

*2-Mehdi EL ARBI, **Sami AIFA**, Pascal PIGEON, Siden TOP, Meral GÖRMEN et Gérard JAOUEN (2011). Fort potentiel antiseptique des citrates de diaryl butènes portant deux chaînes aminoalkyle. TN2011/0604.*

*3-Jihene Elloumi, **Sami Aifa** (2012). Développement d'un système de criblage des inhibiteurs de tyrosine kinase à partir de l'expression de GST-EGFR chez E.coli. TN2012/0409*

*4-Sami Mnif, Marwa Jardak, **Sami Aifa** (2018). Développement d'un kit de nettoyage du biofilm mono-espèce de Macrococcus caseolyticus. TN2018/0364*

*5-Haithem Elleuch, Dhouha Msalbi, **Mohamed Sami Aifa**, Farhat Rezgui (2021)*
Nouveaux sels de phosphoniums de Morita-Baylis-Hillman : Synthèse et propriétés anticancéreuses. TN2021/0084.

*6- Emna Choura, Farès ElGhali, **Sami Aifa** et Fakher Chabchoub (2022)*
*Dérivés benzochroménopyrimidiniques pour l'inhibition de la prolifération des cellules du cancer du côlon. **TN 2022/0148.***

*7- Manel Ben Hsan, Dhouha Msalbi, **Sami Aifa**, Fakher Chabchoub (2022)*
*Composés triazolo[4,3-a] pyrimidiniques utilisés comme agents antiprolifératifs du cancer du sein. **TN2022/0202**.*

*8- Emna Choura, Farès ElGhali, **Sami Aifa** et Fakher Chabchoub (2023)*
Cytotoxicité des dérivés 3-phénylbenzochroménopyrimidiniques synthétisés contre les
*Lignées tumorales du colon et du poumon. **TN2023/0110**.*

*9- Sami Mnif, Hajer Jlidi, Sami Aifa, Ikram Jemel, Yosra Msaed (2025). Kit de détection des biofilms bactériens par traitement enzymatique couplé à la nanotechnologie. **TNP2025/0081**.*

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	x Yes ☐ Maybe ☐ No capacity to send staff on secondment

Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	x Only Researchers ☐ Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	x Academic only ☐ Non-academic only ☐ Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	Visa and insurance

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
In silico screening of drugs*: we are able to screen chemical libraries and select, by high-throughput docking and molecular dynamics, the most potent compounds that are able to target a specific protein.* Cellular cytotoxicity evaluation: we are culturing tumor and non-tumor cell lines in the presence and in the absence of the tested compounds and follow cell viability, necrosis, apoptosis while determining the corresponding IC50 compared to standard compounds. Signal transduction examination: the specific cytotoxicity of some compounds needs the examination of signaling pathways in order to select antagonist for specific pathways through western blot analysis and specific antibodies. Antibiofilm screening: We have developed bacterial biofilm models which we incubate with the tested compounds and evaluate the anti-adhesion and the eradication ability towards biofilms. Quorum sensing examination: the antibiofilm compounds are further screened for their ability to interfere with bacteria quorum sensing mechanisms.

Capabilities or expertise you would especially like to access through the consortium
We have a zebra fish lab and we need to more training in this model to update our skills in more experienced host lab within the consortium. We need to learn the CRISPR-Cas-based strategies in mammalian cells and/or zebrafish. We have a need to learn about the 3-D cell culture We intend to install the organoid as a model for drug testing and we need training in this approach. We are involved in a submitted national consortium of precision medicine and we need to learn more by training in experienced team host. We try to integrate the multiomics approach in our investigation and we need more training in this field (proteomics, transcriptomics, metabolomics with genomics*)* *We intend to build a sustainable collaboration with the different teams of the present consortium for better sharing of experiences and methodologies.*

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	x Yes ☐ No
x We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
None

Internal use and confidentiality statement

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X Understood

Name: Sami Aifa
Date completed: 06/06/2026

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)

InCELLia Participant Profile

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organization (SMEs, NGOs , patient organizations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	INCELLIA IKE
Short name / acronym used in projects	InCELLia
Participant Identification Code (PIC), if available	888772686
Website(s)	https://incellia.com/
Country	Greece
Organisation type	☐ Academic ☑ Non-academic ☐ Hospital / clinic ☑ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Dimitriou Gounari 96, Amarousion, 15125, Athens Greece
Department(s), institute(s), research group(s) or unit(s) that would be involved	InCELLia research team in mathematical modelling, biomedical data analysis and experimental model integration, led by Dr Georgios Lolas.
Additional institutional resources relevant to collaboration	Computational models and analysis workflows; expertise in transparent fish models, including Danionella cerebrum; collaborations linking modelling with biological experiments. Access to specific partner facilities or datasets would require a separate agreement.

Main contact person for this profile	Dr Georgios Lolas, Founder and Legal Representative; applied mathematician specialising in developing mathematical models for biological processes (with special focus on cancer); scientific contact for InCELLia.
Main contact email	glolas@incellia.com
Other contact person(s), if relevant	N/A

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	☐ Retina
☐ Inherited retinal dystrophies (IRDs)	☐ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
☐ Advanced therapies / regenerative medicine	☐ Gene therapy / cell therapy / tissue engineering
☑ **Inflammation / biomarkers / neuroprotection**	☑ **Drug delivery / formulations / nanomedicine**
☑ **Drug discovery / screening / repurposing**	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☑ **AI-assisted diagnosis / software / digital tools**
☐ Clinical trials / translational validation	☑ Other: mechanistic disease modelling; Bayesian inference; longitudinal data analysis; transparent fish models (*danionella cerebrum*). Eye-health applications would be developed with specialist partners.

Free Keywords for internal searchable database (please provide up to 10)	Mechanistic modelling; machine learning; Bayesian inference; Danionella cerebrum; zebrafish; drug-response prediction; biomarkers; longitudinal imaging; drug-delivery modelling; experimental design.
Geographies of interest for exchanges (optional)	International collaboration, including IOBA in Spain. Destinations for any outgoing secondments would be agreed according to scientific fit and staff availability.

Overview of your research / innovation focus
InCELLia is a Greek research SME combining mathematical modelling, machine learning and experimental model integration to study disease mechanisms, toxicity and treatment response. Its main application area is oncology, including primary tumour growth, metastatic progression and the evaluation of therapeutic candidates. The team develops ordinary and partial differential equation models and structured population models, fits them to experimental observations, and uses Bayesian and statistical methods to quantify uncertainty and biological variability. InCELLia connects these computational approaches with research using transparent fish models, including Danionella cerebrum, and with longitudinal measurements from collaborating experimental teams. Its contribution to the ALADDIN project includes mathematical models for therapeutic target identification and preclinical validation of nanobodies. For an eye-health consortium, InCELLia could adapt these methods to questions selected with ophthalmology partners, such as inflammation, tissue damage and repair, drug exposure and treatment response. It could integrate imaging, biomarker and molecular measurements to compare biological hypotheses, identify useful measurements and support experimental planning. These would be jointly developed research applications, with disease-specific validation provided through the consortium.

Why your team could be relevant to a translational eye-health and vision consortium
InCELLia could help turn experimental and clinical observations into quantitative explanations and testable predictions. Models of inflammation, cell survival and tissue responses could be adapted to ocular disease, while transport and exposure models could support the comparison of drug-delivery strategies. Machine learning could help assess candidate biomarkers and combine imaging with molecular measurements, subject to sufficient data and independent validation. Its experience with transparent fish models could also support feasibility studies for longitudinal measurements and treatment screening in collaboration with eye-research specialists. The main added value is a repeated exchange between modelling and experiments: observations inform the model, predictions guide the next experiment, and new results test those predictions.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute
Mechanistic modelling of biological dynamics using ordinary and partial differential equations, structured population models and spatial models. Potential applications include disease progression, inflammatory signalling, cell interactions and treatment response. Bayesian inference, parameter estimation, mixed-effects analysis, model comparison, sensitivity analysis and uncertainty quantification. These methods help assess which conclusions are supported by the available measurements and which additional data would be useful. Machine learning and statistical analysis of biomarker, molecular, imaging-derived and longitudinal data, including feature selection and predictive model assessment. Integration of mechanistic models with data-driven approaches where this improves interpretation or prediction. Drug-exposure, transport and response modelling, including physiologically based pharmacokinetic and population balance approaches where appropriate. Design and interpretation of studies using transparent fish models, with model predictions compared against experimental observations.

Infrastructure, equipment, facilities or datasets relevant to consortium work
InCELLia can contribute computational models, numerical simulation and data-analysis workflows, together with expertise in Danionella cerebrum research. Specific experimental resources, imaging access and study datasets would be defined with the participating partners. Clinical cohorts, ocular samples and specialist ophthalmic facilities would need to be provided through those collaborations. This profile does not commit external facilities or datasets to the consortium.

Training capacity and knowledge-transfer potential
InCELLia **cannot** host incoming staff. It can contribute knowledge transfer through agreed outgoing secondments, workshops at partner institutions and remote mentoring. Topics include translating biological questions into mathematical models, fitting models to data, evaluating uncertainty, using machine learning responsibly, and linking experimental design with computational predictions. Training can use shared case studies and reproducible analysis workflows.

What your team would hope to gain from participating
InCELLia would seek ophthalmology expertise, disease-specific experimental models, longitudinal ocular imaging and biomarker data, and access to relevant validation studies through collaboration. The team would benefit from learning which outcomes matter to patients and clinicians, how ocular therapies are assessed, and how processes like angiogenesis (that are similar to cancer development) contribute to treatment response/success and how its modelling and experimental approaches can be adapted to these needs. Joint publications and durable academic-industry research partnerships are also of interest.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Dr Georgios Lolas	Founder and Legal Representative; scientific lead	InCELLia research team	Researcher	Experienced researcher	glolas@incellia.com Orcid: 0000-0002-2236-5209
Aris Chatzichristos	Physics modeler, Data analyst	InCELLia research team	Researcher	Junior Researcher	

Main disciplines represented in your team	Applied mathematics, mathematical biology, computational oncology, statistics, machine learning and biomedical data integration, linked with experimental biology through collaborative research.
Experience in EU / international / multi-partner collaborative projects	Beneficiary in ALADDIN, a Horizon Europe EIC Pathfinder project, contributing mathematical modelling for target identification and preclinical validation within an international multidisciplinary consortium.
Experience with clinical, translational or patient-facing research (if relevant)	☑ Yes ☐ No ☐ Not applicable Translational and preclinical research through ALADDIN and model-experiment integration. Any patient-facing work would be undertaken by clinical partners.

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

1. Gavriliadis P, Lolas G, Matsoukas T. Age and generation-based model of metastatic cancer: From micrometastases to macrometastases. Journal of Theoretical Biology 623 (2026), 112392. Demonstrates expertise in structured population modelling and the quantitative interpretation of disease progression. https://doi.org/10.1016/j.jtbi.2026.112392
2. Kavidopoulou A.,…., Lolas G. (2022), AI and Big Data for Drug Discovery. In: Sakly, H., Yeom, K., Halabi, S., Said, M., Seekins, J., Tagina, M. (eds) Trends of Artificial Intelligence and Big Data for E-Health. Integrated Science, vol 9. Springer, Cham. A review on the latest advancements on AI and Big Data for Drug Discovery
3. Dlamini, Z., Syrigos K.N., Evangelou G., Vamvakaris I., Lolas G. (2022). AI and Big Data for Cancer Segmentation, Detection and Prevention. In: Sakly, H., Yeom, K., Halabi, S., Said, M., Seekins, J., Tagina, M. (eds) Trends of Artificial Intelligence and Big Data for E-Health. Integrated Science, vol 9. Springer, Cham. A review on cancer segmentation using AI approaches.
4. Ali Z, ... G, Lolas G, Syrigos KN, Jensen LD. Zebrafish patient-derived xenograft models predict lymph node involvement and treatment outcome in non-small cell lung cancer. J Exp Clin Cancer Res. 2022 Feb 9;41(1):58. doi: 10.1186/s13046-022-02280-x. A collaboration effort between NKUA, INCELLIA and LiU to justify that ZTX platform can provide fast accurate and clinical treatment outcomes using NSCLC tumor biopsies (patient derived) send from Greece.
5. Lolas G, Bianchi A, Syrigos KN. Tumour-induced neoneurogenesis and perineural tumour growth: a mathematical approach. Sci Rep. 2016 Feb 10;6:20684. doi: 10.1038/srep20684. PMID: 26861829; PMCID: PMC4748234.

Contribution to ALADDIN modelling workflows for therapeutic target identification and preclinical validation. The public consortium profile documents InCELLia involvement in mathematical modelling across these activities. https://aladdin-project.eu/consortium/

Up to 5 relevant previous projects, grants, networks or collaborations

1. ALADDIN — Accelerated Discovery Nanobody Platform. Funding: European Union, Horizon Europe, EIC Pathfinder Open. Grant Agreement 101130574. Duration: 1 January 2024 to 31 December 2027. Role: beneficiary/partner in an international consortium. InCELLia contributes mathematical models for target identification and preclinical validation, linking computational analysis with experimental evaluation of therapeutic candidates. https://cordis.europa.eu/project/id/101130574
2. ''Emerging mechanisms of tumor-lymphangiogenesis in metastasis: bridging the gaps with a Systems Medicine approach for improve preventive strategies". Our ultimate goal is to develop mathematical and computational models that highlight the crucial role of the lymphatic system in metastasis. In collaboration with National Kapodistrian University of Athens (NKUA).
3. Detection of lung cancer cells on Whole Slide Images using Deep Convolution Neural Networks" We assist the clinical team of the Oncology Unit of NKUA to develop a deeplearning- based CAD method for lung biopsy tissue classification, to optimise and assist histopathological imaging outcomes. In collaboration with UOA.
4. Slow-dividing E.coli competes with cancer cell lines for Arginin"
   Highlight: identify, fey extracellular nutrients can be exploited in tumor therapy: depletion of nutrients, degradation of enzymes, and inhibition of nutrient uptake and utilization. In collaboration with UOA.
5. "Development of an Artificial Intelligence Predictive model for Metastasis based on patient derived Zebrafish xenografting imaging data", a joined work with INCELLiA, UOA and LiU. The aim of this joined work is to develop a predictive model based on patient derived xenograft imaging data in order to identify the metastatic propensity and aggressiveness of the initial primary tumor. The tumor under consideration is NSCLC.

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)
Shareable know-how includes mathematical model formulations, calibration methods and computational analysis workflows for biological systems and therapeutic response. Access to existing code, models and project data would be agreed for each collaboration, with background intellectual property and confidentiality clearly defined. Any ophthalmology-specific application would require dedicated development and validation.

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organization's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organization/team second staff?	☐ Yes ☑ Maybe ☐ No capacity to send staff on secondment Outgoing secondments can be discussed. Staff numbers, duration and total person-months are not yet committed.
Could your organization second staff categories beyond researchers (where relevant to R&I activities)?	☐ Only Researchers ☐ Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff ☐ Other profiles Research staff in modelling and data analysis are the envisaged profile. Final staff categories remain to be agreed.
Could your organization host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only ☐ Both ☑ No capacity to host InCELLia cannot host incoming staff. Incoming secondment allocation: 0 staff and 0 person-months.
Administrative / HR / legal considerations that may affect mobility	Any outgoing mobility depends on staff availability, continuity of company activities, an agreed research and training plan, and suitable arrangements at the receiving organisation. Employment, insurance, access, confidentiality and intellectual-property matters would be settled before an exchange. No incoming secondments should be assigned to InCELLia.

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
Mechanistic models that explain disease dynamics and treatment response; Bayesian and machine-learning methods that combine biological data with quantified uncertainty; experience connecting computational predictions with longitudinal observations in transparent fish models.

Capabilities or expertise you would especially like to access through the consortium
Ophthalmology and ocular biology teams; specialists in inflammation and neuroprotection; drug-delivery and nanomedicine groups; ocular imaging and biomarker datasets; and experimental or clinical partners able to evaluate jointly developed models against relevant outcomes.

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	☑Yes ☐ No

☐ We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
InCELLia cannot host incoming staff. Its contribution should therefore be planned around research collaboration, agreed outgoing secondments and knowledge-transfer activities delivered with partner organisations. Any outgoing staff allocation remains non-binding until the scientific tasks, timing and resources are agreed. The proposed ocular applications build on transferable modelling, data-analysis and experimental expertise. Specific disease models, datasets, experimental responsibilities and validation endpoints should be defined jointly with IOBA and the relevant partners. Access to background models, code and data, as well as ownership and use of jointly developed results, should be agreed before the work starts.

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

☐ **Understood**

Name: Dr Georgios Lolas

Date completed: 17 September 2026

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)

Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
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Important:
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- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	International Iberian Nanotechnology Laboratory
Short name / acronym used in projects	INL
Participant Identification Code (PIC), if available	If no PIC is available yet, please indicate 'not yet available'.
Website(s)	Inl.int
Country	Portugal
Organisation type	☐ Academic x Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body x Other
Main address	Avenida Mestre José Veiga s/n, 4715-330 | Braga, Portugal
Department(s), institute(s), research group(s) or unit(s) that would be involved	Pastrana Research Group Conference, Communication and marketing Unit
Additional institutional resources relevant to collaboration	https://inl.int/services/research-core-facilities/

Main contact person for this profile	Lorenzo Pastrana, Professor, Research Group Leader
Main contact email	Lorenzo.pastrana@inl.int
Other contact person(s), if relevant	Marian Debs

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	☐ Retina
☐ Inherited retinal dystrophies (IRDs)	☐ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
☐ Advanced therapies / regenerative medicine	☐ Gene therapy / cell therapy / tissue engineering
☐ Inflammation / biomarkers / neuroprotection	x Drug delivery / formulations / nanomedicine
☐ Drug discovery / screening / repurposing	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	☐ Other (please specify):

Free Keywords for internal searchable database (please provide up to 10)	Biopolymers, responsive, functionalization, micro and nanoencapsulation, morphological and functional characterization, controlled release
Geographies of interest for exchanges (optional)	No limits

Overview of your research / innovation focus
Pastrana's Research Group pioneers innovative nanotechnology-based solutions to address global challenges and build a healthier, sustainable, fair, and digital food system. This group covers TRL 2 to 7 by developing new active and smart functionalities using biomaterials based on their rheological/textural properties to engineering controlled and smart drug delivery systems by using a large variety of micro and nanoencapsulation technologies including nanospraydryer and ultrasonic atomization among others. In addition, the morphological (TEM, SEM CryoTEM, XPS, SAXS) and functional characterization of the micro and nanoparticles is also a research line. The group develop as well organ on a chip microfluidic devices for assessing digestibility of nanomaterials

Why your team could be relevant to a translational eye-health and vision consortium
Pastrana's Research Group at INL could bring strong added value to a consortium focused on visual health and translational ophthalmology through its expertise in nanotechnology, biomaterials, controlled delivery systems and advanced micro/nanoencapsulation technologies. Although the group's core application area is the development of healthier, sustainable and digital food systems, many of its enabling technologies are highly transferable to ophthalmology, particularly where the field requires new strategies for protecting, stabilising and delivering sensitive bioactive compounds. A key contribution would be the group's experience in engineering biomaterial-based systems with tailored rheological, textural and functional properties. This expertise could support the design of advanced formulations, soft materials or biointerfaces relevant to ocular applications, where viscosity, stability, adhesion,

controlled release and patient acceptability are critical parameters. The group's work on active and smart functionalities could also inspire new approaches for responsive or multifunctional ocular delivery platforms.

The team could also contribute its strong know-how in micro- and nanoencapsulation, including technologies such as nanospray drying and ultrasonic atomisation. These approaches could be explored for the encapsulation, protection and controlled release of therapeutic molecules, antioxidants, anti-inflammatory compounds, peptides, nucleic acids or other sensitive agents relevant to visual health. This could help address common translational barriers in ophthalmology, such as poor stability, limited bioavailability, short residence time and inefficient delivery across biological barriers.

Another important added value is the group's capacity for advanced morphological and functional characterisation of micro- and nanoparticles, using techniques such as TEM, SEM, CryoTEM, XPS and SAXS. This expertise could support the rational design and validation of nanostructured systems by linking particle morphology, surface properties, internal structure and functional performance.

Finally, the group covers a broad translational range from TRL 2 to TRL 7, which is particularly relevant for a consortium aiming to move from research concepts towards patient benefit. Its interdisciplinary background in nanotechnology, biomaterials, formulation engineering and functional materials could open new routes for ophthalmic innovation, especially in areas such as smart ocular formulations, sustained delivery systems, protective bioactive carriers, diagnostic-enabling materials or sustainable biomaterial platforms.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute
Responsive and smart bioactive delivery systems based in biopolymers *Functional modification of biopolymers properties by self-assembling methodologies* *Organ on a chip microfluidic devices for assessing physiological behaviour* *Advanced micro and nanoencapsulation technologies (nanospraydryer, ultrasonic atomization)* *Morphological characterization of micro and nanoparticles* *Functional characterization of biopolymers*

Infrastructure, equipment, facilities or datasets relevant to consortium work
Electron Microscopy and X-Rays *The Electron Microscopy and X-Rays (EMX) facility is a core multi-user facility that features cutting-edge instrumentation, techniques and expertise required for the characterisation of samples in the physical and life sciences.* *This facility focuses on materials and biological research, the development of novel techniques and instrumentation, as well as providing training, technical support and consultation in the areas of electron microscopy and spectroscopy. The centre facility houses several electron microscopes that can probe the physical, electronic, biological, and chemical structure of matter down to the atomic scale.* *These instruments are coupled with advanced in-situ holders, in which the environment is controlled to match near-realistic conditions of operation, and the sample's behaviour is recorded dynamically in real time. State-of-the-art support facilities also are available, including standard specimen preparation equipment and an image analysis laboratory.* • *Transmission Electron Microscopy (TEM)* • *Cryogenic Transmission Electron Microscopy (Cryo-TEM)* • *Scanning Electron Microscopy (SEM)* • *X-Ray Photoelectron Spectroscopy (XPS)* • *X-Ray Diffraction (XRD)* • *Small-Angle X-Ray Scattering (SAXS)*

- *Sample preparation for electron microscopy*
- *In Situ TEM Holders*
- *Software for imaging analyses*

Material Testing

The Material Testing facility uses complementary techniques to determine the physical and chemical properties of the different types of materials.

Thermogravimetric analysis (TGA) is used to measure changes in the weight of a sample as a function of temperature while the sample is subjected to a controlled temperature program in a controlled atmosphere. It is commonly used to characterize the thermal stability, composition, and decomposition kinetics of materials. Nitrogen adsorption-desorption isotherms at -196 °C are used to characterise the porosity and surface area of materials. ICP-OES (Inductively Coupled Plasma Optical Emission Spectroscopy) is used to determine the elemental composition of materials. UV-accelerated weathering tester is used to simulate the effects of sunlight exposure on materials. These chambers are designed to subject samples to controlled levels of ultraviolet (UV) radiation, temperature, and humidity, replicating the conditions that materials experience when exposed to outdoor environments over extended periods. Salt spray testing is a standardised method used to evaluate the corrosion resistance of materials and coatings.

- Thermogravimetric Analysis / Differential Scanning Calorimetry (TGA/DSC)
- Optical Emission Spectrometry (ICP – OES)
- Nitrogen adsorption-desorption isotherms at -196 ºC
- Taber Abrasion Testing
- Salt Spray and Cyclic Corrosion Testing
- Accelerated Weathering Testing
- Universal Materials Testing
- Rheology of Soft Materials

Core fabrication areas

- (Thin) film deposition and Material Growth
- Etching, Ashing and Micromachining
- Optical and E-beam Lithography
- Advanced Packaging, Annealing and Back-End Processes
- Metrology, Inspection and Wafer-Scale Device Testing

Nanophotonics & Bioimaging

The Nanophotonics & Bioimaging (NBI) facility features cutting-edge optical instrumentation and the know-how required for the characterisation of material and life sciences samples.

Techniques available

Advanced optical microscopes

In-house custom-developed microscopes and spectroscopy solutions

Materials, particles, molecules and cells analysis

Optical spectroscopy solutions

Nonlinear microfabrication

Data analysis for microscopy, spectroscopy and flow cytometry

- Advanced Optical Microscopy
- Optical Spectroscopy
- Nanoparticle, Protein and Cell analysis
- Custom Multi-Modal Microscopy and Microfabrication Solutions

Training capacity and knowledge-transfer potential
Scientific: The technologies to manipulate biopolymers to obtain new functionalities The technologies for micro and nanoencapsulation The methodologies for assessing bioactive delivering

Technical:

The INL Postgraduate School is designed especially for Phd candidates pursuing their degree at INL, to equip them with essential skills beyond the lab, offering tailor-made training to enhance their academic and professional growth. The INL Postgraduate School offers complementary and transversal training designed to enhance specific skills that support researchers in their future careers – whether in academia, industry, or entrepreneurship. These training activities help PhD candidates build a well-rounded professional profile, empowering them with a versatile skill set to succeed across diverse career paths. Throughout their time at INL, PhD candidates receive personalised guidance to support their personal and professional development. This includes a structured series of workshops, masterclasses, and peer discussions tailored to each stage of their PhD journey. In the first year, the focus is on career planning, self-assessment, and navigating the early stages of the PhD. The second year emphasises time management, emotional intelligence, and networking skills. In the final year, candidates are prepared for the job market with targeted training on CV writing, interview techniques, and exploring both academic and non-academic career paths.

Others:

The INL Postgraduate School also offer courses in the following topics:

Communication
Data Management
Leadership
Teamwork
Ethics and Responsible Research
Entrepreneurship
Innovation
Career Management

What your team would hope to gain from participating

Examples: access to methods, models, technologies, patient-oriented validation, translational pathways, industrial insight, clinical exposure, data science, regulatory know-how, joint publications or longer-term strategic collaboration. ***Again, please think broadly and beyond ophthalmology.***

Pastrana Research Group would hope to gain access to a highly interdisciplinary environment where its expertise in biomaterials, micro- and nanoencapsulation, smart functional systems and advanced particle characterization can be translated beyond its current core areas and explored in the context of visual health and patient-oriented innovation. Participation in the consortium would provide the group with valuable exposure to ophthalmology-related needs, clinical challenges and biological models, helping to identify where its enabling technologies could contribute to new delivery platforms, protective biomaterials, responsive formulations or advanced diagnostic-supporting materials. This would be particularly relevant given the group's experience in smart, active and sustainable material systems, micro- and nanostructures, and multifunctional material design.
The team would also benefit from access to translational pathways that are less present in food and biomaterials research, including patient-oriented validation, clinical requirements, regulatory expectations, safety assessment and industrial routes towards medical or health-related applications. Such knowledge would help the group better understand how technologies developed at early or mid TRL levels can be adapted, validated and de-risked for use in more demanding biomedical contexts.
In addition, the consortium would offer opportunities to learn from partners with complementary expertise in ophthalmic models, clinical data, regulatory strategy, device or formulation development, and market-oriented innovation. This could open new research directions for the group, particularly in the adaptation of encapsulation technologies, biomaterial platforms and advanced characterization tools to health-related applications.

Finally, Pastrana Research Group would expect to gain long-term strategic value through joint publications, shared methodologies, cross-sector training, new project ideas and durable collaborations with clinical, academic and industrial partners. Beyond ophthalmology, the consortium could help the group expand its translational capacity and position its nanotechnology-based platforms as versatile enabling technologies for broader health and sustainability challenges.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Lorenzo Pastrana	Research group leader		researcher	experienced	Lorenzo.pastrana@inl.int
Mariam Debs	Officer	Communication	managerial	experienced	Mariam.debs@inl.int

Main disciplines represented in your team	biomaterials, encapsulation, bioactive delivery, assessment methodologies, marketing and commercialisation.
Experience in EU / international / multi-partner collaborative projects	Participation in the following projects: Crystal 3, 3DNenonet and coordinate FODIAC
Experience with clinical, translational or patient-facing research (if relevant)	x Yes ☐ No ☐ Not applicable | Short note: Participation in the European project Bugs for urate

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

Please prioritise achievements most relevant to your possible contribution. Please include DOI, URL or identifier.

Pirouni, A.; Drosou, C.; Koskinakis, S.E.; Stergiopoulos, C.; Amado, I.R.; Fuciños, P.; Pastrana, L.; Mishra, P.; Krokida, M. Development of Antimicrobial Textile Coatings Through Encapsulation of ZnO in Electrospun PLA Fibers. *Coatings* 2026, *16*, 672. https://doi.org/10.3390/coatings16060672

Artur J. Martins, Lara Perdigão, Catarina Gonçalves, Isabel R. Amado, Cristiano S. Abreu, António A. Vicente, Rosiane L. Cunha, Lorenzo M. Pastrana, Miguel A. Cerqueira,
Beta-carotene-loaded Oleogels: Morphological analysis, cytotoxicity assessment, in vitro digestion and intestinal permeability, Food Chemistry, Volume 465, Part 2, 2025, 142085, ISSN 0308-8146, https://doi.org/10.1016/j.foodchem.2024.142085.

Marta Vinha Vieira, Shaba Noore, Brijesh Tiwari, Colm O'Donnell, Catarina Gonçalves, Lorenzo M. Pastrana, Pablo Fuciños, Enhancing the stability and functionality of phycobiliproteins as natural food colourants through microparticle formulation, Food Chemistry, Volume 465, Part 2, 2025, 142077, ISSN 0308-8146, https://doi.org/10.1016/j.foodchem.2024.142077

Calero, V., Rodrigues, P.M., Dias, T. et al. A miniaturised semi-dynamic in-vitro model of human digestion. Sci Rep 14, 11923 (2024). https://doi.org/10.1038/s41598-024-54612-w

Ana M. Pinto, Raquel Pereira, Artur J. Martins, Lorenzo M. Pastrana, Miguel A. Cerqueira, Sanna Sillankorva,
Designing an antimicrobial film for wound applications incorporating bacteriophages and ε-poly-l-lysine,
International Journal of Biological Macromolecules, Volume 268, Part 2, 2024, 131963, ISSN 0141-8130,
https://doi.org/10.1016/j.ijbiomac.2024.131963.

Up to 5 relevant previous projects, grants, networks or collaborations

Please provide detailed information (Title, Acronym, funding agency, GA-number, dates). Please indicate your role (coordinator / beneficiary / partner / service provider / collaborator). Indicate if international.

Title: COVICOAT Antiviral edible coating for fresh food
Funded by: EIT Food

Partners: International Iberian Nanotechnology Laboratory and EROSKI
Date: 2020 -2020
Coordinator: Lorenzo Pastrana

Title: TrustEat Building a trusty future food system by using blockchain tech
Funded by: H2020
Partners: International Iberian Nanotechnology Laboratory, WUR and IBM,
Date: 2020 -2023
Coordinator: Lorenzo Pastrana

Title: CRYSTAL3 Commercial & Research Opportunity for Cysteinyl Leukotriene Signalling in Ocular & CNS Dysfunction, Cancer and Cardiovascular Disease
Funded by: H2020
Partners: International Iberian Nanotechnology Laboratory and more
Date: 2021 -2024
(WP): Lorenzo Pastrana

Title: IBEROS+. Instituto de biofabricación en red para el envejecimiento saludable
Funded by: POCTEP
Partners: International Iberian Nanotechnology Laboratory, Universidad de Vigo (coordinador), Universidade do Porto and more
Date: 2023 -2026
(WP): Lorenzo Pastrana

Title: FEAST Fostering European cellular Agriculture for Sustainable Transition Solutions
Funded by: H2020
Partners: International Iberian Nanotechnology Laboratory Universidade de Lisboa, WUR, GFI and more
Date: 2023 -2026
(WP): Lorenzo Pastrana

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)
Please indicate only what can be shared at this stage.

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	x Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	x Only Researchers ☐ Technical Staff/Innovation x Managerial / project support x Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only x Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	e.g. visa, insurance, access to facilities, language, employment status, health & safety induction

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
Please summarise the 2-3 strengths that should be first visible in an internal searchable profile. Encapsulation technologies Biopolymers engineering Organ on a chip devices

Capabilities or expertise you would especially like to access through the consortium
Please indicate the types of teams, methods, infrastructures or translational pathways you would like to connect with in the consortium. Access to cutting-edge infrastructures and specialised technologies Improved translational and methodological standardisation Long-term international collaboration frameworks

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	x Yes ☐ No
☐ We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
Examples: PIC pending approval, institutional approval, legal status, hosting rules, IP constraints, data access, ethics, clinical permissions, staffing continuity, resource availability. The participation has to be approved by the DG

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

x Understood

Name: Lorenzo Pastrana
Date completed: 2 /6 / 2026

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)

Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	INM-Leibniz Institute for New Materials
Short name / acronym used in projects	INM
Participant Identification Code (PIC), if available	If no PIC is available yet, please indicate 'not yet available'.
Website(s)	www.inm-leibniz.inm.de https://www.leibniz-inm.de/en/research/scientific-units/dynamic-biomaterials/
Country	Germany
Organisation type	X Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Campus D2 2; 66123 Saarbrücken, Germany
Department(s), institute(s), research group(s) or unit(s) that would be involved	Dynamic Biomaterials
Additional institutional resources relevant to collaboration	Core Facilities at INM for: • Electron Microscopy, https://www.leibniz-inm.de/en/research/core-facilities/electron-microscopy/ • Fluorescence Microscopy, https://www.leibniz-inm.de/en/research/core-facilities/fluorescence-microscopy/ • Accelerated Research Foundry, https://www.leibniz-inm.de/en/research/core-facilities/accelerated-research/ • Materials Physical Characterization • Bioprinting Laboratory

Main contact person for this profile	Aránzazu del Campo, Prof. Scientific Director and Department Head Dynamic Biomaterials
Main contact email	Aranzazu.delcampo@leibniz-inm.de
Other contact person(s), if relevant	Martina Bonnard, Secretary, martina.bonnard@leibniz-inm.de

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	☐ Retina
☐ Inherited retinal dystrophies (IRDs)	☐ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
X Advanced therapies / regenerative medicine	☐ Gene therapy / cell therapy / tissue engineering
☐ Inflammation / biomarkers / neuroprotection	X Drug delivery / formulations / nanomedicine
☐ Drug discovery / screening / repurposing	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	☐ Other (please specify):

Free Keywords for internal searchable database (please provide up to 10)	Ocular devices, sustained drug delivery, contact lenses, dry eye, biomaterials,
Geographies of interest for exchanges (optional)	

Overview of your research / innovation focus
The Research Department *Dynamic Biomaterials* at INM develops bioactive materials to encapsulate and instruct living cells. With our competences in hydrogel synthesis, photoswitches, biofabrication technologies, and cell biology, we create synthetic cellular microenvironments with dynamic properties and study how living and non-living matter interact. We exploit such interactions to regulate cellular functions by materials design. In cooperation with synthetic biologists and clinicians around the world, we contribute to the field of wearable/implantable living therapeutic devices for sustained and sustainable treatment of chronic diseases, both from fundamental research and application perspectives. These devices contain material-encapsulated drug biofactories that support on-site (in the body) production of the drug using energy sources from the environment. They represent a new concept for zero-waste pharmacology. We work on (i) understanding how material design affects bioactivity of biofactories, (ii) the parallelization of in vitro testing of living devices in simulated physiological conditions to predict bioactivity variability and changes in disease conditions, and on (iii) the design of advanced prototypes for technology transfer, with living drug-eluting contact lenses as advanced prototypes towards innovation.

Why your team could be relevant to a translational eye-health and vision consortium
- Expertise in biomaterials for application in ophthalmic devices

- Broad experience in cell-materials interactions and their impact in biocompatibility
- Experience in the design (from TRL 0) of biomaterials-based ophthalmic devices for the ocular surface
- Networked with contact lens industry and ocular pharma
- Currently going up the technology transfer ladder with an innovative drug eluting contact lens. Experience and network with innovators in ocular market
- Interested in networking with ophthalmic practitioners and partners in the ophthalmic market

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

- Expertise in biomaterials synthesis and purification: organic and hydrogel chemistry
- Expertise in hydrogel processing: soft-lithography, biofabrication (extrusion printing, volumetric printing, laser-assisted printing, ink-jetting)
- Expertise in soft materials characterization: mechanical, thermal, diffusion and optical properties
- Network of innovation experts and innovators in ocular devices

Infrastructure, equipment, facilities or datasets relevant to consortium work

- **Biofabrication equipment**
  - **Extrusion bioprinting:** CELLINK BIO X, GeSiM Bioscaffolder, RegenHU 3D Discovery, customized multimaterial thread fabrication station
  - **Electrospinning:** Bioinica Fluidnatek LE50
  - **Microfluidics:** Atrandi ONYX, Fluigent Raydrop
  - **Film coating and disposition:** Ossila Slot Die Coater
  - **Automated liquid handling:** Hamilton Microlab Nimbus
  - **Volumetric Printing:** Readily 3D Tomolite Performance 2.0
  - **Two-Photon lithography:** Nanoscribe Photonic Professional GT2
- **Optical property characterization equipment**
  - **Optical coherence tomography (OCT):** Thorlabs Ganymed OCT
  - **Refractive index measurement:** Anton Paar Abbemat Multiwavelength Refractometer
- **Mechanical characterization equipment**
  - **Rheological characterization:** Anton Paar MCR 702e, TA Instruments DHR3 Rheometer
  - **Nano-mechanical testing:** Optics11 Pavone Nanoindenter
  - **Dynamic mechanical analysis (DMA):** TA Discovery DMA 850
- **Fluorescence microscopy equipment**
  - **Total Internal Reflection (TIRF) microscopy:** Nikon TiE-TIRF
  - **Automated live-cell imaging and high-content microscopy:** ZEISS Celldiscoverer 7
  - **Light-sheet fluorescence microscopy:** ZEISS Lightsheet Z.1
  - **Super-resolution and nonlinear optical microscopy:** ZEISS LSM 880 Airyscan NLO
  - **Live-cell imaging and photomanipulation:** ZEISS Observer Z1 with Rapp OptoElectronic system
- **Electron microscopy equipment**
  - **Environmental scanning electron microscopy (ESEM):** FEI Quanta 250 FEG, FEI Quanta 400 FEG
  - **Focused ion beam-scanning electron microscopy (FIB-SEM):** FEI Versa 3D
  - **Transmission electron microscopy (TEM):** JEOL JEM-2100
- **Accelerated research facility**
  - **High-throughput liquid handling:** Echo 650 Acoustic Droplet Ejection System, Hamilton STAR
- **Electronic Documentation**
  - Aligned Elements
  - RSpace

Training capacity and knowledge-transfer potential

INM has a dedicated facility with specialized infrastructure for automating living therapeutic materials prototype fabrication and upscaling, including highly qualified personnel to run the interdisciplinary facilities and to support the establishment of a quality management system.

What your team would hope to gain from participating
- *Ophthalmology practitioners to provide feedback and eventually test our prototypes* - *In vitro and in vivo models of ocular tissues and pathological states* - *Experience in patients surveys, contact to patients and patient associations* - *Industrial insight* - *Regulatory knowhow* - *Long term strategic collaboration partners in the field*

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)

Main disciplines represented in your team	Biomaterials
Experience in EU / international / multi-partner collaborative projects	Participant of FET-OPEN and ITN Projects
Experience with clinical, translational or patient-facing research (if relevant)	☐ Yes X No ☐ Not applicable | Short note:

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements
Z. Kafrashian, J. Feng, and A. del Campo, *Fabrication Technologies for Soft, Multimaterial Optical Fibers for In Vivo Diagnostics and Phototherapy, With a Focus on Extrusion Printing.* *Advanced Materials Technologies 2026, 11, 9, e01789* *https://doi.org/10.1002/admt.202501789* *K. Desai, S. Sankaran, A. del Campo, S. Trujillo* *A screening setup to streamline in vitro engineered living material cultures with the host* *Materials Today Bio 2025, 30, 101437* *https://doi.org/10.1016/j.mtbio.2024.101437* *M. Puertas-Bartolomé, I. Gutiérrez-Urrutia, L. L. Teruel-Enrico, C. N. Duong, K. Desai, S. Trujillo, C. Wittmann, A. del Campo,* *Self-Lubricating, Living Contact Lenses.* *Advanced Materials 2024, 36, 2313848* *https://doi.org/10.1002/adma.202313848* *A. Rodrigo-Navarro, S. Sankaran, M.J. Dalby, A. del Campo, M. Salmeron-Sanchez,* *Engineered Living Biomaterials* *Nat Rev Mater 2021, 6, 1175* *https://doi.org/10.1038/s41578-021-00350-8* *S.Bhusari, S.Sankaran, A.delCampo* *Regulating Bacterial Behavior within Hydrogels of Tunable Viscoelasticity.* *Adv. Sci. 2022, 9, 2106026* *https://doi.org/10.1002/advs.202106026*

Up to 5 relevant previous projects, grants, networks or collaborations
DFG-SPP2451, Living Materials with Adaptive Functions (2025-28), PI and Coordinator *Leibniz SAW-Transfer, Transfer Readiness of a New Technology for Sustained Ocular Drug Delivery, PI and Coordinator* *BMFTR-MaterialVital Hub, MaterialVital Digital, PI* *Leibniz Science Campus Living Therapeutic Materials (2020-28), PI and Coordinator*

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)
Selbst-schmierende Kontaktlinsen. Filed on 09.11.2023 **(102023131356.8)** Lebendes Kompositmaterial für medizinische Produkte / Biokompatibles Polymer. Filed on 09.11.2023 **(102023131358.4)** Neuartige Hydrogele / Hydrogele Thio-methylsulfonyl Netzwerke für Biomaterialien, **EP 3994202 A1, JP 2022539551 A, KR 1020220031906 A, US 20220235316 A1** Leitfähige funktionalisierbare Nanokomposite / Leitfähige Tinten zur Herstellung von Elektroden, die spezifisch mit bioaktiven Molekülen, Zellen oder Geweben wechselwirken. **20129135645 A1, JP 2022539551 A, KR 1020220031906 A, US 20220235316 A1**

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	X Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	X Only Researchers X Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	X Academic only ☐ Non-academic only ☐ Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
- Biofabrication - Contact Lenses

Capabilities or expertise you would especially like to access through the consortium
- Clinical workflows - In vitro and in vivo models - Patient surveys - Manufacturing of ocular devices

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	X Yes ☐ No

☐ We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.

Internal use and confidentiality statement

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X Understood

Name: Aránzazu del Campo
Date completed: 27.08.2025

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)

Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	Linköpings Universitet
Short name / acronym used in projects	LIU
Participant Identification Code (PIC), if available	999852236
Website(s)	www.liu.se
Country	Sweden
Organisation type	☒ Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Campus Valla, 581 83 Linköping, Sweden
Department(s), institute(s), research group(s) or unit(s) that would be involved	Department of Health, Medical and Caring Science, Linköping University, 581 83 Linköping, Sweden
Additional institutional resources relevant to collaboration	Pre-clinical: Zebrafish facilities and lab. Facilities for research using mouse, rat, rabbit and pig models. Facilities for molecular biology including qPCR, DNA/RNA sequencing, array analysis, proteomics and genome editing. Clinical: Advanced biobanks and clinical research units that support sample handling, recruitment of additional centers to multi-center studies, and oversight of clinical studies in all aspects from statistics, monitoration, legal and ethical aspects, and data management etc.

Main contact person for this profile	Lasse Jensen, Senior Associate Professor, PI, department of health, medical and caring sciences, division of diagnostics and specialist medicine, unit of cardiovascular medicine
Main contact email	Lasse.jensen@liu.se
Other contact person(s), if relevant	Lena Tasse, EU-project management support, lena.tasse@liu.se Rami Alzukimy, Economist, rami.alzukimy@liu.se

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	☒ Retina
☐ Inherited retinal dystrophies (IRDs)	☒ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
☐ Advanced therapies / regenerative medicine	☐ Gene therapy / cell therapy / tissue engineering
☒ Inflammation / biomarkers / neuroprotection	☐ Drug delivery / formulations / nanomedicine
☒ Drug discovery / screening / repurposing	☒ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	☐ Other (please specify):

Free Keywords for internal searchable database (please provide up to 10)	Zebrafish; vasculature; blood-retinal barrier; choriocapillaris; edema; hypoxia
Geographies of interest for exchanges (optional)	

Overview of your research / innovation focus
We exploit embryonic, larval and adult zebrafish models of retinal or choroidal vascular dysregulation (induced mainly via exposure to hypoxia) to understand molecular and cellular mechanisms involved in vascular leakage, growth (both sprouting and intussusception), regression, effects on the surrounding tissue, and efficacy of newly identified targets or drug candidates. In addition to hypoxia, we have zebrafish models of proliferative diabetic retinopathy, inflammation- or neurodegeneration-induced vascular disruption, and vascular disruption associated to congenital eye diseases (e.g. FEVR, coloboma). *We are proficient in genetic engineering of zebrafish to create knockouts or knock-ins using Cas9:CRISPR. We are also routinely using regular or time-lapse confocal microscopy in longitudinal studies of vascular development or disruption under controlled conditions of retinal/sub-retinal disease progression. We have extensive experience in probing VEGF-pathways including an understanding of the importance of different VEGF-family members and receptors in different eye compartments.*

Why your team could be relevant to a translational eye-health and vision consortium
We can offer expertise and resources mainly in two areas of research: 1) Novel, innovative disease models and 2) Knowledge of and experience in vascular biology. *Zebrafish are becoming increasingly popular and relevant for eye-health and vision research due to their human-like, diurnal physiology (having both rods and cones, similar eye anatomy and molecular regulatory pathways), while also having increased regenerative capacity and pathophysiological relevance such as hypoxia- and diabetes-induced retinal angiogenesis and vascular*

disruption phenotypes associated with certain disease-related loss-of-function mutations. Furthermore, zebrafish embryos and larvae develop rapidly and are transparent which allows longitudinal visualization of any aspect of eye development, and the role of genes or drug in that process. Our experience allow both the development of new models to exploit the benefits of zebrafish in ophthalmology research and the use of our existing models for collaborative projects.

Vascular biology is studied also outside of the eye of the zebrafish (e.g. in cancer and in regulation of the blood-brain-barrier), as well as in other model organisms (the lab has extensive experience in mouse models). Blood vessels are key for many aspects of eye disorders, and the broad and deep knowledge in this area can benefit our collaborators to better study and understand the type of vascular disruption and its pathogenic roles.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

- *Zebrafish disease models*
- *Tools for and experience in studies of vascular biology*
- *Longitudinal imaging (e.g. time-lapse confocal microscopy)*
- *Microdissection (of zebrafish tissues including eyes)*
- *Knowledge of the innovation system including spin-off company creation and development*
- *Knowledge of and experience from translational studies incorporating patient material and information*

Infrastructure, equipment, facilities or datasets relevant to consortium work

- *Zebrafish facility and lab (including microinjection and microdissection systems)*
- *Cell culture labs*
- *Labs for molecular biology (qPCR, western blotting, sequencing, etc)*
- *Microscope facilities including various epi-fluorescence, confocal, light sheet, spinning disk, and two-photon microscopes*

Training capacity and knowledge-transfer potential

- *Experienced in training secondees on both short and long secondments*
- *Secondees will gain knowledge in zebrafish models, cell-based (in vitro) models, or associated techniques (microdissection, microinjection, cell culture, visualization)*
- *Secondees will gain knowledge in vascular biology both broadly (vessel development, structure, function, etc) and specifically how its coupled to the collaborative project*
- *Secondees will be in a translational environment highly used to closely collaboration with both clinicians and pharma/biotech companies*
- *Secondees will gain access to our tools and infrastructure, which can be exploited for adding data to ongoing projects*

What your team would hope to gain from participating

- *Access to complementary technologies such as cell- and/or mouse models or knowledge of how to analyze other aspects of zebrafish eye function)*
- *Complementary knowledge e.g. on how vascular disruption is affecting other parts of the eye and how vascular disruption can be avoided using innovative treatments*
- *Experience from other research environments which broaden the view on research for our early stage researchers*
- *Knowledge of and opportunities for alternative career-development paths for our early stage researchers e.g. at other European universities or in a non-academic sector.*

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Jingdan Shen	Ph.D student	HMV/DISP	Researcher	D	Jingdan.shen@liu.se
Ashitha Vivekanandan	Ph.D student	HMV/DISP	Researcher	D	Ashitha.vadassery.vivekanandan@liu.se

Zhu Pengcheng	Ph.D student	HMV/DISP	Researcher	D	Zhu.pengcheng@liu.se

Main disciplines represented in your team	Vascular biology/pathology, zebrafish models, disease model engineering, fluorescence visualization, hypoxia studies
Experience in EU / international / multi-partner collaborative projects	Participation in three prior EU-staff-exchange programs, an EIC-Pathfinder consortium, two Eurostars consortia and as an early-stage researchers in ERC Advanced grant project and an FP-7 project. In addition, I have participated in six other, international multi-partner projects, of which two as the coordinator.
Experience with clinical, translational or patient-facing research (if relevant)	☒ Yes ☐ No ☐ Not applicable | Short note: Mainly within cancer

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

*Lee C, Chen R, Sun G, Liu X, Lin X, He C, Xing L, Liu L, **Jensen L**, Kumar A, Langer H, Ren X, Zhang J, Huang L, Yin X, Kim J, Zhu J, Huang G, Li J, Lu W, Chen W, Liu J, Hu J, Sun O, Lu W, Fang L, Wang S, Kuang H, Zhang Y, Tian G, Mi J, Kang BA, Narazaki M, Prodeus A, Schoonjans L, Ornitz D, Gariepy J, Eelen G, Dewerchin M, Yang Y, Ou JS, Mora A, Yao J, Zhao C, Liu Y, Carmeliet P, Cao Y, and Li X, "VEGF-B prevents excessive angiogenesis by inhibiting FGF2/FGFR1 pathway", Signal Transduction and Targeted Therapy, 8(1):305*

*Karjosukarso DW, Ali Z, Peters TA, Zhang JQC, Hoogendoorn ADM, Garanto A, van Wijk E, **Jensen LD*** and Collin RWJ*, "Modeling ZNF408-associated FEVR in zebrafish results in abnormal retinal vasculature", Invest. Ophthalmol. Vis. Sci. 61(2):39*

*Ali Z, Zhang JJ, Lagali N, Schmitner N, Salvenmoser W, Mukwaya A, Neuhauss SCF, **Jensen LD*** and Kimmel RA*, "Photoreceptor degeneration accompanies vascular changes in a zebrafish model of diabetic retinopathy", Invest. Ophthalmol. Vis. Sci. 61(2):43*

*Ali Z, Mukwaya A, Biesemeier A, Ntzouni M, Ramsköld D, Giatrellis S, Mammadzada P, Cao R, Lennikov A, Marass M, Gerri C, Hildesjö C, Taylor M, Deng Q, Peebo B, del Peso L, Kvanta A, Sandberg R, Schraermeyer U, Andre H, Steffensen JF, Lagali N, Cao Y, Kele J and **Jensen LD** (2019), "Intussusceptive vascular remodeling precedes pathological neovascularization", Arterioscler. Throm. Vasc. Biol., 39(7), pp. 1402-18.*

*Ali Z, Yang Y, Tracey-White D, Vazquez-Rodriguez G, Moosajee M, Ju R, Li X, Cao Y and **Jensen LD** (2019), "Synchronized tissue-scale vasculogenesis and ubiquitous lateral sprouting underlie the unique architecture of the choriocapillaris", Dev. Biol. 457(2), pp. 206-14.*

Up to 5 relevant previous projects, grants, networks or collaborations

ALADDIN: EIC-PATHFINDER-OPEN project aiming to develop a new humanized zebrafish tumor xenograft platform for accelerated development of nanobody therapeutics in cancer (international)

EVEREST: MSCA-SE project wherein LIU is collaborating with 20+ EU partners to advance research in EV biology including applications within cancer research and novel therapies (international)

ROBO-FISH: A Eurostars project in which Dr. Jensen worked with LSM to develop new injection and visualization robots for creation and analysis of larval zebrafish tumor xenograft models. (international)

MECHANO-CHIP: A Eurostars project in which Dr. Jensen was collaborating with Dutch partners to develop microfluidic systems for cancer research (international)

Swedish Research Council: A highly competitive Sweden-China collaboration grant to in collaboration with Prof. Xuri Li at Sun Yat-Sen Ophthalmic Center in Guangzhou, China, establish shared capabilities for research on the role of PDGF-C and VEGF-B in ophthalmology

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)

Please indicate only what can be shared at this stage.

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	☒ Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	☐ Only Researchers ☒ Technical Staff/Innovation ☒ Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only ☒ Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	Periodically difficult to get independent access to facilities (due to delays in administration), Proficiency in English language is required.

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
Experts in eye vascular biology *Zebrafish models for eye diseases (focus on the vasculature)* *Translational research profile and experience working with pre-clinical methodologies in clinical studies/contexts*

Capabilities or expertise you would especially like to access through the consortium
OCT-A capabilities and knowledge *Clinical samples and/or multi-OMICs data from patients with eye diseases driven by vascular disruption (e.g. PDR, AMD, FEVR, etc)* *Mouse models of PDR and/or AMD to validate findings from zebrafish models*

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	☒ Yes ☐ No
☒ We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
Examples: PIC pending approval, institutional approval, legal status, hosting rules, IP constraints, data access, ethics, clinical permissions, staffing continuity, resource availability.

Please provide the following details (as many entries as needed).

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

☐ **Understood**

Name:

Date completed:

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)

Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	Linköpings universitet
Short name / acronym used in projects	LiU
Participant Identification Code (PIC), if available	999852236
Website(s)	www.liu.se https://liu.se/en/employee/neina50 https://liu.se/en/research/cornea-research
Country	Sweden
Organisation type	x Academic ☐ Non-academic x Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Campus Valla, Linköping, 581 83
Department(s), institute(s), research group(s) or unit(s) that would be involved	Division of Ophthalmology, Department of Biomedical and Clinical Sciences
Additional institutional resources relevant to collaboration	Full core facility including omics, molecular biology, sequencing, microscopy, animal care and housing, access to supercomputer for computational work, full ophthalmology research facility including molecular biology and cell culture labs, biobank, femtosecond laser surgical suite for preclinical research, in vivo diagnostic and imaging equipment for preclinical research, full bioinformatics expertise

Main contact person for this profile	Neil Lagali, Professor, PI of eye diseases research group, Division of Ophthalmology, Dept of Biomedical and Clinical Sciences
Main contact email	neil.lagali@liu.se
Other contact person(s), if relevant	Petros Moustardas, Research Associate, Co-supervisor and mentor, research and technical support: petros.moustardas@liu.se

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
X Ocular surface	☐ Retina
☐ Inherited retinal dystrophies (IRDs)	X Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	X Optometry / visual behaviour / rehabilitation
X Advanced therapies / regenerative medicine	X Gene therapy / cell therapy / tissue engineering
X Inflammation / biomarkers / neuroprotection	X Drug delivery / formulations / nanomedicine
X Drug discovery / screening / repurposing	X Ocular vascular disease
X Infection / trauma-related ocular disease	X Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	X AI-assisted diagnosis / software / digital tools
X Clinical trials / translational validation	X Other (please specify): ocular imaging, ocular omics, rare ocular diseases, ocular medical devices

Free Keywords for internal searchable database (please provide up to 10)	Ocular surface disease, limbal stem cells, rare eye diseases, AI-based decision support, corneal regenerative medicine, drug repurposing, transcriptomics, proteomics, tear film, neuropathic ocular pain
Geographies of interest for exchanges (optional)	all

Overview of your research / innovation focus
The ocular research group at Linköping University has the following research focuses: nAMD: Longitudinal OCTA studies of nAMD patients receiving anti-VEGF injections, including proteome profiling (plasma and tears) and miRNome profiling long-COVID studies: eye symptoms in long-COVID, tests of ocular and autonomic nerve/brain function, neuroinflammatory pathways and mechanisms, tear proteome profiling, visual training and rehabilitation Corneal nerves and neuropathic pain/allodynia: corneal nerve imaging in patients, corneal nerve regeneration studies with experimental therapies (preclinical models), experimental treatment of neuropathic pain (clinical studies) Limbal stem cell deficiency: preclinical models, scRNA-seq and spatial transcriptomics analyses in mouse models, evaluation of experimental therapies (drug repurposing), PAX6 mutations and role in corneal development, homeostasis and development of aniridia-associated keratopathy, effects of p63 mutations including preclinical model of AEC-EEC syndrome Corneal angiogenesis: preclinical studies of mechanisms of angiogenesis/neovascularization in the cornea including experimental therapy evaluation

Diabetes studies: ocular impact of type 2 diabetes including corneal nerves, diabetic retinopathy, long-term longitudinal corneal imaging studies in diabetes cohorts, scRNA-seq of blood, proteomics of blood and tears

Corneal regeneration and biomaterials: evaluation of new biomaterials and medical devices for implantation in the cornea to treat degenerative and refractive disorders, including laboratory validation, ex vivo studies, development of novel surgical methods (femtosecond surgical laser), preclinical large-animal models, clinical translation

Deep learning and corneal imaging: collaborative work with confocal microscopy images and their use in disease diagnosis, deep-learning assisted pipelines for image processing and analysis, clinical decision support

Infectious keratitis: epidemiological studies of various IK, including clinical examinations, retrospective analyses, evaluation of treatment, diagnostics, etc.

Why your team could be relevant to a translational eye-health and vision consortium

Please explain the specific added value your organisation / team could bring to a consortium centred on visual health, translational ophthalmology and related enabling technologies. ***Please think broadly: the contribution does not need to be directly ophthalmology-related. Interdisciplinarity is strongly encouraged, and expertise from apparently distant fields may open unexpected routes to innovation, new paradigms, and creative solutions to complex challenges that currently limit progress from research to patient benefit****.*

Our research team develops new therapies, devices, treatments and diagnostics from a basic science research level through to preclinical validation and into first human clinical trials. We also have projects exclusively focused on mechanisms of disease and while others are clinically-focused on improving understanding and therapy of eye diseases. We have a broad area of interest and deep expertise in cornea and ocular surface disease models and mechanisms, experimental therapy evaluation and imaging. We bring a long track record of ocular surface disease research to the consortium including various research tools, methods and platforms.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

Please provide bullet points covering your strongest capabilities, methods, models, analytical approaches, translational assets or innovation strengths.

- transgenic and acquired mouse models of limbal stem cell deficiency
- mouse models of ocular inflammation, neurodegeneration, neovascularization
- rabbit models of corneal inflammation, neurotrophic keratopathy, neovascularization
- ocular drug delivery approaches
- biomaterials use in the eye, slow release drug delivery
- experimental femtosecond laser surgery
- ocular imaging, in vivo diagnostics, corneal biomechanics
- transcriptomics (bulk, single cell, spatial) of cornea tissue, blood, proteomics of blood and tear film
- bioinformatics analysis and data visualization for large datasets
- in vivo confocal microscopy imaging, image processing, AI-based analysis and diagnostic tools
- translational research and surgery facility

Infrastructure, equipment, facilities or datasets relevant to consortium work

Examples: specialised laboratories, clean rooms, imaging platforms, -omics, biobanks, clinical cohorts, animal models, software tools, rehabilitation platforms, regulatory support or prototyping capacities.

- full basic science and translational ophthalmology labs including cell culture lab
- core facility infrastructures (sequencing, microscopy, animal work, omics)
- bioinformatics expertise

- femtosecond laser, biomechanical analyzer, OCT, in vivo confocal microscopy, meibomography, topography, dynamic pupillometry, etc.
- corneal image datasets (published and non-published)
- omics datasets (some analyzed, others un-analyzed)
- facilities to host researchers (office space, IT support, lab space, etc.)
- biobank of patient-derived blood and tear samples

Training capacity and knowledge-transfer potential

Please indicate what inbound staff could realistically learn in your organisation (scientific, technical, clinical, translational, regulatory, entrepreneurial or managerial skills).

- scientific skills in various molecular biology and translational workflows
-data analysis and processing using existing datasets including bioinformatics
- observation of clinical data collection, imaging
- learning image analysis and AI-based diagnostic tools
- skills in cell and biomaterial laboratory work
- collaborative research, performing laboratory work and analyses as part of collaborative research projects

What your team would hope to gain from participating

Examples: access to methods, models, technologies, patient-oriented validation, translational pathways, industrial insight, clinical exposure, data science, regulatory know-how, joint publications or longer-term strategic collaboration. ***Again, please think broadly and beyond ophthalmology.***

- new collaborations to move forward translational research towards clinical use
  - we can provide expertise and methods, resources, etc., that extend beyond ophthalmology and are applicable to molecular biology generally
  - we have projects involving COVID-19, type 2 diabetes, Parkinson's disease, etc.
  - we have experience in designing and conducting clinical studies and trials, including ethics, protocol and documentation requirements

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Neil Lagali	PI	Ophthalmology	Professor	senior	neil.lagali@liu.se
Petros Moustardas	Supervision, training	Ophthalmology	Research Associate	Mid-stage	petros.moustardas@liu.se
Ava Dashti	Collaboration, training, supervision	Ophthalmology	Doctoral candidate	Early-stage	ava.dashti@liu.se

Main disciplines represented in your team	Ophthalmology, optometry, cell biology, genetics, omics, biomaterials, AI, medical devices, imaging, biomarkers, clinical trials, preclinical models.
Experience in EU / international / multi-partner collaborative projects	Current partner in Horizon Europe project consortium, past coordinator of 2 EU project consortia, past MC member of 2 EU COST Actions (1 as Action Chair), current member of 2 EU COST Actions, past member of 1 Marie Curie RISE consortium, current member of 1 Marie Curie Doctoral Network
Experience with clinical, translational or patient-facing research (if relevant)	X Yes ☐ No ☐ Not applicable | Short note: clinical observational and interventional studies, translation from bench to clinic

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

Please prioritise achievements most relevant to your possible contribution. Please include DOI, URL or identifier.

Romano D, Romano V, Daruich A, Ferrari G, Robert M, Semeraro F, Lagali N, Bremond-Gignac D; ANIRIDIA-NET Clinical Guidelines Working Group; Aniridia-Europa and Geniris Working Group. Congenital aniridia: European COST action ANIRIDIA-NET guidelines for diagnosis, management and care. Acta Ophthalmol. 2026 May;104(3):267-279. doi: 10.1111/aos.17587. Epub 2025 Sep 2. PMID: 40892309; PMCID: PMC13058686.

Dina Javidjam, Meri Vattulainen, Neil Lagali, Petros Moustardas. Decoding murine corneal epithelial specification and homeostasis by single-cell spatial transcriptomics with scRNA-seq enrichment.
bioRxiv 2026.05.08.723186; doi: https://doi.org/10.64898/2026.05.08.723186

Ji M, Song Y, Roth J, Dashti A, Lazo J, Lincke A, Macedo AFT, Löwe W, Lagali N. Deep learning-based segmentation and density estimation of corneal nerves and dendritic cells from In Vivo confocal microscopy images. Sci Rep. 2026 Jan 13;16(1):1620. doi: 10.1038/s41598-025-34412-6. PMID: 41530230; PMCID: PMC12800303.

Javidjam D, Moustardas P, Abbasi M, Dashti A, Rautavaara Y, Lagali N. A human-like model of aniridia-associated keratopathy for mechanistic and therapeutic studies. JCI Insight. 2024 Dec 3;10(2):e183965. doi: 10.1172/jci.insight.183965. PMID: 39625791; PMCID: PMC11790027.

Moustardas P, Abbasi M, Javidjam D, Asamoah CS, Schweitzer-Chaput A, Cisternino S, Bremond-Gignac D, Aberdam D, Lagali N. Duloxetine enhances PAX6 expression and suppresses innate immune responses in murine LPS-induced corneal inflammation. Ocul Surf. 2024 Oct;34:225-234. doi: 10.1016/j.jtos.2024.08.008. Epub 2024 Aug 8. PMID: 39127390.

Up to 5 relevant previous projects, grants, networks or collaborations

Please provide detailed information (Title, Acronym, funding agency, GA-number, dates). Please indicate your role (coordinator / beneficiary / partner / service provider / collaborator). ***Indicate if international.***

Stem Cell Therapy Development for Corneal Epithelium Regeneration, STEM-CORE, Marie Curie Doctoral Network, 101226474, 2025-2028. Role: Beneficiary (International)

Novel advanced and repurposed therapeutics for vision restoration in a group of severe rare ocular surface diseases: from validation to first clinical investigations, RESTORE VISION, EU Horizon Europe, 101080611, 2024-2027, Role: Beneficiary (International)

Networking to address an unmet medical, scientific, and societal challenges, ANIRIDIA-NET, EU COST Action CA18116, European Union COST Programme, 2019-2023, Role: Coordinator/Action Chair (International)

Aniridia – Novel therapeutic tools to treat or prevent progressive cornea opacification, AAK-INSIGHT, European Joint Programme on Rare Diseases (EJPRD), EJPRD20-135, 2020-2024, Role: Coordinator (International)

Advanced Regenerative and RESTtorative Therapies to combat corneal BLINDNESS, ARREST BLINDNESS, EU Horizon2020, 667400, 2016-2020, Role: Coordinator (International)

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)

Please indicate only what can be shared at this stage.

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	X Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	☐ Only Researchers x Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff x Other profiles: clinical specialists and residents
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only x Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	Insurance from home country needed

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
Please summarise the 2-3 strengths that should be first visible in an internal searchable profile. cornea, ocular surface disease, regenerative medicine, omics, stem cells, imaging

Capabilities or expertise you would especially like to access through the consortium
Please indicate the types of teams, methods, infrastructures or translational pathways you would like to connect with in the consortium. - complementary expertise, whatever makes sense

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	X Yes ☐ No
☐ We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
A form is needed to be signed by the division head prior to receiving guest researchers. Guests will undergo lab safety training.

Internal use and confidentiality statement

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x Understood

Name: Neil Lagali
Date completed: May 29, 2026

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)

Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	University of New South Wales (UNSW, Sydney)
Short name / acronym used in projects	UNSW, Sydney (NSW)
Participant Identification Code (PIC), if available	999862809
Website(s)	www.unsw.edu.au/medicine-health/our-schools/optometry
Country	Australia
Organisation type	X Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Level 5, Rupert Myers Building, North Wing, Gate 14, Barker St, UNSW, Sydney, 2052
Department(s), institute(s), research group(s) or unit(s) that would be involved	Ocular Surface Research Group; Eye Research Group; Retinal Disease and Clinical Imaging Research Group; Retinal Research Group; Ocular Pathology Research Group; and Optical Imaging and Visualisation (OIV) Lab.
Additional institutional resources relevant to collaboration	Mark Wainwright Analytical Centre (MWAC): https://www.unsw.edu.au/research/facilities-and-infrastructure/find-a-facility/mwac Ramaciotti Centre for Genomics: https://www.unsw.edu.au/science/ramaciotti 3D Culture Facility: https://www.unsw.edu.au/research/facilities-and-infrastructure/find-a-facility/3dcf

Main contact person for this profile	A/Prof Nicole Carnt, Director of Research, School of Optometry and Vision Science, UNSW, Sydney
Main contact email	n.carnt@unsw.edu.au
Other contact person(s), if relevant	A/Prof Maria Markoulli, Co-Director of Teaching, School of Optometry and Vision Science, UNSW, Sydney

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
x Ocular surface	x Retina
☐ Inherited retinal dystrophies (IRDs)	x Age-related ocular disease
x Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
x Advanced therapies / regenerative medicine	x Gene therapy / cell therapy / tissue engineering
x Inflammation / biomarkers / neuroprotection	x Drug delivery / formulations / nanomedicine
x Drug discovery / screening / repurposing	☐ Ocular vascular disease
x Infection / trauma-related ocular disease	x Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	x AI-assisted diagnosis / software / digital tools
x Clinical trials / translational validation	☐ Other (please specify):

Free Keywords for internal searchable database (please provide up to 10)	corneal organoids; ocular surface disease; ocular allergy; tear biomarkers; retinal biomarkers; oculomics; multimodal ocular imaging; artificial intelligence; clinical trials; ocular infection
Geographies of interest for exchanges (optional)	

Overview of your research / innovation focus
The participating UNSW teams undertake multidisciplinary translational vision science across ocular surface disease, retinal disorders, ocular pathology, advanced imaging, artificial intelligence, biomarker discovery, regenerative medicine and public health. The overarching research line is to advance the prevention, diagnosis and treatment of ocular disease by integrating discovery science, clinically relevant models, enabling technologies and patient-centred validation. Research strengths include ocular infection, inflammation and allergy; dry eye disease and contact lens-related ocular health; tear, tissue and imaging biomarkers; ocular pain; retinal degeneration and age-related macular degeneration; and therapeutic discovery and evaluation. Core platforms include human donor tissues, human corneal organoids and advanced 3D ocular tissue models, tissue explants, cell culture and animal models, genomics and molecular profiling, clinical cohorts, multimodal ocular imaging and clinical trials. Emerging oculomics research applies AI-enabled analysis of ocular imaging and retinal biomarkers to investigate ocular disease and associations with systemic health. Across these areas, the teams connect mechanistic discovery with biomarker and technology development, clinical validation and patient-centred research, supporting more precise diagnostics, therapeutics and prevention strategies.

Why your team could be relevant to a translational eye-health and vision consortium
UNSW offers a collaborative translational ecosystem that integrates discovery science, advanced disease modelling, biomarker development, clinical research and patient-centred validation. Complementary expertise across ocular surface disease, retinal disorders, ocular pathology, multimodal imaging, artificial intelligence and regenerative medicine enables research questions to be addressed from mechanism through to clinical evaluation.

Our added value lies in linking human donor tissues, human corneal organoids and advanced 3D ocular tissue models, preclinical models, multi-omics, clinical cohorts, imaging and trials. Collaboration across academia, healthcare, industry, regulatory stakeholders and health consumers helps align research priorities and translation pathways with scientific evidence, clinical need and lived experience. Health consumers can contribute to priority setting, study design, interpretation, dissemination and implementation.

Beyond ophthalmology, the teams contribute expertise in immunology, microbiology, neuroscience, ageing, bioinformatics, data science, oculomics and public health. This breadth supports interdisciplinary staff mobility, reciprocal skills transfer and new approaches at the interface of eye health and systemic health.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

- Ocular surface disease, infection, inflammation, dry eye disease and ocular allergy
- Retinal disease mechanisms, neurodegeneration, ageing and neuroprotection
- Human donor tissue, ocular pathology and tissue-explant research
- Human corneal organoids and advanced 3D ocular tissue models
- Cell culture, preclinical and animal models for disease mechanisms and therapeutic evaluation
- Tear, tissue, molecular and imaging biomarker discovery and validation
- Genomics, transcriptomics, multi-omics and bioinformatics
- Quantitative multimodal ocular imaging, hyperspectral retinal imaging and retinal biomarker development
- Oculomics and AI-enabled analysis for ocular and systemic health research
- Clinical cohorts, clinical trials, patient-reported outcomes and translational validation
- Health consumer engagement, co-design and patient-centred research
- Drug discovery, screening and therapeutic development

Infrastructure, equipment, facilities or datasets relevant to consortium work

- UNSW Mark Wainwright Analytical Centre (MWAC): shared microscopy, imaging, mass spectrometry, flow cytometry and complementary analytical capabilities
- UNSW Ramaciotti Centre for Genomics: next-generation sequencing, gene-expression and transcriptomic analysis, single-cell approaches and bioinformatics
- UNSW 3D Culture Facility: organoid, explant and advanced cell-culture platforms for disease modelling and therapeutic discovery
- Advanced anterior and posterior ocular imaging facilities, including hyperspectral retinal imaging and Spectralis OCT/OCTA platforms
- Specialised ocular surface, retinal and ocular pathology laboratories, with access to human donor ocular tissues through collaborating eye banks
- Clinical cohorts in ocular surface disease, infection, allergy, dry eye disease, retinal disorders and age-related eye disease
- Biobanked tear, tissue, imaging and multi-omics datasets, with image-analysis, AI and bioinformatics workflows
- Clinical trial capability and health consumer networks supporting patient-centred validation

Training capacity and knowledge-transfer potential

Inbound staff could undertake reciprocal scientific, technical, clinical and translational training across:

- Human donor tissue, ocular pathology, tissue explants, human corneal organoids and advanced 3D ocular tissue models
- Cell culture, preclinical and animal models of ocular disease
- Tear and tissue biomarker collection, molecular profiling, genomics, transcriptomics and bioinformatics
- Multimodal ocular imaging, hyperspectral retinal imaging, OCT/OCTA, quantitative image analysis and oculomics
- AI and machine learning for imaging, biomarker discovery and clinical decision support
- Clinical research methods, trial design, patient-reported outcomes and translational validation
- Health consumer engagement, co-design, priority setting and patient-centred research
- Research ethics, governance and regulatory considerations relevant to clinical and translational studies
- Grant development, project management, scientific communication, research leadership and international collaboration

What your team would hope to gain from participating
Building on successful informal staff exchanges and international collaborations over the past five years, which have contributed to Q1 journal publications, higher degree research theses, research training and leadership development, the consortium would provide a structured mechanism to expand staff mobility, shared training and joint supervision. We seek access to complementary disease models, advanced therapeutic and regenerative-medicine platforms, patient-derived validation cohorts, multi-omics and data-science methods, and specialised imaging and computational infrastructure. We also aim to strengthen expertise in gene and cell therapies, neuroprotection, drug delivery, bioengineering, biomaterials, regulatory science, health technology assessment, implementation research and commercialisation. Engagement with academic, clinical, industry and consumer partners would support cross-sector exposure, patient-oriented validation and more effective translation. The intended outcomes are sustainable international partnerships, joint publications and supervision, collaborative funding, stronger research capability and longer-term pathways from discovery to clinical and societal benefit.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Nicole Carnt	Associate Professor/Director of Research	Ocular Surface Research Group	Academic	Mid	
Maria Markoulli	Associate Professor Co-Director of Teaching	Ocular Surface Research Group	Academic	Mid	
Fiona Stapleton	Scientia Professor/Clinical Teaching Director, UNSW	Eye Research Group	Academic	Senior	
Jackie Tan	Associate Professor Director Eye Research Group	Eye Research Group	Academic	Mid	
Mark Willcox	Professor	Eye Research Group	Academic	Senior	
Isabelle Jalbert	Professor	Ocular Surface Research Group	Academic	Senior	
Suki Jaiswal	Post Doctoral Scientist	Ocular Surface Research Group	Academic	Early	
Lisa Nivison-Smith	Senior Lecturer	The Retinal Disease and Clinical Imaging Research Group	Academic	Mid	
Daisy Shu	Senior Lecturer	Retinal Research Group	Academic	Early	
Michele Madigan	Associate Professor	Ocular Pathology Group	Academic	Senior	
Jerome Ozkan	Senior Lecturer	Ocular Surface Research Group	Academic	Mid	
Maitreyee Roy	Associate Professor	Ocular Imaging and Visualisation Group	Academic	Mid	

Main disciplines represented in your team	Optometry; vision science; cell and molecular biology; immunology; microbiology; ocular pathology; genomics and bioinformatics; biomaterials and tissue engineering; clinical research; data science and artificial intelligence; public health; and regulatory science.

Experience in EU / international / multi-partner collaborative projects	Experience in international and multi-partner collaborations across academia, healthcare and industry includes participation in COST Action CA24112, Implementation of Tear Fluid Biomarkers in Precision Medicine, and partnership in the FLAIR consortium proposal under Horizon Europe call HORIZON-HLTH-2025-01-TOOL-03 on multimodal generative AI in biomedical research.
Experience with clinical, translational or patient-facing research (if relevant)	x Yes ☐ No ☐ Not applicable | Clinical cohorts, industry-sponsored trials, patient-reported outcomes, health consumer engagement and patient-centred validation across ocular surface and retinal disease.

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

1. Shrestha GS; Vijay AK; Stapleton F; White A; Pickford R; Carnt N, 2023, 'Human tear metabolites associated with nucleoside-signalling pathways in bacterial keratitis', *Experimental Eye Research*, 228, pp. 109409, http://dx.doi.org/10.1016/j.exer.2023.109409

2. Khan S; Markoulli M; Nureen L; Di Girolamo N; Willcox M, 2026, 'Effects of Sitagliptin and Celery Seed Extract on Corneal Nerve Morphology and Sensory Dysfunction in Diabetic Rats', *Nutrients*, 18, http://dx.doi.org/10.3390/nu18142243

3. Phu J; Khuu SK; Nivison-Smith L; Kalloniatis M, 2025, 'Standard automated perimetry for glaucoma and diseases of the retina and visual pathways: Current and future perspectives', *Progress in Retinal and Eye Research*, 104, http://dx.doi.org/10.1016/j.preteyeres.2024.101307

4. McLellan FC; Liang YG; Jin Y; Madigan MC; Ng PQ; White AJR; Williams PA; Shu DY, 2026, 'Synergistic Cytokine Signaling Drives Angiofibrotic Gene Pathways in Primary Human Retinal Endothelial Cells', *American Journal of Pathology*, 196, pp. 2237 - 2255, http://dx.doi.org/10.1016/j.ajpath.2026.06.007

5. Sarwat S; Willcox M; Stapleton F; Roy M, 2026, 'Biophysical Interactions of Silicon Quantum Dots With Human Tears and Dry Eye Drops: A Langmuir Trough Study', *Translational Vision Science and Technology*, 15, http://dx.doi.org/10.1167/tvst.15.1.10

Up to 5 relevant previous projects, grants, networks or collaborations

Examples on the diversity and breadth of research projects, networks and collaboration are detailed below:

1. Advancing Corneal Infection Research with Human Organoids
- Funding agency: UNSW Global Research Impact Program (GRIP), Round 3 (2025)
- Role: Coordinator / Lead Investigator (CIA)
- Investigators: A/Prof Nicole Carnt (CIA), A/Prof Ashok Tiwari, Prof Virender Sangwan, Prof Mark Willcox, Dr Anna Ferguson
- International: Yes (Australia-India collaboration)
- Focus: Development of human corneal organoid models to advance translational research in corneal infection.

2. AZR-MD-001 for Meibomian Gland Dysfunction (MGD)
- Funding agency: Azura Ophthalmics Pty Ltd (Industry-sponsored clinical trial)
- Role: Clinical trial site / research partner
- Investigators: A/Prof Jackie Tan, Scientia Prof Fiona Stapleton
- International: Yes (industry networks spanning Australia, USA, UK, Japan and China)
- Focus: Vehicle-controlled clinical trial evaluating safety and tolerability of AZR-MD-001 for meibomian gland dysfunction.

3. Understanding and Managing the Impact of Smoke on the Eyes of Australian Bush Firefighters
- Funding agency: Natural Hazards Research Australia, Core Research Program
- Role: Research partner

- Investigators: Dr Suki Jaiswal, Prof Isabelle Jalbert
- National collaboration: Australia
- Focus: Environmental impacts on ocular surface health and strategies to reduce vision-related effects of smoke exposure.

4. From Fumarates to Microtubules: Novel Anti-angiogenic and Anti-fibrotic Therapies for AMD

- Funding agency: BrightFocus Foundation (USA), New Investigator Grant Program
- Role: Project lead
- Investigator: Dr. Daisy Shu (CIA)
- International: Yes (USA-Australia)
- Focus: Development of novel therapeutic approaches for age-related macular degeneration.

5. Understanding Inner Retinal Remodelling to Improve Clinical Outcomes for Patients with Age-Related Macular Degeneration

- Funding agency: National Health and Medical Research Council (NHMRC) Investigator Grant (2020-2025)
- Role: Project lead
- Investigator: Dr. Lisa Nivison-Smith (CIA)
- National collaboration: Australia
- Focus: Mechanisms of retinal remodelling and translation of findings to improve outcomes for patients with AMD.

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)
• Granted patent family for Antimicrobial Compounds (2016-079), with national-phase protection in the United States (Patent No. 10,988,444; granted 2021), Australia (granted 2022) and New Zealand (Patent No. 754200; granted 2025). Inventors: R. Kuppusamy, D. Black, N. Kumar and M.D.P. Willcox • Human corneal organoid, tissue-explant and advanced 3D ocular tissue-model platforms • Clinical trial capabilities and protocols for ocular surface disease, dry eye disease, contact lenses and meibomian gland dysfunction • Established tear biomarker collection, ocular surface phenotyping and patient-reported outcome methods • Clinical imaging datasets and AI-enabled analysis pipelines for detection, monitoring and risk stratification • Ocular surface and retinal disease cohorts with linked clinical, imaging, biological and patient-reported data • Health consumer engagement frameworks for co-design, priority setting, dissemination and translation

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	x Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	☐ Only Researchers x Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only x Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	Researchers could be onboarded through the Research Practicum pathway: https://www.unsw.edu.au/study/international-students/study-abroad#researchpracticum

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
Translational disease models: human corneal organoids, tissue explants, cell culture systems and animal models for mechanistic research and therapeutic evaluation. Ocular biomarkers and oculomics: tear, tissue, retinal and imaging biomarkers for disease detection, monitoring and risk stratification. Advanced clinical imaging and AI: multimodal anterior and posterior ocular imaging, quantitative image analysis and AI-enabled diagnostic and predictive modelling.

Capabilities or expertise you would especially like to access through the consortium
• Advanced therapies, regenerative medicine and gene/cell therapy platforms • Novel organoid systems, patient-derived models and validation cohorts • Multi-omics, systems biology and computational biology • AI, digital health and clinical decision-support technologies • Drug delivery, biomaterials and bioengineering • Regulatory science, health technology assessment and implementation research • Industry partnerships, commercialisation pathways and innovation ecosystems • International patient networks, longitudinal cohorts and real-world evidence platforms

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	x Yes ☐ No
x We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
Examples: PIC pending approval, institutional approval, legal status, hosting rules, IP constraints, data access, ethics, clinical permissions, staffing continuity, resource availability.

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

x Understood

Name: Nicole Carnt
Date completed: 2 Oct 2026

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)

Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	OCUPHARM DIAGNOSTICS, S.L
Short name / acronym used in projects	OCUPHARM
Participant Identification Code (PIC), if available	934947693
Website(s)	www.ocupharm.com
Country	SPAIN
Organisation type	☐ Academic X Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company X Research infrastructure ☐ NGO / foundation ☐ Public body X Other
Main address	Calle Gamonal, 16-20. Planta 3. Ofi. S 28031 MADRID.
Department(s), institute(s), research group(s) or unit(s) that would be involved	Ocupharm Diagnostics – All Departments. Biochemistry, ophthalmological, CRO, I+D+I and regulatory consulting.
Additional institutional resources relevant to collaboration	OCUPHARM DIAGNOSTICS, S.L / San Carlos Health Research Institute / Complutense University of Madrid

Main contact person for this profile	RUBÉN ÁLVAREZ DE BLAS (CEO)
Main contact email	ralvarez@coupharm.com
Other contact person(s), if relevant	

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
X Ocular surface	☐ Retina
☐ Inherited retinal dystrophies (IRDs)	☐ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	X Optometry / visual behaviour / rehabilitation
☐ Advanced therapies / regenerative medicine	☐ Gene therapy / cell therapy / tissue engineering
☐ Inflammation / biomarkers / neuroprotection	X Drug delivery / formulations / nanomedicine
☐ Drug discovery / screening / repurposing	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
X Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
X Clinical trials / translational validation	X Other (please specify): Specialized in visual sciences and eye pharmacology

Free Keywords for internal searchable database (please provide up to 10)	Spin-off / Molecular Biochemistry / Custom Trials / Regulatory / in-vivo experimentation / Clinical Trials / Post-Market / CRO / Patent/ ISO 9001 and ISO 13485 Certifications and internal audits.
Geographies of interest for exchanges (optional)	No preferences

Overview of your research / innovation focus
Our research and innovation focus centres on providing specialised ophthalmological expertise to support the research, development, and regulatory advancement of ophthalmic products. As a team of healthcare professionals with specialised knowledge in ophthalmology, we provide scientific and clinical support throughout the product development lifecycle, helping to facilitate the translation of innovative solutions into safe, effective and clinically relevant products. Our activities encompass comprehensive, end-to-end Contract Research Organisation (CRO) services tailored to the specific needs of ophthalmological development programmes. By combining clinical expertise with an understanding of research and regulatory requirements, we support the planning, execution and management of studies, while contributing to the generation of robust scientific and clinical evidence. In parallel, we provide regulatory and quality management support to organisations involved in the development and commercialisation of ophthalmological products and medical devices. Our services include consultancy and auditing activities in accordance with ISO 9001 and ISO 13485 standards, supporting organisations in the implementation, maintenance and continuous improvement of their quality management systems. These activities contribute to ensuring compliance with applicable quality and regulatory requirements throughout the product lifecycle. Through the combination of specialised healthcare knowledge, ophthalmological expertise, end-to-end CRO capabilities and quality and regulatory consultancy, our work aims to facilitate the efficient development of innovative ophthalmological products and support their successful progression towards clinical application and patient care. This integrated approach enables us to provide comprehensive support across clinical research, product development, regulatory processes and quality management.

Why your team could be relevant to a translational eye-health and vision consortium
We are a multidisciplinary team with specialised expertise in ophthalmology, providing integrated support across the research, development and clinical translation of innovative ophthalmological products and medical devices. Our capabilities encompass the design, coordination and execution of preclinical and clinical research activities, combining scientific and clinical expertise with a strong focus on quality, regulatory compliance and translational applicability. Our extensive experience in ophthalmological research enables us to support development programmes in accordance with internationally recognised quality standards and regulatory frameworks, including ISO standards, Good Laboratory Practice (GLP) and Good Clinical Practice (GCP). This integrated expertise allows us to generate robust and reliable evidence while supporting the efficient progression of innovative ophthalmological solutions from research and development towards clinical applications.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute
CLINICAL TRIALS: We design, coordinate and conduct customised clinical studies across the product development lifecycle, from early-stage clinical investigation through post-market clinical follow-up. Our ophthalmology-focused expertise supports the generation of high-quality clinical evidence aligned with applicable regulatory requirements and Good Clinical Practice (GCP). **PRECLINICAL STUDIES:** We design and perform tailored preclinical studies to evaluate the safety, performance, efficacy and biocompatibility of ophthalmological products and medical devices. Study design is adapted to the specific characteristics and intended use of each product, supporting evidence generation and risk mitigation prior to clinical development. **IN VIVO STUDIES:** We conduct in vivo research using relevant animal models to assess the safety, performance and biological effects of ophthalmological products. All studies are performed in accordance with applicable regulatory requirements, Good Laboratory Practice (GLP), and rigorous animal welfare and ethical standards. **CRO SERVICES:** As a specialised ophthalmology-focused Contract Research Organisation (CRO), we bridge clinical practice and clinical research by facilitating access to relevant patient populations, clinical expertise and high-quality clinical data. We provide end-to-end operational and scientific support for the planning, management and execution of ophthalmological research programmes. **CONSULTANCY AND REGULATORY SUPPORT:** We provide specialised consultancy on technical documentation, regulatory strategy and quality requirements for ophthalmological products and medical devices. Our services include support in interactions with Regulatory Authorities and Notified Bodies, as well as guidance throughout the CE marking process, helping clients navigate regulatory pathways and strengthen their readiness for market access. **QUALITY MANAGEMENT CONSULTANCY AND AUDITING:** We provide consultancy and auditing services for Quality Management Systems in accordance with ISO 9001 and ISO 13485. Our support covers system implementation, assessment and continuous improvement, helping organisations strengthen quality processes, regulatory compliance and overall product development readiness.

Infrastructure, equipment, facilities or datasets relevant to consortium work
Access to university and clinical research infrastructure: As a UCM spin-off company, we have access to and collaborate with advanced university research infrastructure and state-of-the-art equipment, providing complementary technical capabilities to support ophthalmological R&D, preclinical and clinical research activities. • **University Research Infrastructure:** Access to specialised laboratories, research facilities and advanced scientific equipment available through the Universidad Complutense de Madrid (UCM), enabling the implementation of research and development activities according to the specific technical requirements of each project. • **San Carlos Health Research Institute:** Through established agreements with the Instituto de Investigación Sanitaria San Carlos, we have access to its research and clinical facilities, enabling collaborative research activities and the use of specialised infrastructure and technical expertise relevant to ophthalmological R&D.

Training capacity and knowledge-transfer potential
Incoming staff could gain hands-on experience in ophthalmological research and clinical development within a highly interdisciplinary environment, covering scientific, technical, clinical, computational and translational aspects of visual health. They would receive training in the design and execution of preclinical and clinical research activities, as well as in the evaluation of ophthalmological products and medical devices. From a technical perspective, they could become familiar with advanced experimental methodologies for ophthalmological research, including the development and application of tailored in vitro and in vivo models, experimental design, safety and performance assessment, and the generation and analysis of clinically relevant data. In addition, incoming researchers could gain experience in regulatory and quality aspects of ophthalmological product development, including Good Laboratory Practice (GLP), Good Clinical Practice (GCP), ISO standards, Quality Management Systems under ISO 9001 and ISO 13485, technical documentation, regulatory strategy and interactions with Regulatory Authorities and Notified Bodies. Training would also include exposure to CRO project management, clinical research coordination, translational development, technology evaluation and innovation processes, as well as collaboration with academic, clinical and industrial partners. This combination of scientific, clinical, operational, regulatory and quality expertise would provide incoming staff with a comprehensive understanding of the pathway from ophthalmological R&D to clinical application and regulatory readiness.

What your team would hope to gain from participating
The team would hope to gain access to complementary expertise, advanced technologies and innovative translational approaches that strengthen its research lines in retinal diseases and visual health, particularly in neuroprotective compounds, gene and cell therapies, multimodal imaging, Artificial Intelligence and advanced analytical platforms. It would also value opportunities to integrate complementary datasets, methodologies and experimental strategies, improving the robustness, reproducibility and clinical relevance of preclinical and patient-oriented research, as well as gaining exposure to infrastructures and technologies not currently available within the organisation. In addition, the team would benefit from international mobility and training opportunities for PhD students, senior researchers and technical staff, fostering the acquisition of new methodologies, analytical tools and multidisciplinary approaches, while strengthening scientific and professional development. Finally, it would seek to establish long-term international collaborations through joint research projects, coordinated funding applications, scientific communications and high-impact publications, promoting sustained knowledge transfer, innovation and translational impact across participating groups.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Rubén Álvarez de Blas	Chief Executive Officer (CEO)	Executive Management	Executive Management	Senior/Executive	ralvarez@ocupharm.com
M. Luisa González Rubio	Technical Director	Technical Department	Management/ Technical Management	Senior / Management	mlgonzalez@ocupharm.com
Elena Fernández Jiménez	Technical Director	Technical Department	Management/ Technical Management	Senior / Management	efernandez@ocupharm.com
Bryan Marcelo Espin Valverde	Head of Studies	Biochemistry Department	Management	Management	bespin@ocupharm.com
Kathlyn Quispitongo Oblitas	Head of Studies	Reulatory / Pre clinical Department	Management	Management	koblitas@ocuphar.com
Paloma Porras Ángel	Head of Studies	Clinical Trial Department	Management	Management	pangel@ocupharm.com
Irene Alonso Gil	Head of Clinical Operations	CRO Department	Management	Management	ialonso@ocupharm.com
Álvaro Xavier Tintín Castillo	Head of Clinical Operations	CRO Department	Management	Management	atintin@ocupharm.com
Isabel Lorenzo Sánchez	Administrative	Purchasing and Administration Department	Administrative	Administrative	ilorenzo@ocupharm.com

Main disciplines represented in your team	Optometry, Biochemistry, Regulatory, CRO, Post-market, ISO 9001 and ISO 13485, consulting and auditing.
Experience in EU / international / multi-partner collaborative projects	Participation in: CRYSTAL3 Project ID **101007931** | Programme **H2020 MSCA-RISE-2020** 3D NEONET Project ID **734907** | Programme **H2020 MSCA-RISE-2016** ORBITAL Project ID **813440** | Programme **H2020 MSCA-ITN-2018** DryEye Project ID **673533** | Programme **H2020 PHC-12-2014-1 SME instrument phase 1**
Experience with clinical, translational or patient-facing research (if relevant)	X Yes ☐ No ☐ Not applicable | Short note:

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements
Our collaboration with the academic community provides us with up-to-date knowledge of innovations in the sector, enabling us to develop pioneering and effective solutions to our clients' challenges. This expertise facilitates the design

and validation of new procedures and protocols that comply with the requirements of current regulations for medical and pharmaceutical products. For example:

- Efficacy studies on the removal of lipid and protein deposits from contact lenses.
- Detection of hyaluronic acid by high-performance liquid chromatography (HPLC).
- Development of a needle sheath for use in suprachoroidal space puncture.

Up to 5 relevant previous projects, grants, networks or collaborations

OC-IOT-0725-PMCS - CRO Myopia progression with freeform myopia control spectacles (International – We are coordinator)

Ocupharm Diagnostics offers full coordination and execution of the clinical trial, including the following services:

- **Study Planning and Start-up:** Definition of study objectives, review and validation of the protocol and development of timelines.
- **Ethical and Regulatory Documentation:** Preparation of all required documentation for submission to relevant Ethics Committees (IRB/EC), competent health authorities, public trial registries (e.g., EU CTR, ClinicalTrials.gov), and assurance companies. Ongoing follow-up and response to queries or requests for clarification.
- **Study Monitoring:** Coordination of site activation and initial training, combined with both remote and on-site monitoring as per the clinical monitoring plan. This includes verification of protocol adherence, source data verification (SDV), identification of protocol deviations, and supervision of data quality.
- **Adverse Event Management:** Identification, documentation, follow-up, and reporting of any adverse events occurring during the trial, ensuring full compliance with applicable regulatory requirements.
- **Regular Meetings with the Sponsor:** Ongoing review of study progress, discussion of operational challenges, preliminary data review, and adjustment of project plans as needed.
- **Data and Quality Management:** eCRF configuration and oversight, ongoing data review, validation and cleaning, generation of quality reports, and issuance and resolution of queries.
- **Study Closure and Final Delivery:** Site close-out visits, archiving study documentation in an electronic Trial Master File (eTMF), preparation of final reports, and handover of all documentation in audit-ready format.

OC-OL-0325-ISO:9+13 - Implementation of an Integrated Quality and Medical Device Management System

Ocupharm Diagnostics propose methodology consists of the following stages:

- **Establishment of the Working Team.**
- **Identification and definition of the Organisation's key processes.**
- **SWOT (Strengths, Weaknesses, Opportunities and Threats) analysis.**
- **Definition, review and optimisation of the system documentation.**
- **Review, management and optimisation of resources.**
- **Identification of applicable legal and regulatory requirements.**
- **Analysis and definition of Medical Devices.**
- **Measurement and monitoring of the Quality Management System and Medical Devices.**

- **Conducting an internal audit.**
- **Remote participation in the external audit and preparation of the Corrective Action Plan (CAP).**

OC-DI-0925-ENZYMEB2 – DETERMINATION OF THE PROTEIN-CLEANING CAPACITY OF THE HYDRO HEALTH PRODUCT

Ocupharm Diagnostics offers full coordination and execution of the assessment of the protein-cleaning capacity:
The selected lenses will undergo simulated use by incubation in artificial tears enriched with proteins. The lenses will subsequently be cleaned with the test product according to the instructions provided by the manufacturer. Protein levels will be measured using a commercially available Biuret assay. The cleaning capacity will be determined by comparing the relative amounts of protein before and after cleaning.

OC-AV-0126-101A - DETERMINATION OF THE PHYSICAL COMPATIBILITY OF THE GP CONDITIONER PRODUCT WITH CONTACT LENSES

Ocupharm Diagnostics offers full coordination and execution of the clinical trial, including the following services:
Physical compatibility testing in accordance with ISO 11981:2017 for the **GP CONDITIONER** product. The assessment includes parameters such as base curve radius, diameter, power and spectral transmittance in contact lenses treated with the test solution.

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)
ES2932995 - Ocular parasite detection system ES2529369 - Use of the compound calcium 2,5-dihydroxybenzenesulfonate (calcium dobesilate) for the preparation of a medicament for the treatment of achondroplasia ES2735642 - Food product for ocular hypertension and glaucoma ES2469290 - Improvement in the topical application of ocular drugs through the administration of nucleotides ES2724933 - Combination therapy with melatonin for reducing intraocular pressure ES2597827 - Use of melatonergic compounds for treating the ocular surface ES2650981 - Preparation and use of an Artemia salina extract for treating the ocular surface US20190275089 - Preparation and use of an extract of Artemia salina to treat the ocular surface EP3488855 - PREPARATION AND USE OF AN EXTRACT OF ARTEMIA SALINA TO TREAT THE OCULAR SURFACE US20190275089 - Preparation and use of an extract of Artemia salina to treat the ocular surface

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	X Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	X Only Researchers X Technical Staff/Innovation X Managerial / project support X Admin Staff X Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only X Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
• Development and preclinical validation of innovative ophthalmological products and therapies, combining specialised ophthalmology expertise with experimental and translational approaches to assess safety, performance, efficacy and clinical relevance. • Translational research capabilities focused on bridging preclinical findings with clinical application, including the design and evaluation of clinically relevant studies, generation of robust evidence, and integration of scientific, clinical and regulatory considerations. • Experience in collaborative R&D projects involving clinical and industrial partners, supporting technology transfer, product development and the translation of innovative ophthalmological solutions towards clinical and market applications.

Capabilities or expertise you would especially like to access through the consortium
Access to cutting-edge infrastructures and specialised technologies not currently available in our centre, including advanced imaging, high-throughput screening, AI-driven analytical tools and large-scale computational approaches that could maximise the value and translational relevance of our datasets and experimental results. Improved translational and methodological standardisation, refining key experimental and analytical aspects that facilitate robust, reproducible and clinically meaningful extrapolation of preclinical findings. Effective communication and dissemination strategies, aiming not only to reach the scientific community through high-impact publications, but also to improve outreach to society, increasing awareness and understanding of ocular diseases and emerging therapies. Long-term international collaboration frameworks, fostering durable partnerships supported by joint research projects, coordinated funding applications, and co-authored scientific outputs. Exchange of expertise and methodologies, enabling mutual learning and integration of innovative techniques that enhance experimental capabilities across participating groups.

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	X Yes ☐ No
X We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
None

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

X **Understood**

Name: Ruben Álvarez de Blas
Date completed: 18/09/2026

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)

Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	Universidad Nacional del Sur
Short name / acronym used in projects	UNS
Participant Identification Code (PIC), if available	PIC de CONICET 998619754 (to be confirmed)
Website(s)	https://inibibb.conicet.gov.ar/
Country	Argentina
Organisation type	X Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Camino La Carrindanga Km7, Edificio E1, 8000, Bahía Blanca, Argentina.
Department(s), institute(s), research group(s) or unit(s) that would be involved	Lipid and Signal Transduction Laboratory, Instituto de Investigaciones Bioquímicas de Bahía Blanca (INIBIBB), Consejo Nacional de Investigaciones Científicas y Tecnologicas (CONICET), Universidad Nacional del Sur (UNS). *(The INIBIBB is research institute affiliated with both, CONICET and UNS).* https://inibibb.conicet.gov.ar/laboratorio-de-lipidos-y-transduccion-de-senales/
Additional institutional resources relevant to collaboration	Facilities and equipment of the INIBIBB-CONICET-UNS: https://inibibb.conicet.gov.ar/equipos/ Animal Facilities: CONICET–UNS Absolute Barrier Animal Facility. https://inibibb.conicet.gov.ar/bioterio/ Other services available at UNS: Flow cytometry services. https://inbiosur.conicet.gov.ar/citometria-de-flujo/

Main contact person for this profile	**Melina Valeria Mateos,** Independent Researcher INIBIBB-CONICET. Associate Professor, Pharmacology and Toxicology, Department of Biology, Biochemistry and Pharmacy, UNS.
Main contact email	mvmateos@criba.edu.ar melinavaleriamateos@gmail.com
Other contact person(s), if relevant	Name, role, email

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
X Ocular surface	X Retina
☐ Inherited retinal dystrophies (IRDs)	X Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
☐ Advanced therapies / regenerative medicine	☐ Gene therapy / cell therapy / tissue engineering
X Inflammation / biomarkers / neuroprotection	☐ Drug delivery / formulations / nanomedicine
X Drug discovery / screening / repurposing	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	☐ Other (please specify):

Free Keywords for internal searchable database (please provide up to 10)	Retinal pigment epithelium, retina, inflammation, oxidative stress, lipid signaling, pre-clinical models
Geographies of interest for exchanges (optional)	Spain, Germany, France

Overview of your research / innovation focus
Please describe the main focus of the team / unit that could participate. Keep it concise but specific (recommended: 150-250 words). **Lipid and Signal Transduction Lab at INIBIBB (PI: Melina V. Mateos, Members: Vicente Bermúdez, María Sol Echevarría, Paula E. Tenconi).** Our overall goal is to investigate novel molecular mechanisms that may modulate the inflammatory process in ocular and retinal degenerative diseases that can lead to vision loss, including uveitis, dry eye disease, age-related macular degeneration and diabetic retinopathy, among others. The pharmacological therapeutic options currently available for these inflammatory ocular diseases remain limited, that is why it is essential to deepen our understanding of the underlying molecular mechanisms involved in order to identify novel therapeutic targets. Over the past 13 years, our laboratory has characterized the role of the Phospholipase D (PLD) signaling pathway, which generates bioactive lipid second messengers, in the inflammatory response of retinal pigment epithelium (RPE) cells exposed to inflammatory injury. Pharmacological inhibition of the PLD pathway not only prevented the inflammatory response and oxidative stress but also preserved RPE cell function and viability. These findings position classical PLD isoforms (PLD1 and PLD2) as potential therapeutic targets to simultaneously counteract and/or prevent the inflammatory response and oxidative stress, two key factors in the pathophysiology of the above mentioned inflammatory retinal diseases. Currently, we are investigating the effects of PLD inhibitors in an *in vivo* model of uveitis as well as in an *ex vivo* model of neurodegeneration using porcine neural retinal explants. Our preliminary data from the endotoxin-induced uveitis rat model show that the PLD2 inhibitor is able to reduce clinical signs, total protein concentration in the aqueous humor and cellular infiltration in both the anterior and posterior segments of the eye. As future projects, we aim to investigate the role of the PLD pathway in ocular surface diseases, such as dry eye disease, and in the physiopathology of retinoblastoma (RB).

Why your team could be relevant to a translational eye-health and vision consortium

Please explain the specific added value your organisation / team could bring to a consortium centred on visual health, translational ophthalmology and related enabling technologies. ***Please think broadly: the contribution does not need to be directly ophthalmology-related. Interdisciplinarity is strongly encouraged, and expertise from apparently distant fields may open unexpected routes to innovation, new paradigms, and creative solutions to complex challenges that currently limit progress from research to patient benefit****.*

Our approach may lead to new opportunities for the development of safer and more effective pharmacological therapies for inflammatory ocular diseases. In the future, the implementation of innovative therapeutic strategies could not only significantly improve patients' quality of life but also contribute to strengthening public health systems. Regarding RB, in Argentina is often diagnosed at a later stage, and survival rates are lower in the less developed regions. As a result, these patients may progress to more aggressive forms of RB and have a higher probability of recurrence or resistance to conventional therapy. Therefore, despite major advances in treatment, these cases remain difficult to manage with currently available pharmacological options, and most require enucleation. In this context, the search for new compounds with therapeutic activity against retinoblastoma has become a priority.

Our laboratory has extensive expertise in a wide range of cell and molecular biology techniques, as well as microscopy. All members of the laboratory have completed training at UNS (equivalent to a FELASA Function A and B of the Functions framework defined in EU Directive 2010/63/EU) for working with laboratory animals. We have experience in both basic and preclinical research, using *in vitro* cell culture models and the endotoxin-induced uveitis (EIU) rat model. We have the infrastructure and trained personnel required to establish and implement several animal models using specific pathogen-free (SPF) rodents. We are currently establishing a rat model of dry eye disease and in the near future we will be implementing behavioral assays to assess visual function, including the visual cliff test and the looming test.

In addition to the members of our laboratory, the team participating in this consortium includes researchers from other laboratories at INIBIBB whose research lines, while not directly related to vision science, provide expertise that we believe will make a valuable contribution to the consortium:

-**Dr. Gerardo Martín Oresti,** together with his principal collaborator, **Dr. Florencia Santiago Valtierra**, investigate **lipid metabolism** and membrane remodeling during cellular differentiation, with particular emphasis on polyunsaturated fatty acids (PUFAs), glycerophospholipids, and highly unsaturated sphingomyelin and ceramide species containing very-long-chain PUFAs. Their research combines *in vivo*, *ex vivo*, and *in vitro* models with chromatographic (GC-MS and HPLC), biochemical, and molecular approaches.

-**Dr. Daniel Alejandro Peñalva** has expertise in assessing the **biophysical properties of membranes**. His expertise includes the development and study of model membrane systems, such as liposomes, giant unilamellar vesicles, monolayers, and supported lipid bilayers, and the assessment of the biophysical properties of both biological and model membranes.

Our team has extensive experience primarily in the development and management of preclinical models. Given our background in biochemistry, molecular biology and biophysics, together with our expertise in describing molecular mechanisms, we can contribute with scientific rigor to the preclinical stages of research projects, supporting their progression toward translational applications. Our principal added value lies in the integration of ocular preclinical models with lipid signaling, lipidomic and membrane biophysics approaches. We can provide mechanistic and functional validation of candidate therapeutic targets in RPE cells, retinal explants and rodent models of ocular inflammatory diseases.

In summary the team can contribute to:

- preclinical evaluation of anti-inflammatory and neuroprotective compounds
- lipid and fatty-acid profiling of ocular tissues, cells and biological fluids
- characterization of membrane remodeling associated with oxidative stress and inflammation
- development of lipid-based biomarkers and pharmacodynamic readouts
- training in experimental design, animal models, lipid analysis and membrane biophysics

These capabilities complement clinical, organoid, imaging, drug-delivery and patient-oriented partners by providing a mechanistic and preclinical bridge between target discovery and translational validation.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

Please provide bullet points covering your strongest capabilities, methods, models, analytical approaches, translational assets or innovation strengths.

-Molecular and cell biology: design, development, optimization, and execution of cell and molecular biology techniques (evaluation of cell viability, cell cycle, gene expression at the RNA and protein levels, oxidative stress, and inflammatory responses). Evaluation of metabolic pathways and enzyme activities using radiolabeled substrates (^{3}H, ^{14}C, ^{32}P). The INIBIBB is authorized by Argentina's Nuclear Regulatory Authority (ARN) to handle radioisotopes.

-Lipid biochemistry analyses: lipid and fatty-acid analysis using gas chromatography, GC–MS, HPLC, thin-layer chromatography, and complementary molecular techniques. Quantitative and qualitative analysis of lipids and saturated, monounsaturated, and polyunsaturated fatty acids (omega-3, omega-6, and omega-9), including both *cis* and *trans* isomers. Lipid and fatty-acid profiling, data analysis and interpretation.

- Analysis of membrane biophysical properties: study of model membrane systems, such as liposomes, giant unilamellar vesicles, monolayers, and supported lipid bilayers.

- Laboratory animal experimentation conducted by specialized technical and professional staff: Animal handling, restraint, identification, and sex determination techniques. Substance administration and sample collection procedures. Design and implementation of sedation, analgesia, and anesthesia protocols. Performance of surgical procedures. Conduct of euthanasia procedures and necropsies. Monitoring of the genetic quality of laboratory animals: sample collection and genotyping using molecular biology techniques. Implementation of specialized diets and oral administration of substances through drinking water. Performance of behavioral tests in dedicated rooms equipped for proper execution, including behavioral monitoring and video recording systems.

Infrastructure, equipment, facilities or datasets relevant to consortium work

Examples: specialised laboratories, clean rooms, imaging platforms, -omics, biobanks, clinical cohorts, animal models, software tools, rehabilitation platforms, regulatory support or prototyping capacities.

INIBIBB offers well-established infrastructure and recognized expertise in biochemical research.

Facilities and equipment of the INIBIBB-CONICET-UNS: https://inibibb.conicet.gov.ar/equipos/

-Four cell culture rooms, equipped with laminar flow hoods, incubators, water baths, light microscopes, centrifuges, liquid nitrogen tanks.
-80°C and -20°C freezers.
-Centrifuges and ultracentrifuge
- Cryostat microtome (Leica CM1860)
-RT-PCR
- G:BOX Chemi XRQ imaging system with data processing software
-GC/MS Gas Chromatograph (Agilent 8890/5977C GC/MSD system. Configuration: Split/splitless (S/SL) injection port and a single quadrupole mass spectrometer equipped with a diffusion pump).
-High-Performance Liquid Chromatograph (HPLC). Agilent 1260 Infinity with a UV detector.
- Inverted fluorescence microscope: Nikon Eclipse TE2000-S equipped with a high-sensitivity monochrome digital camera (Nikon DS-Qi2 model).
-Confocal Microscope. LSM 900 Confocal Microscope with two confocal detectors and an Axio Observer 7 inverted microscope.
-Multiskan GO microplate reader (UV/visible spectrofluorometer).
-Spectrofluorometers (Agilent Cary Eclipse and AMINCO SLM 4800)
-Equipment for the formation of Giant Unilamellar Vesicles (Vesicle Prep Pro, Nanion Technologies GmbH)
-Extrusion system for the generation of Large Unilamellar Vesicles (LiposoFast, Avestin Inc.).
Animal Facilities: CONICET–UNS Absolute Barrier Animal Facility. Production and supply of SPF rodents: C57BL/6 mice, BALB/c mice, and Wistar rats. Rodent housing in individually ventilated cage (IVC) racks and conventional cages. Laboratory animal experimentation services provided by specialized technical and professional staff.
https://inibibb.conicet.gov.ar/bioterio/
Other services available at UNS: Flow cytometry services. https://inbiosur.conicet.gov.ar/citometria-de-flujo/

Training capacity and knowledge-transfer potential

Please indicate what inbound staff could realistically learn in your organisation (scientific, technical, clinical, translational, regulatory, entrepreneurial or managerial skills).

Inbound staff could acquire advanced knowledge and practical expertise in multidisciplinary biomedical research approaches, integrating molecular and cell biology, lipid biochemistry, membrane biophysics, and *in vitro* and *in vivo* experimental models. They would learn how to design, optimize, and perform experimental strategies to investigate cellular and molecular mechanisms, including the evaluation of cell viability, cell cycle regulation, gene and protein expression, oxidative stress, inflammatory responses, metabolic pathways and signal transduction, and enzymatic activities.
They could also develop expertise in lipid and fatty-acid profiling, including the analysis and interpretation of lipid composition and metabolism, as well as the study of how lipid alterations influence membrane structure and function through model membrane systems.
The INIBIBB was established in 1970 at the UNS. Over the years, INIBIBB has become a leading center for research in Biochemistry and Neuroscience. Committed to both scientific research and academic training, the Institute continues to contribute to the advancement of knowledge in these fields. Additionally, our laboratory has more than 13 years of experience conducting research in vision sciences.

What your team would hope to gain from participating

Examples: access to methods, models, technologies, patient-oriented validation, translational pathways, industrial insight, clinical exposure, data science, regulatory know-how, joint publications or longer-term strategic collaboration. ***Again, please think broadly and beyond ophthalmology.***

Through participation in this consortium, our team will benefit from access to cutting-edge technologies, multidisciplinary expertise, and innovative translational strategies that will enhance our research capabilities and expand our current research in retinal diseases and visual health. We are particularly interested in learning about induced pluripotent stem cell (iPSC) technology and retinal organoids.
Furthermore, this program will provide valuable opportunities for knowledge exchange, collaborative research, and capacity building, enabling the implementation of new methodologies. We have also a strong interest in establishing new collaborations as well as in strengthening the ongoing ones.
Since our research line is predominantly focused on basic science, we are interested in gaining knowledge of the regulatory, legal, and patient-related aspects involved in clinical trial design. This knowledge will be essential to support the future translation of our findings from fundamental research into clinically relevant applications.
Participation in this program will make a significant contribution to the training of PhD students and postdoctoral fellows, enabling them to learn new methodologies and, above all, giving them the opportunity to gain research and academic experience in another country. For senior researchers the exchange program would give them the opportunity to acquire and implement novel methodologies, analytical tools, and experimental strategies that could be implemented at our institution. In addition, the program will also provide training opportunities for faculty members at UNS, which will greatly enrich the university's graduate academic programs.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Melina Valeria Mateos	Independent Researcher. Associate Professor.	INIBIBB-CONICET-UNS	Researcher	Experienced	mvmateos@crib.edu.ar melinavaleriamateos@gmail.com
Vicente Bermúdez	Assistant Researcher	INIBIBB-CONICET-UNS	Researcher	Experienced	bermudez@inibibb-conicet.gob.ar
Gerardo Martín Oresti	Independent Researcher. Associate Professor.	INIBIBB-CONICET-UNS	Researcher	Experienced	gmoresti@criba.edu.ar

Daniel Alejandro Peñalva	Associate Researcher. Associate Professor.	INIBIBB-CONICET-UNS	Researcher	Experienced	dpenalva@criba.edu.ar
María Sol Echevarría	PhD Student	INIBIBB-CONICET-UNS	Researcher	Early-stage	msechevarria@inibibb-conicet.gob.ar
Paula Estefanía Tenconi	Academic staff member	UNS	Researcher	Early-stage	petenconi@inibibb-conicet.gob.ar
TBD	2 x PhD students	INIBIBB-CONICET-UNS	Researcher	Early-stage	

Main disciplines represented in your team	Biochemistry, cell biology, molecular biology, signal transduction, biochemistry of lipids, biophysics
Experience in EU / international / multi-partner collaborative projects	Please indicate the most relevant experience in 1 or 2 lines (no specific details here)
Experience with clinical, translational or patient-facing research (if relevant)	☐ Yes X No ☐ Not applicable | Short note:

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

Please prioritise achievements most relevant to your possible contribution. Please include DOI, URL or identifier.

1. “Shielding retinal pigment epithelium cells from high glucose-induced oxidative stress: the protective effect of Phospholipase D (PLD) pathway inhibition”. Echevarría, MS; Tenconi, PE; Bermúdez, V; Calandria, JM; Bazán, NG, Mateos, MV*. *Biochimica et Biophysica Acta- Molecular Cell Research* (2026) 1873, 120121. (SCIMAGO: Q1). doi: 10.1016/j.bbamcr.2026.120121.

2. “Targeting Phospholipase D Pharmacologically Prevents Phagocytic Function Loss of Retinal Pigment Epithelium Cells Exposed to High Glucose Levels”. Bermúdez, V; Tenconi, PE; Echevarría, MS; Asatrian, A; Calandria, JM; Giusto, NM; Bazan, NG; Mateos, MV*. *International Journal of Molecular Sciences* (2022) 23, 11823. (SCIMAGO: Q1). doi: 10.3390/ijms231911823.

3. “New Retinal Pigment Epithelial Cell Model to Unravel Neuroprotection Sensors of Neurodegeneration in Retinal Disease”. Asatryan, A; Calandria, JM; Kautzmann, MA; Jun, B; Gordon, WC; Do, K; Bhattacharjee, S; Pham, TL; Bermúdez, V; Mateos, MV; Heap, J and Bazan, NG. *Frontiers in Neuroscience* (2022) 16-926629. (SCIMAGO Q2). doi: 10.3389/fnins.2022.926629. eCollection 2022.

4. "High glucose-induced phospholipase D activity in retinal pigment epithelium cells: new insights into the molecular mechanisms of diabetic retinopathy". Tenconi, PE; Bermúdez, V; Oresti, GM; Giusto, NM; Salvador, GA and Mateos, MV*. *Experimental Eye Research* (2019) 184, 243-257. (SCIMAGO: Q1 Ophthalmology). doi: 10.1016/j.exer.2019.04.028.

5. “The Phospholipase D pathway mediates the inflammatory response of the retinal pigment epithelium”. Mateos, MV*; Kamerbeek, CB; Giusto, NM and Salvador, GA. *International Journal of Biochemistry & Cell Biology* (2014) 55, 119-128. (SCIMAGO: Q1). doi: 10.1016/j.biocel.2014.08.016.

Up to 5 relevant previous projects, grants, networks or collaborations

Please provide detailed information (Title, Acronym, funding agency, GA-number, dates). Please indicate your role (coordinator / beneficiary / partner / service provider / collaborator). *Indicate if international.*

National grants:

- -PIP 2021-2023 (11220200101062CO) granted by National Scientific and Technical Research Council (CONICET), Argentina. Period December 2021- December2025. Project title: "Classic PLD isoforms as potential therapeutic targets for retina inflammatory pathologies" (Role PI).
  -PGI 24/B368 granted by Secretaría General de Ciencia y Tecnología, Universidad Nacional del Sur, Argentina (SGCyT-UNS). Period January 2025-December 2028. Project title: "Canonical Phospholipase D (PLD) isoforms as potential therapeutic targets for uveitis and inflammatory diseases of the ocular surface"" (Role PI).

International collaborations and scientific networks:

- Thanks to a ***2018 Developing Country Eye Research Fellowship (DCERF)*** (granted by ARVO Foundation to Melina Mateos) and to a ***2019 ARVO Foundation Collaborative Research Fellowship*** (granted to Vicente Bermúdez) our group began a collaboration with Dr. Nicolás Bazán's laboratory (Neuroscience Center of Excellence, School of Medicine, Louisiana State University Health New Orleans, New Orleans, USA). During this collaboration, we contributed to the characterization of a novel human retinal pigment epithelium (RPE) cell line, designated ABC cells, which exhibits phenotypic, genetic, and functional traits closely resembling native RPE. Also, we were able to study the role of the PLD pathway in this novel experimental model. As a result of this collaboration, we have published 3 research papers outlined in the previous section.
- Thanks to a ***2023 ARVO Foundation Collaborative Research Fellowship*** (granted to Paula Estefanía Tenconi) our laboratory began a collaboration with Dr. Wai Kit Chu (Deparment of Ophthalmology and Visual Sciences, The Chinese University of Hong Kong). This collaboration enabled us to investigate the effects of PLD inhibitors in the endotoxin-induced uveitis rat model and to initiate the PLD-retinoblastoma project.
- During 2025, an ***International Union of Biochemistry and Molecular Biology (IUBMB) Fellowship*** awarded to María Sol Echevarría, a doctoral fellow in our laboratory, enabled her to complete a research stay in Dr. Iván Fernández Bueno's laboratory (IOBA, Universidad de Valladolid, Spain), where she was trained in the porcine neuroretinal explant model. This fellowship marked the beginning of a new collaboration for our laboratory.
- From May 2026 to May 2028, **Melina Mateos serves as President of the Argentine Association for Vision and Ophthalmology Research (AIVO)**. AIVO is the international chapter affiliated with ARVO (https://www.arvo.org/members/international-chapter-affiliates/aivo) and is an interdisciplinary scientific society composed of ophthalmologists, biochemists, biologists, pharmacologists, and bioengineers, among others, engaged in research on vision and ophthalmology. AIVO also works in close collaboration with the Argentine Council of Ophthalmology (CAO), the Argentine Society for Ocular Surface (SASO), the Argentine Society of Ophthalmology (SAO) and the Argentine Society for Research in Biochemistry and Molecular Biology (SAIB).

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)
None.

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	X Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	X Only Researchers ☐ Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only X Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	Visa processing, if required.

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
Please summarise the 2-3 strengths that should be first visible in an internal searchable profile. • Preclinical evaluation of anti-inflammatory, anti-oxidants and neuroprotective compounds: • Lipid and fatty-acid profiling of ocular tissues, cells and biological fluids: lipid and fatty-acid analysis using gas chromatography, GC–MS, HPLC, thin-layer chromatography, and complementary molecular techniques. Characterization of membrane remodeling associated with oxidative stress and inflammation • *In vitro* experimental design, *in vivo* animal models.

Capabilities or expertise you would especially like to access through the consortium
Please indicate the types of teams, methods, infrastructures or translational pathways you would like to connect with in the consortium. • Access to state-of-the-art infrastructures and specialized technologies beyond the current capabilities of our institute. This possibility would significantly strengthen our research capacity (advanced imaging platforms, omics technologies, high-throughput screening approaches, AI-driven analytical tools). • Acquire training in advanced human retinal modelling using iPSC technology. • Acquire training in organotypic neural retina culture platform. • *In vivo* preclinical models of retinal disease: acquire expertise in animal models (mainly rodent models) of ocular diseases, with the aim of subsequently implementing and further developing these models at INIBIBB. • Acquire expertise in electroretinography for the evaluation of visual function, enabling the implementation of these approaches at INIBIBB. • Development of long-term international collaboration frameworks to build durable partnerships, supported by joint research programs, funding applications, and the generation of high-impact co-authored scientific outputs. • Exchange of expertise and methodologies.

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	X Yes ☐ No
X We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
None.

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organization / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organization. Submission of this form confirms that the contributor agrees to this internal use. Organizations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

X Understood

Name: Melina Valeria Mateos
Date completed: July 18, 2026

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)

Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

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Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	The University of Auckland
Short name / acronym used in projects	UoA
Participant Identification Code (PIC), if available	999613131
Website(s)	https://profiles.auckland.ac.nz/i-rupenthal; www.botu.nz (note, group website is currently undergoing update and may not be available)
Country	New Zealand
Organisation type	x Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Private Bag 92019, Auckland 1142, New Zealand
Department(s), institute(s), research group(s) or unit(s) that would be involved	Buchanan Ocular Therapeutics Unit, Department of Ophthalmology
Additional institutional resources relevant to collaboration	Cell culture suite, confocal microscope, animal facility, access to human tissues via the New Zealand National Eye Bank

Main contact person for this profile	Ilva Rupenthal, Professor, Director of the Buchanan Ocular Therapeutics Unit, Department of Ophthalmology, Faculty of Medical and Health Sciences
Main contact email	i.rupenthal@auckland.a.cnz

Other contact person(s), if relevant	Dr Priyanka Agarwal, pharmaceutical formulation lead, p.agarwal@auckland.ac.nz Dr Lola Mugisho, biomedical lead, lola.mugisho@auckland.ac.nz

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	☐ Retina
☐ Inherited retinal dystrophies (IRDs)	x Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
x Advanced therapies / regenerative medicine	☐ Gene therapy / cell therapy / tissue engineering
x Inflammation / biomarkers / neuroprotection	x Drug delivery / formulations / nanomedicine
x Drug discovery / screening / repurposing	x Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	x Other (please specify): in vitro and ex vivo models

Free Keywords for internal searchable database (please provide up to 10)	Drug delivery; drug penetration; formulation; inflammation; inflammasome; extracellular vesicles; human tissue explants; in vitro and ex vivo models
Geographies of interest for exchanges (optional)	NA

Overview of your research / innovation focus
The BOTUT team focusses on translational research integrating pharmaceutical and biomedical expertise to develop innovative ocular drug delivery systems with improved safety and efficacy with current projects on dry eye, diabetic retinopathy and uveitis treatment. We have worked on nanoparticulate, polymeric, in situ gelling and semifluorinated alkane systems for the delivery of various therapeutics but specifically inflammasome blockers to reduce inflammatory processes in various ocular diseases. We also have extensive expertise in preclinical models used to translate novel ocular therapeutics from bench to bedside, including ocular drug penetration models as well as in vitro disease models using cells and tissue explants.

Why your team could be relevant to a translational eye-health and vision consortium
Our team would bring specific expertise in developing and testing ocular delivery systems, evaluating the efficacy of novel treatments in our inflammatory in vitro and ex vivo models, providing expertise in inflammasome biology and most importantly offering clinically relevant human explant models that support clinical translation of novel therapies. We use our ocular models as a window to neurodegenerative diseases and have started to investigate the involvement of the gut-eye-axis in chronic inflammatory eye diseases. Finally, we have extensive expertise in research commercialization being inventors and income sharing scientist on patents as well as being involved in two university spinout companies.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute
• *Ex vivo ocular drug penetration models and analytical capabilities (fluorescence/confocal microscope and HPLC)* • *In vitro cell models of diabetic retinopathy and other inflammatory retinal conditions*

- *Human organotypic retinal culture (sandwich model including retina, choroid and sclera) to investigate disease pathways and novel drug molecules*
- *Human choroidal explant model to investigate neovascularisation processes and novel therapies*
- *Animal models of ocular and neurodegenerative diseases (although limited)*

Infrastructure, equipment, facilities or datasets relevant to consortium work

- *Cell culture suite including a CELLCYTE X live cell imager and analyzer*
- *Fluorescence and confocal microscope*
- *Navicyte Ussing chamber with electrodes*
- *Access to human tissues from the New Zealand National Eye Bank*
- *In depth grading system for experimental autoimmune uveitis signs (and collaboration on AI algorithms for automated grading)*
- *Animal facility with limited ocular and neurodegenerative disease models*
- *Access to clinical cohorts via our clinicians*

Training capacity and knowledge-transfer potential

- *Technical/scientific/translational: In vitro and ex vivo models for drug penetration and anti-inflammatory drug efficacy*
- *Entrepreneurial: patenting and founding a start-up company*

What your team would hope to gain from participating

Sharing our models to be used by others; access to other models to test our inflammasome therapeutics; joint publications and longer-term strategic collaboration; insight into the European research culture and funding system; industrial insight and regulatory know-how

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Ilva Rupenthal	Professor and Director of the Buchanan Ocular Therapeutics Unit	Ophthalmology	Researcher	Experienced	i.rupenthal@auckland.ac.nz
Lola Mugisho	Senior Research Fellow	Ophthalmology	Researcher	Experienced	lola.mugisho@auckland.ac.nz
Priyanka Agarwal	Senior Research Fellow	Ophthalmology	Researcher	Experiences	p.agarwal@auckland.ac.nz

Main disciplines represented in your team	Pharmaceutical sciences, biomaterials, cell biology, immunology and inflammation, commercialization
Experience in EU / international / multi-partner collaborative projects	Alexander von Humboldt fellowship (Cologne, Germany), DAAD PPP grant (Marburg, Germany), ERASMUS+ (Tarsus, Turkey) and FCMH (Marburg/Giessen. Germany) fellowships; multi-partner Health Research Council of New Zealand project and program grant involvement
Experience with clinical, translational or patient-facing research (if relevant)	☐ Yes x No ☐ Not applicable | Short note: Translational/preclinical research; patient facing clinical work only via clinical collaborators

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements
1. *SHOME A, MUGISHO OO, AGARWAL P, NIEDERER RL,* ***RUPENTHAL ID*** *(2026) Blocking the inflammasome pathway: An effective way to treat inflammation in experimental autoimmune uveitis, European Journal of Pharmacology 1011:178442 DOI: 10.1016/j.ejphar.2025.178442*
2. *NASCIMENTO BARBALHO G, BRUGGER S, GRATIERI T, KECK CM, RUPENTHAL ID, AGARWAL P (2026) Influence of simulated tear flow on ocular penetration of transfersomes, European Journal of Pharmaceutics and Biopharmaceutics, DOI: 10.1016/j.ejpb.2026.115154*
3. *LOUIE HH, MUGISHO OO, CHAMLEY LW, RUPENTHAL ID (2025) Small extracellular vesicles released from ARPE-19 cells grown under diabetic retinopathy conditions promote NLRP3 inflammasome activation, ACS Biomaterials Science & Engineering 11(20):6134-6149 DOI: 10.1021/acsbiomaterials.5c00718*
4. *MUGISHO OO, ARYAL J, SHOME A, LYON H, ACOSTA ML, GREEN CR, RUPENTHAL ID (2023) Orally delivered connexin43 hemichannel blocker, tonabersat, inhibits vascular breakdown and inflammasome activation in a mouse model of diabetic retinopathy, International Journal of Molecular Sciences 24(4):38764 DOI: 10.3390/ijms24043876*
5. *KUO YJ, LOUIE HH, RUPENTHAL ID, MUGISHO OO (2021) Characterization of a novel human organotypic retinal culture technique, Journal of Visualized Experiments 172:e620464 DOI: 10.3791/62046*

Up to 5 relevant previous projects, grants, networks or collaborations
1. *Solving chronic and recurrent inflammation: Beyond corticosteroid therapy, Health Research Council Explorer Grant, (NZ$150,000, principal investigator, 2026-2028)* national
2. *Blocking the inflammasome pathway in neurodegenerative diseases, Buchanan Charitable Foundation (NZ$1,800,000, co-principal investigator, 2025-2031)* national with international collaborations in the future
3. *Gut-eye axis – is the inflammasome the missing link? Safe Sight Society Research Grant (NZ$17,135, principal investigator, 2024-2025)* national with international collaboration
4. *Humboldt Research Fellowship for Experienced Researchers, Tackling the vicious circle of dry eye disease (€37,210, principal investigator, 2022-2024)* international with the University of Cologne
5. *Health Research Council Project Grant, Tackling the vicious circle of dry eye disease (NZ$1,195,447, principal investigator, 3 years)* national with international collaboration

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)
1. *SHOME A, MUGISHO OO, RUPENTHAL ID. Improved treatment of inflammation. International Patent Application No. PCT/IB2025/059738 (filed 25th September 2025)*
2. *COUTINHO FP, RUPENTHAL ID, GREEN CR. Methods of treatment and novel constructs. International Patent Application No. PCT/NZ2018/050059 (submitted 28th April 2018); now at National Phase US11466069B2, CN111201247A, EP3615575A4, WO2018199777A1, CA3061738A1, JP2024059834A*

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	x Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	x Only Researchers ☐ Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only x Both ☐ No capacity to host

Administrative / HR / legal considerations that may affect mobility	visa

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
• *Expertise in developing and characterizing ocular formulations, especially for poorly soluble drugs* • *Extensive in vitro and ex vivo models to test anti-inflammatory drugs with strong inflammasome biology expertise* • *Access to human donor tissues to investigate certain disease pathways*

Capabilities or expertise you would especially like to access through the consortium
• *Access to other relevant animal models to test anti-inflammasome drugs* • *Gut-eye-axis research capabilities (e.g. microbiome or gut tissue preparation expertise)* • *Immunology expertise (e.g. flow cytometry analysis of draining lymph nodes)* • *Long-term international collaborations*

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	x Yes ☐ No
x We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
None

9. Additional collaborators you would like us to consider

If there are any additional organisations, teams or contacts - current collaborators, previous collaborators or desired future collaborators - that you think could be relevant for this MSCA-SE consortium, please list them below. The coordination team may contact them to explore their potential interest in joining the consortium-building process.

Please provide the following details (as many entries as needed).

Type of organisation: ☐ Academic ☐ Clinical ☐ SME/industry ☐ NGO/foundation ☐ Public body ☐ Other

Organisation / team name:

Country:

Website:

Existing link with your team: ☐ Current collaborator ☐ Previous collaborator ☐ Desired collaborator ☐ Other

Suggested contact person/email (if known):

Why could they be relevant for this MSCA-SE consortium?

Any additional comments or sensitivities before contacting them?

Universidad de Valladolid

Internal use and confidentiality statement

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X Understood

Name: Ilva Rupenthal
Date completed: 22/08/2026

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)

Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	DE ROTOS Y DESCOSIDOS
Short name / acronym used in projects	RYD
Participant Identification Code (PIC), if available	879476109
Website(s)	https://www.derotosydescosidos.com
Country	Spain
Organisation type	☐ Academic x☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body x☐ Other
Main address	Conde de Benavente, 3, 1A. 47003 Valladolid. Spain
Department(s), institute(s), research group(s) or unit(s) that would be involved	Communication, comercialization
Additional institutional resources relevant to collaboration	

Main contact person for this profile	Silvia Fernandez. CEO.
Main contact email	silvia@derotosydescosidos.com
Other contact person(s), if relevant	

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	☐ Retina
☐ Inherited retinal dystrophies (IRDs)	☐ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
☐ Advanced therapies / regenerative medicine	☐ Gene therapy / cell therapy / tissue engineering
☐ Inflammation / biomarkers / neuroprotection	☐ Drug delivery / formulations / nanomedicine
☐ Drug discovery / screening / repurposing	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	**X** Other (please specify): Digital Marketing, Commercialisation , Communication, Web development

Free Keywords for internal searchable database (please provide up to 10)	Science communication; EU project dissemination; stakeholder engagement; commercialisation strategies; market research and exploitation
Geographies of interest for exchanges (optional)	

Overview of your research / innovation focus
RYD's innovation focus lies in transforming complex biomedical research into clear, engaging and impact-oriented communication, dissemination and exploitation strategies. Building on its participation in European MSCA projects such as CRYSTAL3 and EVEREST, RYD contributes expertise in digital outreach, stakeholder engagement, science communication and market-oriented positioning to strengthen knowledge transfer and societal impact.

Why your team could be relevant to a translational eye-health and vision consortium?
RYD is not contributing eye-health research directly, but is very relevant for translation, communication and impact. The team can support dissemination, stakeholder engagement, digital visibility and exploitation-oriented messaging, helping ensure that scientific advances in vision research are better understood, shared and positioned for real-world uptake. Beyond supporting communication activities, RYD can also contribute more strategically to the consortium by collaborating on market research, business-oriented analysis, exploitation planning, and the development of digital tools to facilitate visibility, stakeholder engagement and future commercialisation pathways.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute
• Strategic communication and dissemination of EU R&I projects, aligned with Horizon Europe requirements. • Translation of complex scientific content into clear, impactful narratives for diverse audiences. • Multi-channel outreach (digital, media, events) to maximise visibility, engagement, and impact. • Stakeholder engagement bridging academia, industry, and society to enhance knowledge transfer. • Development of high-quality communication, training, and visual materials. • Impact-oriented approach: KPIs, analytics, and reporting for EU deliverables.

- Support to exploitation and market uptake through targeted communication and positioning.
- Creative, agile methodologies (storytelling, design thinking) to boost outreach effectiveness.

Infrastructure, equipment, facilities or datasets relevant to consortium work

- Dedicated communication and dissemination office with capacity to coordinate EU project outreach and stakeholder engagement.
- Professional digital infrastructure: web development tools, content management systems, and social media management platforms.
- In-house design and multimedia production capabilities (graphic design, video, branding, digital content creation).
- Tools for analytics and impact monitoring (KPIs, engagement metrics, reporting dashboards).
- Established networks and databases of stakeholders (scientific, industry, media) supporting dissemination and exploitation.
- Facilities to support organisation of events, workshops, and training activities (online and hybrid formats).

Training capacity and knowledge-transfer potential

- Training in science communication for complex biomedical topics (e.g. EVs, biomarkers, therapeutics).
- Skills in translating research outputs into accessible content for scientific, clinical, industry, and public audiences.
- Hands-on experience in dissemination strategies within Horizon Europe projects (planning, execution, reporting).
- Digital communication skills: social media, web content, multimedia production, and branding.
- Stakeholder engagement and networking across academia, industry, and society.
- Exposure to impact-driven approaches: KPIs, analytics, and evaluation of communication effectiveness.
- Insight into exploitation, innovation uptake, and market-oriented communication strategies.
- Project management and coordination skills in international, interdisciplinary consortia.

What your team would hope to gain from participating

- Direct exposure to cutting-edge research methods, technologies, and clinical/translational workflows.
- Insight into scientific and industrial innovation pathways, including product development and commercialisation.
- Strengthening collaboration with leading academic and biotech partners across Europe.
- Access to new case studies and high-impact content to enhance science communication strategies.
- Improved capacity to support exploitation, market uptake, and innovation positioning.
- Opportunities for joint outputs (publications, dissemination materials) and long-term strategic collaborations.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Silvia Fernandez	CEO	Communication	Managerial	Leading / Expert	silvia@derotosydescosidos.com
María del Pilar Orihuela	Communication Specialist	Communication	Technical	Experienced / Senior	maria@derotosydescosidos.com
Sandra Pilar Soria	Communication Specialist	Communication	Technical	Mid-career	sandra@derotosydescosidos.com
Jesus Molpeceres	Multimedia & Content Production	Communication	Technical	Experienced / Senior	jesus@derotosydescosidos.com

Main disciplines represented in your team	• Science communication • Digital communication & media • Graphic design and visual storytelling • Stakeholder engagement and outreach • Marketing and dissemination for R&I projects
Experience in EU / international / multi-partner collaborative projects	Extensive experience in communication, dissemination, and stakeholder engagement within EU-funded, multi-partner research and innovation projects (including Horizon Europe / MSCA), supporting coordination, outreach, and impact activities across international consortia.
Experience with clinical, translational or patient-facing research (if relevant)	☐ Yes ☐ No x☐ Not applicable | Short note:

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

Up to 5 relevant previous projects, grants, networks or collaborations
RYD has participated as a non-academic beneficiary in two European MSCA projects: • CRYSTAL3, H2020-MSCA-RISE, GA No. 101007931, 01/01/2021–31/12/2024; and • EVEREST, Horizon Europe MSCA Staff Exchanges, GA No. 101183034, 01/01/2025–31/12/2028. Through these projects, RYD has contributed expertise in communication, dissemination, digital visibility, stakeholder engagement and commercialisation-oriented support for international, intersectoral biomedical research consortia

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	x☐ Yes ☐ Maybe ☐ No capacity to send staff on secondment Sending capacity details: 1-3 staff; 3-6 person-months; typical duration(s): 1-2 month; staff categories: Technical.
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	☐ Only Researchers x☐ Technical Staff/Innovation x☐ Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only x☐ Both ☐ No capacity to host Hosting capacity details: 1-3 staff; 4-12 person-months; typical duration(s): 1-12 months; staff categories: Technical.
Administrative / HR / legal considerations that may affect mobility	

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
• Science communication and dissemination for EU-funded R&I projects (Horizon Europe / MSCA). • Translation of complex biomedical research into impactful, accessible content for diverse audiences. • Digital outreach and multi-channel engagement (web, social media, multimedia, events). • Stakeholder engagement and bridging academia–industry–society to enhance knowledge transfer and impact.

Capabilities or expertise you would especially like to access through the consortium
• Access to cutting-edge EV research teams working on biomarkers, therapeutics, and clinical applications. • Exposure to advanced methodologies and infrastructures (e.g. omics, imaging, biomanufacturing, clinical validation workflows). • Insight into translational pathways from research to clinical and market applications. • Collaboration with industry partners to understand innovation, product development, and commercialisation processes. • Access to real case studies, data, and scientific outputs to strengthen impact-driven communication and exploitation strategies.

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	x☐ Yes ☐ No
Previous visibility indication in the submitted form: ☐ Coordination team only X☐ Searchable by other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.

Internal use and confidentiality statement

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☐ **Understood**

Name:
Date completed:

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)

Name: Silvia Fernandez

Date completed: 27/04/2026

Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	South East Technological University
Short name / acronym used in projects	SETU
Participant Identification Code (PIC), if available	886107126
Website(s)	https://www.setu.ie/ ; https://www.setu.ie/research-innovation/institute-centres-and-groups/research-groups/science/ocular-therapeutics-research-group-otrg ; https://innovision.setu.ie/ ; https://pmbrc.org/
Country	IRELAND
Organisation type	☐ Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	SETU Cork Road Campus, Waterford, Ireland X91KE0K
Department(s), institute(s), research group(s) or unit(s) that would be involved	Faculty of Science and Computing, Department of Science, Ocular Therapeutics Research Group
Additional institutional resources relevant to collaboration	Materials characterization, chromatography, mass spectrometry, spectroscopy, microscopy, formulation and pre-formulation, molecular biotechnology, biomedical, pre clinical in vitro and ex-vivo platforms, as well as access to PPI Ignite for Patient Involvement in Research

Main contact person for this profile	Name, title, position, department / unit
Main contact email	

Other contact person(s), if relevant	Laurence Fitzhenry, Head of Faculty of Science and Computing, PI, Ocular Therapeutics Research Group, (Laurence.Fitzhenry@setu.ie) Tess Ames, Research Group Manager (Tess.Ames@setu.ie) Marcos Mariano, Scientific Lead, OTRG (Marcos.Mariano@setu.ie)

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	☐ Retina
☐ Inherited retinal dystrophies (IRDs)	☐ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
☐ Advanced therapies / regenerative medicine	☐ Gene therapy / cell therapy / tissue engineering
☐ Inflammation / biomarkers / neuroprotection	☐ Drug delivery / formulations / nanomedicine
☐ Drug discovery / screening / repurposing	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	☐ Other (please specify):

Free Keywords for internal searchable database (please provide up to 10)	Ocular Drug Delivery Ophthalmic Therapeutics Nanomedicine Nanoparticle Drug Delivery Systems Lipid Nanoparticles (LNPs) Nanostructured Lipid Carriers (NLCs) Advanced Drug Delivery Systems Pharmaceutical Formulation Science Translational Medicine Ocular Pharmacokinetics Controlled Drug Release
Geographies of interest for exchanges (optional)	

Overview of your research / innovation focus
The Ocular Therapeutics Research Group (OTRG) at South East Technological University is a multidisciplinary research group focused on the development of innovative therapeutic technologies to address unmet clinical needs in ophthalmology and related healthcare sectors. The group's primary expertise lies in ocular drug delivery, pharmaceutical nanotechnology, biomaterials, and translational medicine, with a particular emphasis on overcoming biological barriers that limit the effectiveness of conventional treatments. OTRG develops advanced drug delivery systems, including lipid nanoparticles, nanostructured lipid carriers, polymeric delivery platforms, and controlled-release technologies designed to improve therapeutic efficacy, patient adherence, and clinical outcomes. Research spans the full translational pathway, from formulation design and physicochemical characterisation through to in vitro, ex vivo, and preclinical evaluation. This integrated approach enables the group to bridge the gap between fundamental pharmaceutical science and real-world clinical application. A key strength of OTRG is its commitment to translational and impact-driven research. The group works closely with academic institutions, healthcare providers, industry partners, patient organisations, and regulatory stakeholders to ensure that research outputs are aligned with patient and societal needs. Public and Patient Involvement (PPI) is embedded throughout the research lifecycle, supporting the co-design of research questions, methodologies, and impact pathways.

Beyond ophthalmology, OTRG applies its expertise in nanomedicine, advanced therapeutics, and drug delivery technologies to broader challenges within human and animal health, reflecting a growing One Health perspective. Through international collaborations, European-funded projects, industry engagement, and researcher training initiatives, OTRG aims to accelerate the translation of innovative therapeutic technologies from laboratory discovery to patient benefit.

Why your team could be relevant to a translational eye-health and vision consortium

Please explain the specific added value your organisation / team could bring to a consortium centred on visual health, translational ophthalmology and related enabling technologies. ***Please think broadly: the contribution does not need to be directly ophthalmology-related. Interdisciplinarity is strongly encouraged, and expertise from apparently distant fields may open unexpected routes to innovation, new paradigms, and creative solutions to complex challenges that currently limit progress from research to patient benefit****.*

The Ocular Therapeutics Research Group (OTRG) at South East Technological University (SETU) brings expertise that extends beyond traditional ophthalmic formulations and encompasses translational medicine, advanced therapeutic technologies, public and patient involvement, innovation management, and multidisciplinary research integration.
At its core, OTRG specialises in the development of novel drug delivery systems, nanomedicine platforms, biomaterials, pharmaceutical formulation, and translational therapeutic technologies. These capabilities support the development of next-generation treatments capable of overcoming biological barriers, improving therapeutic efficacy, and accelerating progression from laboratory discovery to clinical application.
However, OTRG's contribution to a consortium centred on visual health extends beyond therapeutic development. The group has extensive experience coordinating large international research networks, managing complex multi-partner projects, and facilitating collaboration between academia, healthcare providers, industry, patient organisations, regulators, and policymakers. This systems-level perspective enables OTRG to contribute to the entire innovation pathway, from early-stage discovery through to implementation and patient impact, as well as provide support and training in project and programme management, communication and community outreach, and engagement.
A distinctive strength of the group is its leadership in Public and Patient Involvement (PPI), citizen engagement, and co-design methodologies. OTRG has pioneered approaches that embed patients and end-users throughout the research lifecycle, ensuring that technological innovation remains aligned with real-world clinical needs and societal priorities.
The group is particularly interested in interdisciplinary collaborations that bring together expertise from fields such as artificial intelligence, psychology, behavioural science, implementation science, engineering, digital health, social sciences, health economics, veterinary medicine, environmental sciences, and policy research. Such collaborations have the potential to generate new paradigms for understanding disease, improving healthcare access, enhancing treatment adherence, developing innovative diagnostic and therapeutic technologies, and addressing challenges that currently limit translation into patient benefit.
Through its combination of scientific expertise, translational focus, stakeholder engagement, and international network-building experience, OTRG can serve as both a technology development partner and an integrative hub capable of connecting diverse disciplines to accelerate innovation in visual health and beyond.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

Please provide bullet points covering your strongest capabilities, methods, models, analytical approaches, translational assets or innovation strengths.

- Ocular drug delivery and ophthalmic formulation development
- Nanomedicine and nanoparticle-based therapeutic systems
- Lipid nanoparticle (LNP) and nanostructured lipid carrier (NLC) technologies
- Controlled and sustained drug release platforms
- Pharmaceutical formulation design, optimisation and scale-up
- Biomaterials and advanced therapeutic delivery systems
- Physicochemical characterisation of drug delivery systems
- Stability assessment and formulation development under ICH guidelines
- In vitro, ex vivo and preclinical evaluation of ophthalmic therapeutics
- Ocular pharmacokinetics and biodistribution studies
- Development of translational models for ocular surface and retinal disease
- Drug repurposing and reformulation strategies

- Microfluidics-enabled formulation development and process optimisation
- Quality by Design (QbD) and pharmaceutical development methodologies
- Regulatory science and translational development planning
- Technology readiness advancement from discovery through preclinical validation
- Public and Patient Involvement (PPI) and co-design methodologies
- Citizen science and stakeholder engagement in healthcare research
- Implementation of Responsible Research and Innovation (RRI) frameworks
- Research impact planning, pathway-to-impact development and knowledge mobilisation
- Academic-industry collaboration and technology transfer
- Commercialisation strategy and spin-out support
- International research network coordination and consortium management
- Researcher training and capacity building through MSCA, doctoral and postdoctoral programmes
- Open science, research integrity and FAIR data practices
- One Health approaches linking human, animal and environmental health
- Interdisciplinary collaboration across pharmaceutical sciences, biotechnology, engineering, health sciences and social sciences
- Development and management of large-scale European research and innovation projects
- Integration of patient, clinical, academic and industry perspectives to accelerate translation from research to patient benefit

Infrastructure, equipment, facilities or datasets relevant to consortium work

- *Pharmaceutical and molecular biotechnology laboratories supporting formulation development, characterisation, and translational research*
- *Nanomedicine and drug delivery development facilities for lipid nanoparticles (LNPs), nanostructured lipid carriers (NLCs), nanoemulsions, liposomes, and polymeric delivery systems (e.g. high pressure homogenizer and probe ultrasound)*
- *Microfluidics-based formulation development platforms for advanced therapeutic manufacturing and process optimisation (Unchain Labs Sunscreen®)*
- *Particle characterisation infrastructure; including dynamic light scattering (DLS), zeta potential analysis, Laser diffraction particle sizing, X-ray diffraction (XRD) and stability assessment.*
- *Physicochemical characterisation facilities supporting formulation optimisation, quality control, and translational development, including liquids and solid rheology (Rheometer, FT4), calorimetry (ITC, DSC) and thermogravimetric analysis (TGA).*
- *Cell culture laboratories for in vitro therapeutic screening, cytotoxicity assessment, and mechanistic studies*
- *Ex vivo ocular tissue models for evaluating drug penetration, retention, permeability, and tissue interactions (Franz Diffusion cell and Ussing Chamber)*
- *Access to preclinical ophthalmic models through established national and international collaborations*
- *Analytical chemistry capabilities supporting formulation characterisation, impurities and drug quantification including liquid (UPLC, HPLC) and gas chromatography (GC), including mass spectroscopy (UPLC-MS and GC-MS).*
- *Stability testing infrastructure supporting pharmaceutical product development under regulatory frameworks*
- *Expertise in pharmacokinetic, biodistribution, and translational evaluation studies*
- *Public and Patient Involvement (PPI) infrastructure, including established relationships with patient organisations, advisory panels, and engagement networks*
- *Access to patient insight and co-design expertise through collaborations with organisations such as Fighting Blindness, Dry Eye Foundation, IPPOSI, and EUPATI*
- *Research management and innovation support infrastructure for large-scale international collaborative projects*
- *Regulatory, ethics, data management, and open science support through established institutional frameworks*
- *Commercialisation and technology transfer support through established innovation and enterprise networks*
- *European project management infrastructure with extensive experience coordinating Horizon Europe, MSCA, and international collaborative programmes*
- *Training and researcher development platforms supporting doctoral, postdoctoral, academic, industry, and clinical participants*
- *Digital collaboration and knowledge-sharing infrastructure enabling international and intersectoral research networks*
- *Access to multidisciplinary expertise across pharmaceutical sciences, biotechnology, health sciences, engineering, computer science, psychology, social sciences, and public engagement*
- *Strong links to clinical, industry, regulatory, and patient stakeholder communities that support translation from laboratory research to real-world implementation and patient benefit*

Training capacity and knowledge-transfer potential

Please indicate what inbound staff could realistically learn in your organisation (scientific, technical, clinical, translational, regulatory, entrepreneurial or managerial skills).

As part of the Department of Sciences, OTRG is included in a collaborative environment across physics, biology, and chemistry supports pharmaceutical and veterinary education and research. Within this multidisciplinary setting, the group work closely with scientists from diverse scientific backgrounds, providing inbound staff with broad interdisciplinary exposure. Scientific, translational, innovation, and project management skills spanning the full research and innovation lifecycle can be offered. Please, see an extensive list below.

Scientific and Technical Skills:
• Ophthalmic drug delivery and therapeutic development
• Nanomedicine and nanoparticle formulation design
• Lipid nanoparticle (LNP), nanostructured lipid carrier (NLC), and nanoemulsion technologies
• Pharmaceutical formulation optimisation and characterisation
• Microfluidics-enabled manufacturing approaches
• Physicochemical analysis of advanced therapeutic systems
• Stability assessment and pharmaceutical product development
• Experimental design and translational research methodologies
• In vitro and ex vivo model development and evaluation
• Ocular pharmacokinetics and biodistribution studies
• Data analysis and interpretation for preclinical therapeutic development
Translational and Innovation Skills
• Translational medicine and technology development pathways
• Advancing technologies through Technology Readiness Levels (TRLs)
• Research impact planning and pathway-to-impact development
• Public and Patient Involvement (PPI) and co-design methodologies
• Citizen science and stakeholder engagement approaches
• Responsible Research and Innovation (RRI)
• Open science and FAIR data principles
• Academic-industry collaboration and partnership development
• Technology transfer and commercialisation planning
Regulatory and Research Governance Skills
• Research ethics and governance processes
• Data management planning and research integrity
• Regulatory considerations for advanced therapeutic development
• Quality by Design (QbD) principles
• Risk management and quality assurance approaches
Leadership and Management Skills
• International consortium development and management
• Horizon Europe and MSCA project implementation
• Research leadership and multidisciplinary team working
• Stakeholder engagement and network building
• Science communication and public engagement
• Grant development and funding strategy
• Project management within complex international collaborations

Finally, OTRG offers a particularly strong environment for researchers interested in understanding how scientific discoveries are translated into real-world healthcare innovations, combining laboratory science with patient engagement, commercialisation, regulatory awareness, and international research collaboration.

What your team would hope to gain from participating

Examples: access to methods, models, technologies, patient-oriented validation, translational pathways, industrial insight, clinical exposure, data science, regulatory know-how, joint publications or longer-term strategic collaboration. ***Again, please think broadly and beyond ophthalmology.***

Through participation in the programme, OTRG aims to expand its collaborative network by engaging with researchers and institutions that share complementary expertise and aligned translational goals. We expect the programme to facilitate access to new methodologies, advanced technologies, and multidisciplinary perspectives that can strengthen our ongoing research activities. Particular areas of interest include the development of advanced 3D cell culture and disease models, proteomics and biomolecular analysis workflows, access to animal facilities and preclinical ocular disease models, as well as potentially transitioning into clinical trial readiness, advanced microscopy and structural characterisation techniques for nanoparticle systems, as well as artificial intelligence and data-driven approaches for pharmaceutical and biomaterials research. Exposure to complementary translational pathways, regulatory perspectives, and industrially relevant methodologies would also be highly valuable. In addition to scientific and technical advancement, we view the programme as an opportunity to establish long-term strategic collaborations supporting joint publications, future grant applications, infrastructure sharing, researcher mobility, and interdisciplinary training of students and early-career researchers.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Laurence Fitzhnery	Principal Investigator	Science & Computing/OTRG	Principal Investigator	Senior Researcher	Laurence.Fitzhenry@setu.ie
Marcos Mariano	Scientific Lead/Post Doc	OTRG	Post Doc	Mid Level	Marcos.Mariano@setu.ie
Tess Ames	Research Group Manager	OTRG	Administrative Support	N/A	Tess.Ames@setu.ie

Main disciplines represented in your team	• Pharmaceutical Sciences • Drug Delivery and Pharmaceutical Technology • Nanomedicine • Biomaterials • Molecular and Cell Biology • Pharmacology • Ophthalmology and Vision Research • Translational Medicine • Biotechnology • Public and Patient Involvement (PPI) • Innovation and Commercialisation • Research Management and International Project Delivery
Experience in EU / international / multi-partner collaborative projects	Extensive experience coordinating and participating in large-scale European and international collaborative research programmes, including Horizon Europe, MSCA, Staff Exchange, Doctoral Network, and COFUND initiatives involving academic, industry, clinical, and patient organisation partners.
Experience with clinical, translational or patient-facing research (if relevant)	☐ Yes ☐ No ☐ Not applicable | Short note:

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements
Publication

Development and in vitro and ex vivo characterization of a twin nanoparticulate system to enhance ocular absorption and prolong retention of dexamethasone in the eye: from lab to pilot scale optimization. 23 Apr 2025, Nanoscale Advances. 7, 10, p. 3125-3142 18 p. (Published)

Designing biocompatible curcumin-loaded Pluronic F127 micelles for enhanced ocular drug delivery

Klahan, B., O'Reilly, N. J., Sigurdsson, H. H., Chauhan, A., Mering, S., Hakkarainen, J. J. & Fitzhenry, L., 05 Feb 2026, In: Journal of Drug Delivery Science and Technology. 119, 108100.

Corrigendum to "Composite hydrogel contact lens systems for naringenin: A comparison of drug loading and release behaviour in drug carrier systems" [J. Drug Deliv. Sci. Technol. 102-A (2024) 106375]: A comparison of drug loading and release behaviour in drug carrier systems" [J. Drug Deliv. Sci. Technol. 102-A (2024) 106375] (Journal of Drug Delivery Science and Technology (2024) 102(PA), (S177322472401044X), (10.1016/j.jddst.2024.106375))

Faran, S. A., Nguyen, D. C. T., Dowling, J., Ryan, R., McLoughlin, P. & Fitzhenry, L., 01 Feb 2025, In: Journal of Drug Delivery Science and Technology. 104, 106554.

Crosslinked Poly(HEMA)–Based Cellulose Nanocrystal Contact Lenses: Polymer Structure and Nanoparticle Lixiviation

Spilit, A., Faran, S. A., Dufresne, A., Fitzhenry, L. & Mariano, M., 01 Jan 2025, In: International Journal of Polymer Science. 2025, 1, 3707326.

Saoirse Casey-Power, Camila Vardar, Richie Ryan, Gautam Behl, Peter McLoughlin, Mark E. Byrne, Laurence Fitzhenry, NAD+-associated-hyaluronic acid and poly(L-lysine) polyelectrolyte complexes: An evaluation of their potential for ocular drug delivery, European Journal of Pharmaceutics and Biopharmaceutics, Volume 192,2023,Pages 62-78,ISSN 0939-6411,

Up to 5 relevant previous projects, grants, networks or collaborations

ORBITAL – Ocular Research by Integrated Training and Learning Network, H2020 MSCA ITN, GA 813440, 2019 – 2024 (~€4M, Coordinator) – International
INNOVISION – Innovative Vision Research, Horizon Europe MSCA Postdoctoral Cofund, GA 101217199, 2026 – 2030 (~€3.1M, Coordinator) – International
PPI – Ignite, Health Research Board Ireland, 2026 – 2030, GA NA, ~€10M, National, Beneficiary
EVEREST – Extracellular Vesicle Research Exchanges for Advanced Biomarkers and Therapeutics, MSCA Staff Exchange, Horizon Europe, International, Beneficiary2025 – 2028
OcuHealth – Enterprise Ireland Commercialisation Fund, ~€700k, Principal Investigator, 2023-2026

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)

Patents:
Ocular drug delivery (NLC family): US patent granted October 2023, with related EP and WO applications. Waterford Institute of Technology listed as assignee.

Microemulsion for ophthalmic drug delivery (LME family): European patent granted July 2025, with unitary effect registered across multiple EU member states. A divisional EP application is pending. Waterford Institute of Technology listed as assignee.

Artificial tears (NLC family): European patent granted March 2026, with related applications pending in US and JP. Waterford Institute of Technology listed as assignee.

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	☐ Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	☐ Only Researchers ☐ Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only ☐ Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	Will need to expand this out to other researchers in the group, which may require visa considerations, but we have a lot of experience in this so can support that

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
Please summarise the 2-3 strengths that should be first visible in an internal searchable profile. • Advanced drug delivery and nanomedicine expertise, particularly in ophthalmic therapeutics, pharmaceutical formulation, and translational therapeutic development. • Strong capability in translating research from laboratory discovery to patient benefit through industry collaboration, commercialisation, regulatory awareness, and innovation-focused research programmes. • Leadership in Public and Patient Involvement (PPI), stakeholder engagement, and international consortium building, enabling multidisciplinary collaborations that connect science, technology, healthcare, and society.

Capabilities or expertise you would especially like to access through the consortium
Please indicate the types of teams, methods, infrastructures or translational pathways you would like to connect with in the consortium. • Clinical ophthalmology teams, patient cohorts, clinical trial networks, and healthcare implementation partners to strengthen translational pathways from preclinical research to patient care. • Advanced imaging, diagnostics, artificial intelligence, digital health, omics technologies, and data science expertise to support the development of next-generation approaches to visual health research. • Experts in behavioural science, psychology, health economics, social sciences, rehabilitation, regulatory science, and policy development to address barriers to healthcare access, treatment adherence, patient outcomes, and real-world impact. • Industry partners and technology developers with expertise in manufacturing, scale-up, regulatory approval, commercialisation, and market adoption of innovative healthcare technologies.

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	☐ Yes ☐ No
☐ We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.

Examples: PIC pending approval, institutional approval, legal status, hosting rules, IP constraints, data access, ethics, clinical permissions, staffing continuity, resource availability.

Participation is subject to institutional approval, resource availability, and alignment with ongoing research priorities and funding commitments. Any staff exchanges involving access to patient data, clinical environments, human participants, animal studies, or proprietary technologies would require appropriate ethical, legal, regulatory, and data protection approvals.
Specific collaboration activities may also be subject to intellectual property agreements, material transfer agreements, data sharing agreements, and partner-specific requirements. OTRG welcomes discussions with potential partners to identify suitable exchange opportunities and establish appropriate governance arrangements where required.

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

☐ **Understood**

Name:
Date completed:

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)

Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	Technische Universität Dresden
Short name / acronym used in projects	TUD
Participant Identification Code (PIC), if available	999897729
Website(s)	https://tu-dresden.de/ Marius Ader / Ader Lab: https://tu-dresden.de/cmcb/crtd/forschungsgruppen/crtd-forschungsgruppen/ader Additional research-group page: https://www.digs-ils.phd/research/research-groups/marius-ader
Country	Germany
Organisation type	☒ Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Fetscherstr. 105, 01307 Dresden, Germany
Department(s), institute(s), research group(s) or unit(s) that would be involved	Center for Regenerative Therapies Dresden (CRTD), Center for Molecular and Cellular Bioengineering (CMCB), Technische Universität Dresden. Marius Ader group / Cell Replacement in the Mammalian Retina.
Additional institutional resources relevant to collaboration	CRTD/CMCB regenerative-medicine environment; retinal organoid and pluripotent-stem-cell-based research; preclinical retinal degeneration models; broader Dresden expertise in regenerative biology, organoids and model organisms.

Main contact person for this profile	Prof. Dr. Marius Ader, PI / Full Professor, CRTD/CMCB, Technische Universität Dresden
Main contact email	marius.ader@tu-dresden.de
Other contact person(s), if relevant	Jochen Hentschel, Lab manager, jochen.hentschel@tu-dresden.de

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	☒ Retina
☐ Inherited retinal dystrophies (IRDs)	☐ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
☒ Advanced therapies / regenerative medicine	☒ Gene therapy / cell therapy / tissue engineering
☐ Inflammation / biomarkers / neuroprotection	☐ Drug delivery / formulations / nanomedicine
☐ Drug discovery / screening / repurposing	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	☐ Other (please specify):

Free Keywords for internal searchable database (please provide up to 10)	retinal organoids; iPSC; photoreceptor transplantation; RPE replacement; cell replacement; retinal degeneration; photoreceptor integration; retinal regeneration; preclinical models; cell therapy
Geographies of interest for exchanges (optional)	Any

Overview of your research / innovation focus
The Ader Lab at the Center for Regenerative Therapies Dresden focuses on the development of cell-based strategies for the treatment of retinal degeneration, with particular emphasis on replacing lost photoreceptors and retinal pigment epithelial (RPE) cells. The group generates photoreceptor-containing retinal organoids and RPE from pluripotent stem cells, including human induced pluripotent stem cells (iPSCs), and evaluates their potential in preclinical models of retinal degeneration. Research addresses the isolation and enrichment of rod and cone photoreceptors and RPE, transplantation into degenerating retina, and analysis of donor-cell morphology, maturation, synaptic connectivity and functional responses. A major research interest is understanding the cellular and molecular requirements for successful donor-cell interaction with the host retina. The group's work includes studies of photoreceptor material exchange, donor-cell integration and functional repair. Current directions include defining conditions that optimise photoreceptor and RPE replacement and exploring sequential co-transplantation approaches. The overall orientation is preclinical and translational, with the long-term aim of developing cell-based therapeutic strategies for blinding retinal diseases.

Why your team could be relevant to a translational eye-health and vision consortium
The Ader Lab could provide a complementary regenerative-retina and cell-replacement capability within an IOBA-led translational eye-health network. Its expertise in human iPSC-derived retinal organoids, photoreceptor and RPE preparation, transplantation and functional analysis is particularly relevant to consortium activities addressing inherited

and degenerative retinal disease. The group offers an important bridge between human cell-based models and in vivo preclinical validation, allowing disease mechanisms and therapeutic concepts to be assessed in complex retinal systems. The strongest potential complementarity identified with IOBA concerns retinal disease modelling and regenerative therapy: IOBA's work on PROM1-associated retinal degeneration, patient-derived iPSCs, retinal organoids and neuroretinal models could be connected with Dresden expertise in photoreceptor/RPE replacement, transplantation and functional validation. This could create opportunities for knowledge transfer in both directions, from disease modelling and clinically informed research questions towards advanced regenerative validation, and from transplantation and cell-integration expertise towards improved preclinical therapeutic development.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

- Generation and use of retinal organoids and RPE from pluripotent stem cells, including human iPSC-derived material.
- Isolation and enrichment of rod and cone photoreceptors and RPE from stem-cell-derived retinal organoids.
- Retinal transplantation approaches for photoreceptor and RPE replacement in preclinical retinal degeneration models.
- Preclinical mouse models of retinal degeneration and assessment of donor-cell behaviour in degenerating retina.
- Analysis of transplanted donor cells using morphology, immunohistochemistry/immunocytochemistry, expression patterns and assessment of synaptic connections.
- Functional assessment of retinal repair, including light-response measurements and electrophysiological approaches such as electroretinography.
- Investigation of donor-cell integration, maturation and interaction with host retinal cells, including Müller glia and second-order neurons.
- Research into cytoplasmic/material transfer between photoreceptors and its implications for retinal cell-replacement strategies.
- Development of strategies for combined photoreceptor and RPE replacement, including sequential co-transplantation.
- Knowledge transfer on regenerative-retina models, transplantation workflows and preclinical validation of cell-replacement concepts.

Infrastructure, equipment, facilities or datasets relevant to consortium work

- CRTD/CMCB research environment focused on regenerative medicine and cell-based approaches.
- Cell culture and pluripotent-stem-cell-derived retinal organoid/RPE research capacity.
- Preclinical retinal degeneration models, particularly mouse models used for photoreceptor/RPE transplantation studies.
- Platforms and expertise for retinal transplantation and post-transplant donor-cell analysis.
- Immunohistochemical/immunocytochemical and microscopy-based analysis of retinal tissue and transplanted cells.
- Functional retinal assessment, including electroretinography and light-response analysis.
- Broader Dresden ecosystem of regenerative biology, organoid and model-organism expertise, including additional potential complementarities identified during the meeting.
- A range of core facilities including light microscopy, electron microscopy, mouse house, genome center, flow cytometry and stem cell engineering are available in house.

Training capacity and knowledge-transfer potential

- **Hands-on training in retinal cell-replacement research**, from the generation and characterisation of iPSC-derived retinal organoids to the preparation of photoreceptor and RPE populations for transplantation studies.
- **Practical training in preclinical retinal transplantation models**, including experimental design, transplantation procedures and evaluation of donor-cell behaviour in models of retinal degeneration.

- **Training in the assessment of retinal repair**, combining structural and cellular analyses with evaluation of donor-cell maturation, interaction with host retinal cells, synaptic connectivity and functional responses.
- **Exposure to translational development of cell-based therapies**, providing insight into how experimental findings on photoreceptor and RPE replacement can inform the development of therapeutic strategies for retinal degeneration.

What your team would hope to gain from participating
• Access to clinically informed retinal disease models and translational research questions, particularly inherited retinal degeneration and patient-derived iPSC systems. • Exchange with teams working on PROM1 and other genetically defined retinal diseases, retinal organoids and disease modelling. • Exposure to complementary neuroprotection, retinal explant, gene-editing and clinically oriented validation approaches. • Development of longer-term collaboration between Dresden regenerative-retina research and IOBA's translational ophthalmology environment. • Joint research and knowledge-transfer opportunities linking cell replacement, disease modelling and functional retinal validation.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Prof. Dr. Marius Ader	PI / Full Professor	Center for Regenerative Therapies Dresden (CRTD), an institute of the Center for Molecular and Cellular Bioengineering (CMCB); Ader Lab	Research staff / PI	Experienced Researcher	marius.ader@tu-dresden.de

Main disciplines represented in your team	Retinal biology; regenerative medicine; cell replacement; stem-cell biology; retinal organoids; photoreceptor biology; RPE biology; transplantation; preclinical retinal disease models; microscopy/immunohistochemistry; electrophysiology, vision-guided behavioral test (Light/dark box, opto-kinetic tracking)
Experience in EU / international / multi-partner collaborative projects	Marius indicated that he had not been very active in EU projects previously but is becoming more involved. He expressed interest in building international collaboration through MSCA Staff Exchanges and recommended the COST Action 'Retina 4 Future' as a networking route.
Experience with clinical, translational or patient-facing research (if relevant)	☐ Yes ☒ No ☐ Not applicable | Short note:

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements
• **Tessmer K, Hammer J, Gasparini SJ, Schmidtke K, Adhikari T, Michalke L, Loureira LR, Kurtz A, Last A, Chelius N, Johannsen H, Carido M, Ferguson S, Petzold A, Friedrich UA, Kurth T, Dahl A, Feldmann A, Almedawar S, Knöbel S, Ader M (2026).** Extensive monolayer formation after transplantation depends on a subpopulation of human iPSC-derived RPE cells. **iScience, accepted.**

- **Seidemann, S., Salomon, F., Hoffmann, K. B., Kurth, T., Sbalzarini, I. F., Haase, R. & Ader, M. (2024).** Automated quantification of photoreceptor outer segments in developing and degenerating retinas on microscopy images across scales. **Frontiers in Molecular Neuroscience, 17, 1398447.** DOI: **10.3389/fnmol.2024.1398447**

- **Carido, M., Völkner, M., Steinheuer, L. M., et al., including Ader, M. (2023).** *Reliability of human retina organoid generation from hiPSC-derived neuroepithelial cysts.* **Frontiers in Cellular Neuroscience, 17, 1166641.** DOI: **10.3389/fncel.2023.1166641**

- **Gasparini, S. J., Tessmer, K., Reh, M., et al., including Ader, M. (2022).** *Transplanted human cones incorporate into the retina and function in a murine cone degeneration model.* **Journal of Clinical Investigation, 132(12), e154619.** DOI: **10.1172/JCI154619**

- **Heisterkamp, P., Borsch, O., Diaz Lezama, N., et al., including Ader, M. (2022).** Evidence for endogenous exchange of cytoplasmic material between a subset of cone and rod photoreceptors within the adult mammalian retina via direct cell-cell connections. **Experimental Eye Research, 219, 109033.** DOI: **10.1016/j.exer.2022.109033**

Up to 5 relevant previous projects, grants, networks or collaborations
Please provide detailed information (Title, Acronym, funding agency, GA-number, dates). Please indicate your role (coordinator / beneficiary / partner / service provider / collaborator). ***Indicate if international.***   • COST Action 'Retina4Future' – CA24105. Partner. (Oct 2025 – Oct 2029) (https://retina4future.eu/) • BMBF / ERA Net Neuron 01EW2106, Human organoid system based therapy development for age-related macular degeneration. Beneficiary. (2021 – 2025) • DFG SPP2127 Gene and cell based therapies to counteract neuroretinal degeneration (2018 – 2025). Beneficiary. https://www.spp2127.de • BMBF ReSight. Coordinator. (2017 – 2022). https://www.gesundheitsforschung-bmftr.de/de/resight.php.

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)
N/A

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	☐ Yes ☒ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	☒ Only Researchers ☐ Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only ☒ Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	licenses for animal work necessary when doing in vivo work

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
Please summarise the 2-3 strengths that should be first visible in an internal searchable profile. 1. Human iPSC-derived retinal organoids and RPE for regenerative-retina research. 2. Photoreceptor/RPE transplantation and preclinical retinal degeneration models, including functional validation. 3. Expertise in donor-cell maturation, integration/interaction with host retina and cell-replacement mechanisms.

Capabilities or expertise you would especially like to access through the consortium
1. Patient-derived retinal disease models and clinically informed disease phenotyping. 2. Complementary gene-editing, neuroprotection, retinal explant and translational ophthalmology approaches. 3. Clinical validation pathways and potentially industrial/regulatory expertise relevant to cell-based retinal therapies.

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	☒ Yes ☐ No
☒ We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
Examples: PIC pending approval, institutional approval, legal status, hosting rules, IP constraints, data access, ethics, clinical permissions, staffing continuity, resource availability. NDA, institutional approval, ethics needed

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

☒ **Understood**

Name: Marius Ader
Date completed: 04/10/2026

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)

Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	University of Alicante
Short name / acronym used in projects	NEUROVIS
Participant Identification Code (PIC), if available	999895886
Website(s)	https://www.ua.es
Country	Spain
Organisation type	X Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Carretera San Vicente del Raspeig s/n, Alicante, Spain.
Department(s), institute(s), research group(s) or unit(s) that would be involved	Please name the unit(s) most relevant to the potential MSCA-SE action
Additional institutional resources relevant to collaboration	Facilities and equipment's of the University of Alicante https://sstti.ua.es/ https://sstti.ua.es/es/infraestructuras-de-apoyo/animalario.html

Main contact person for this profile	Nicolás Cuenca. Full Professor of Cell Biology. Department of Physiology, Genetics and Microbiology
Main contact email	cuenca@ua.es

Other contact person(s), if relevant	Pedro Lax Zapata. Full Professor of Physiology. Department of Physiology, Genetics and Microbiology. pedro.lax@ua.es

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	X Retina
X Inherited retinal dystrophies (IRDs)	☐ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
X Advanced therapies / regenerative medicine	X Gene therapy / cell therapy / tissue engineering
X Inflammation / biomarkers / neuroprotection	X Drug delivery / formulations / nanomedicine
☐ Drug discovery / screening / repurposing	X Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	☐ Other (please specify):

Free Keywords for internal searchable database (please provide up to 10)	Retinal diseases, animal models, neuroprotection, gene therapy, prime editing, retinal organoids, cell culture, optomotor analysis, ERGs, Multielectrode arrays,
Geographies of interest for exchanges (optional)	Europe

Overview of your research / innovation focus
The NeuroVis Research Group at the University of Alicante investigates the cellular and molecular mechanisms underlying retinal degeneration and develops innovative therapeutic strategies to preserve or restore vision. Our research spans the entire translational pipeline, from fundamental studies of retinal circuitry and disease mechanisms to the preclinical evaluation of advanced therapies. A major focus of our work is the study of inherited retinal degenerations, glaucoma, diabetic retinopathy and retinal changes associated with neurodegenerative disorders such as Parkinson's disease. Using a combination of animal models and human donor retinas, we investigate neuronal remodeling, synaptic alterations, neuroinflammation and glial responses to identify therapeutic targets. The group has extensive expertise in retinal neuroprotection, gene therapy using AAV vectors, CRISPR-based genome editing, stem-cell-based regenerative approaches, and innovative photoactive biomaterials for retinal stimulation and vision restoration. We combine advanced retinal imaging, quantitative morphology, immunohistochemistry, electrophysiology, behavioral analysis and molecular biology to assess therapeutic efficacy at structural and functional levels. An additional research focus is the biology of intrinsically photosensitive melanopsin retinal ganglion cells (ipRGCs), investigating their role in circadian regulation, non-image-forming vision and neurodegenerative diseases, with particular emphasis on their potential as early biomarkers of neurological disorders. Through this multidisciplinary approach, NeuroVis aims to accelerate the translation of basic discoveries into clinically relevant therapies and diagnostic tools that improve visual function and patient quality of life.

Why your team could be relevant to a translational eye-health and vision consortium

The NeuroVis Research Group (University of Alicante) brings a multidisciplinary perspective that combines retinal neuroscience, translational ophthalmology, neurodegeneration, advanced imaging, gene and cell therapies, and bioengineering-oriented therapeutic development. Our added value lies in bridging fundamental mechanisms of neuronal degeneration with the preclinical validation of innovative therapeutic strategies, providing expertise that complements both basic and clinically oriented partners.

A major strength of our team is its internationally recognized expertise in **retinal neuroprotection**. For more than three decades, we have investigated the cellular and molecular mechanisms responsible for neuronal death and retinal remodeling in inherited and acquired retinal diseases. Our work has identified early pathogenic events that precede irreversible degeneration and has contributed to the development of therapeutic approaches aimed at preserving retinal structure and visual function. This expertise provides an essential framework for evaluating emerging therapies and identifying compounds of treatment efficacy.

Our research uniquely integrates **animal models with human pathological tissue**, allowing direct translation of experimental findings into clinically relevant contexts. We have extensive experience working with numerous models of inherited retinal degeneration, glaucoma and retinal injury, together with post-mortem human donor retinas affected by retinal diseases as well as systemic neurodegenerative disorders such as Parkinson's disease and diabetic retinopathy. This comparative approach enables us to identify conserved disease mechanisms, validate therapeutic targets, and discover retinal compounds with potential clinical application for both ocular and neurological diseases.

Another distinctive contribution is our expertise in the biology of **intrinsically photosensitive melanopsin retinal ganglion cells (ipRGCs)**. Our work has significantly advanced the understanding of these specialized neurons and their role in circadian photoentrainment, sleep regulation, cognitive function and non-image-forming vision. Because dysfunction of ipRGCs has emerged as an early indicator of several neurodegenerative disorders, this research opens new opportunities for biomarker discovery, patient stratification and translational studies extending beyond ophthalmology into neuroscience, ageing and precision medicine.

Our team also contributes broad experience in the development and preclinical assessment of **advanced therapeutic technologies**. These include AAV-mediated gene augmentation, CRISPR-based genome editing, stem-cell-based retinal replacement strategies and neuroprotective pharmacological interventions to restore vision after photoreceptor degeneration. We combine high-resolution retinal imaging, quantitative morphology, electrophysiology, behavioral analysis and molecular biology to provide comprehensive validation of therapeutic efficacy from cellular mechanisms to functional outcomes.

Importantly, our expertise extends beyond disease modelling. We offer advanced capabilities in retinal circuit analysis, synaptic connectivity, quantitative image analysis, histopathology and functional evaluation, providing enabling technologies that can support multiple work packages across a collaborative consortium. These complementary skills facilitate the validation of novel therapeutic approaches, medical devices, biomaterials and regenerative strategies developed by other partners.

The interdisciplinary nature of NeuroVis places us at the interface of neuroscience, ophthalmology, regenerative medicine, gene therapy, biomaterials and translational medicine. We believe this broad scientific perspective can stimulate new collaborations across traditionally separated disciplines, helping to generate innovative concepts, identify shared disease mechanisms, and accelerate the translation of laboratory discoveries into effective clinical solutions that improve visual health and patient quality of life.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

NeuroVis is a **multidisciplinary research group** integrating expertise in neuroscience, ophthalmology, cell and molecular biology, gene therapy, regenerative medicine, bioengineering and translational vision research. Our laboratory is fully equipped to perform comprehensive structural, molecular and functional studies, providing an integrated platform for preclinical ophthalmic research. Our main strengths include:

- **Retinal neurodegeneration and neuroprotection**, with more than 30 years of experience investigating the cellular and molecular mechanisms underlying inherited and acquired retinal diseases.

- **Animal models** of inherited retinal dystrophies, glaucoma, diabetic retinopathy and retinal injury for mechanistic studies and preclinical therapeutic evaluation.
- **Human retinal pathology**, including extensive analysis of post-mortem donor retinas from patients with inherited retinal diseases, diabetic retinopathy, Parkinson's disease and other neurodegenerative disorders.
- **Advanced retinal morphology and immunocytochemistry**, including multicolor immunolabelling, high-resolution confocal microscopy, three-dimensional image reconstruction, quantitative morphometry and retinal circuit analysis.
- **Functional phenotyping**, including electroretinography (ERG), multi-electrode array (MEA) recordings and **optomotor behavioral testing** for comprehensive assessment of visual function.
- **Gene therapy and genome editing**, including AAV-mediated gene augmentation, CRISPR-based genome editing and preclinical validation of innovative therapeutic strategies.
- **Cell-based regenerative medicine**, including stem-cell-derived retinal models and retinal cell replacement approaches.
- **Melanopsin retinal ganglion cell (ipRGC) biology**, with internationally recognized expertise in circadian regulation and non-image-forming vision.
- **Integrated translational analyses**, combining structural, molecular, electrophysiological and behavioral data to evaluate disease progression and therapeutic efficacy.
- **Comprehensive research infrastructure**, with dedicated facilities and state-of-the-art equipment to perform all experimental procedures in-house, from retinal histology and immunocytochemistry to confocal microscopy, electrophysiology, behavioral testing and quantitative image analysis.

Infrastructure, equipment, facilities or datasets relevant to consortium work

Facilities and equipment of the NeuroVis Research Group (University of Alicante)

- **Animal research facilities:** Fully equipped laboratories for mouse and rat models of inherited retinal degeneration, glaucoma, diabetic retinopathy and retinal injury, including facilities for behavioral vision testing and experimental therapeutic interventions.
- **Retinal histology and immunocytochemistry laboratory:** Complete infrastructure for tissue processing, cryosectioning, paraffin embedding, multicolor immunocytochemistry, histological staining and quantitative retinal morphometry.
- **Advanced microscopy platform:** Light and electron microscopy. High-resolution laser scanning confocal microscopy, fluorescence microscopy, stereomicroscopy and three-dimensional image reconstruction for detailed structural and cellular analysis of retinal tissues.
- **Electrophysiology platform:** Electroretinography (ERG) and multi-electrode array (MEA) recording systems for functional assessment of retinal activity and evaluation of therapeutic efficacy.
- **Behavioral vision platform:** Optomotor response system for quantitative assessment of visual acuity and contrast sensitivity in experimental animal models.
- **Molecular and cell biology laboratory:** Facilities for DNA/RNA extraction, PCR and RT-PCR, protein analysis, immunoblotting, cell culture, transfection, CRISPR-based genome editing and AAV vector-based gene therapy studies.
- **Preclinical therapeutic platforms:** Infrastructure for evaluating neuroprotective therapies, gene augmentation, genome editing, stem-cell-based regenerative strategies and photoactive biomaterials for retinal stimulation and vision restoration.
- **Human retinal pathology platform:** Access to and analysis of post-mortem human donor retinas from patients with inherited retinal diseases, diabetic retinopathy, Parkinson's disease and other neurodegenerative disorders, enabling direct translational validation of experimental findings.
- **Image analysis and quantitative morphometry:** Advanced software platforms for three-dimensional reconstruction, retinal cell quantification, morphometric analysis and neuronal connectivity studies.
- **Integrated multidisciplinary infrastructure:** NeuroVis possesses the experimental facilities and specialized equipment required to perform the complete workflow of translational retinal research in-house, from molecular and cellular studies to structural, electrophysiological and behavioral assessment of therapeutic interventions.

Training capacity and knowledge-transfer potential

The NeuroVis Research Group provides a highly interdisciplinary training environment that combines retinal neuroscience, translational ophthalmology, molecular biology, electrophysiology, regenerative medicine and preclinical therapeutic development. Our laboratory has a strong commitment to research training and international knowledge exchange, having successfully supervised **18 PhD theses** over the last years. These doctoral projects have involved researchers from diverse disciplines, including **biology, ophthalmology, neurology and veterinary medicine**, reflecting the multidisciplinary nature of our research. The successful training of ophthalmologists, neurologists and veterinarians highlights the **translational scientific trajectory** of our group, demonstrating our ability to bridge basic neuroscience with clinical and biomedical research while fostering collaboration across different disciplines. We have also trained numerous undergraduate, MSc, PhD and postdoctoral researchers from national and international institutions.

Incoming researchers would receive hands-on training in experimental models of inherited and acquired retinal diseases, including retinal degeneration, glaucoma, diabetic retinopathy and neurodegenerative disorders such as Parkinson's disease. They would acquire practical experience in advanced retinal histology, multicolor immunocytochemistry, confocal microscopy, quantitative retinal morphometry, neuronal circuit analysis and molecular biology techniques.

Training also includes functional assessment of retinal therapies using electroretinography (ERG), multi-electrode array (MEA) recordings and optomotor behavioral testing, providing comprehensive expertise in evaluating retinal structure and function. Researchers would also gain experience in AAV-mediated gene therapy, CRISPR-based genome editing, neuroprotective strategies, stem-cell-based approaches and the evaluation of innovative photoactive biomaterials for vision restoration.

A distinctive strength of NeuroVis is the integration of animal models with human donor retinal tissue, allowing trainees to validate experimental findings in clinically relevant samples and to understand the translational pathway from basic discoveries to patient-oriented applications.
Beyond technical skills, visitors would benefit from an international, collaborative and multidisciplinary research environment that promotes scientific exchange, experimental design, critical thinking and the development of collaborative projects, facilitating long-term knowledge transfer and the establishment of sustainable research networks in vision science and translational ophthalmology.

What your team would hope to gain from participating
Participation in this consortium would provide an outstanding opportunity to strengthen the translational impact of our research by integrating complementary expertise, emerging technologies and multidisciplinary approaches that extend beyond our current capabilities. While NeuroVis has extensive experience in retinal neurodegeneration, neuroprotection and preclinical therapeutic development, we are particularly interested in collaborating with partners working in advanced gene and RNA therapies, retinal organoids, biomaterials, medical devices, optogenetics, artificial intelligence, multimodal imaging and precision medicine. A major objective is to establish long-term international collaborations that evolve into sustainable scientific partnerships through joint research projects, coordinated European funding applications, shared infrastructures, co-supervision of students, and high-impact scientific publications. We believe that combining our expertise in retinal neuroscience and translational ophthalmology with complementary disciplines such as bioengineering, computational sciences, nanotechnology and regenerative medicine will create new opportunities for innovation and accelerate the translation of research into clinical practice. The consortium would also provide an exceptional framework for the international training and mobility of our researchers. PhD students, postdoctoral fellows, clinicians and technical staff would benefit from exposure to new experimental models, technologies and research environments, while developing interdisciplinary skills, international networks and collaborative experience that will enhance their scientific careers. For our senior researchers, participation would facilitate the incorporation of novel methodologies, analytical tools and enabling technologies into our research programs, particularly in areas where collaboration with spesialized partners can substantially enhance the quality and translational relevance of our work. At the same time, we are committed to sharing our expertise in retinal neurobiology, retinal circuitry, human retinal pathology and preclinical evaluation of innovative therapies, ensuring a genuine bidirectional exchange of knowledge. Overall, we expect this consortium to become a long-term strategic partnership that promotes bidirectional knowledge transfer, joint publications, coordinated European funding applications, industrial collaboration and the co-development of innovative technologies, ultimately accelerating the translation of research into tangible benefits for patients.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Nicolas Cuenca	Full Professor	Department of Physiology, Genetics and Microbiology	Researcher	Experienced	cuenca@ua.es

Pedro Lax	Full Professor	Department of Physiology, Genetics and Microbiology	Researcher	Experienced	Pedro.lax@ua.es
Xavier Sánchez-Sáez	Assistant professor	Department of Physiology, Genetics and Microbiology	Researcher	Experienced	xsanchez@ua.es
Isabel Ortuño Lizarán	Assistant professor	Department of Physiology, Genetics and Microbiology	Researcher	Experienced	isortliz@ua.es
Carla Sánchez Castillo	Assistant professor	Department of Physiology, Genetics and Microbiology	Researcher	Experienced	carla.sanchez@ua.es
Santiago Milla Navarro	Assistant professor	Department of Physiology, Genetics and Microbiology	Researcher	Experienced	s.milla@gcloud.ua.es
Joaquín Jimenez Candela	PhD student	Department of Physiology, Genetics and Microbiology	Researcher	Early-Stage	joaquin.jimenez@ua.es
Marina Pastor Mas	Technician	Department of Physiology, Genetics and Microbiology	Technician	Experienced	marina.pastor@ua.es
Sergio Fuster Picher	PhD student	Department of Physiology, Genetics and Microbiology	Researcher	Early-Stage	Sergio.fuster@ua.es
Ceren Sahin	PhD student	Department of Physiology, Genetics and Microbiology	Researcher	Early-Stage	ceren.sahin@ua.es

Main disciplines represented in your team	Cell biology, Physiology, Ophthalmology, Neuroscience, Immunology, Gene Therapy, Molecular Biology, Neuroprotection, Animal Models, Inmunohistochemistry Inherited retinal diseases.
Experience in EU / international / multi-partner collaborative projects	Our research group is an active participant in the **European Union's Marie Skłodowska-Curie Actions (MSCA) Doctoral Networks** under the **Horizon Europe** program, including the **RETORNA** project. We also participate in the European **OPTOREMODE** project, funded through the **European Research Area Network (ERA-NET) for Research on Rare Diseases**, and collaborate in research programs supported by **The Michael J. Fox Foundation for Parkinson's Research**, **Bayer**, and **Boehringer Ingelheim**.
Experience with clinical, translational or patient-facing research (if relevant)	☐ Yes ☐ No X Not applicable | Short note:

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements
1. Cuenca N, Ortuño-Lizarán I, Sánchez-Sáez X, Kutsyr O, Albertos-Arranz H, Fernández-Sánchez L, Martínez-Gil N, Noailles A, López-Garrido Ja, López-Gálvez M, Lax P, Maneu V, Pinilla I. Interpretation of OCT and OCTA images from a histological approach: Clinical and experimental implications. Prog Retin Eye Res. 2020 Jan 3:100828. IF 20,198 doi: 10.1016/j.preteyeres.2019.100828 2. Lee JY, Martin-Bastida A, Murueta-Goyena A, Gabilondo I, Cuenca N, Paola Piccini P, Jeon B. Multimodal brain and retina imaging of dopaminergic degeneration in Parkinson's disease. Nature reviews Neurology. 2022 Apr;18(4):203-220. IF 42,937. doi: 10.1038/s41582-022-00618-9

3. Kutsyr O, Noailles A, Martínez-Gil N, Maestre-Carballa L, Martinez-Garcia M, Maneu V, Cuenca N, Lax P. Short-term high-fat feeding exacerbates degeneration in retinitis pigmentosa by promoting retinal oxidative stress and inflammation. Proc Natl Acad Sci U S A. 2021 Oct 26;118(43):e2100566118. IF 11.205. doi: 10.1073/pnas.2100566118
4. Ruiz-Pastor MJ, Sánchez-Sáez X, Kutsyr O, Albertos-Arranz H, Sánchez-Castillo C, Ortuño-Lizarán I, Martínez-Gil N, Vidal-Gil L, Méndez L, Sánchez-Martín M, Maneu V, Lax P, Cuenca N. Prph2 knock-in mice recapitulate human central areolar choroidal dystrophy retinal degeneration and exhibit aberrant synaptic remodeling and microglial activation. Cell Death Dis. 2023 Nov 1;14(11):711. IF 9.00 doi: 10.1038/s41419-023-06243-8
5.Albertos-Arranz H, Martínez-Gil N, Sánchez-Sáez X, Molina-Martín JC, Lax P, Cuenca N. Neuronal Degeneration and Glial Activation in the Absence of Vascular Changes in Human Retinas of Patients with Diabetes. Invest Ophthalmol Vis Sci. 2025 Mar 3;66(3):53. IF 4,7. doi: 10.1167/iovs.66.3.53.

Up to 5 relevant previous projects, grants, networks or collaborations

Selected Research Projects

International

- **RETORNA – Targeting RNA as a Therapeutic Strategy for Retinal Diseases** (Project No. **101073316**)
  Funding: European Union, **Marie Skłodowska-Curie Actions (MSCA) Doctoral Networks**, Horizon Europe (HORIZON-MSCA-2021-DN-01-01).
  Participant: Nicolás Cuenca (NeuroVis Research Group).
  Duration: 2022–2027.
  Total Project Budget: €2,568,427.20.
  Funding to NeuroVis: €503,942.40.
- **OPTOREMODE. Retinitis Pigmentosa Diagnosis and Therapy: Retinal Remodelling and Optogenetic Reactivation of the Degenerated Retina**
  Funding: European Research Area Network for Research on Rare Diseases (E-Rare ERA-NET, JTC 2013), European Community.
  Principal Investigator: Nicolás Cuenca
  Duration: January 2013 – December 2016.

National

- **Optogenetic Therapy for Neurodegenerative Diseases**
  Funding: Generalitat Valenciana – European Regional Development Fund (ERDF/FEDER), Grant IDIFEDER/2017/064.
  Principal Investigator: Nicolás Cuenca
  Duration: February 2017 – December 2017.
  Funding: €414,391.33.
- **Therapy for Hereditary Retinal Dystrophies Using Neuroprotective Agents and CRISPR: A Translational Study from Animal Models to Human Retinal Organoids** (PROMETEO/2021/024)
  Funding: Ministry of Innovation, Universities, Science and Digital Society (Generalitat Valenciana).
  Principal Investigator: Nicolás Cuenca
  Duration: January 2021 – December 2023.
  Funding: €353,865.63.
- **Therapeutic Approaches for PRPH2-Associated Retinal Disorders: Prime Editing and Gene Augmentation for Targeted Gene Therapy** (PID2024-156526OB-I00)
  Funding: Spanish Ministry of Science, Innovation and Universities.
  Principal Investigators: Pedro Lax and Nicolás Cuenca.
  Duration: September 2025 – August 2028.
  Funding: €237,500.

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)

Patents

1. New immortalized cell line MU-PH1 (ACC3187) derived from adult mouse retinal Müller cells for the treatment of retinal neurodegenerative diseases
Inventors: V.E. Maneu Flores, V. Gómez Vicente, M.L. Gil Herrero, D. Gozalbo Flor, **N. Cuenca Navarro**
National patent application: ES201201274
International application: PCT/ES2013/070775
Patent publications: WO2014096479-A1; WO2014096479-A8; ES2478694-A1; ES2478694-B2
Priority country: Europe
Patent granted: 19 February 2015
Patent holders: University of Alicante and University of Valencia
Commercial exploitation: Applied Biological Materials Inc. (Canada)
This patent describes the generation of the immortalized Müller glial cell line **MU-PH1 (ACC3187)**, a valuable experimental platform for retinal research and the development of novel therapeutic strategies for retinal neurodegenerative diseases.

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2. Use of 3-(2-Isothiocyanatoethyl)-5-methoxy-1H-indole for the treatment of retinal degenerative diseases
Inventors: R. León Martínez, L. Campello Blasco, P. Michalska Dziama, A. Noailles Gil, O. Kutsyr, L. Fernández Sánchez, V. Maneu Flores, P. Lax Zapata, A. García García, **N. Cuenca Navarro**
Spanish patent application: P201830704
Priority country: Spain
Priority date: 13 July 2018
Patent holders: University of Alicante (70%), Fundación para la Investigación Biomédica del Hospital Universitario de La Princesa (20%), and Autonomous University of Madrid (10%)
Substantive examination successfully completed: 2 June 2020
This patent covers the therapeutic use of **3-(2-isothiocyanatoethyl)-5-methoxy-1H-indole** as a novel neuroprotective compound for the treatment of retinal degenerative diseases.

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	X Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	X Only Researchers ☐ Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only X Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	None

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
The NeuroVis Research Group offers an integrated preclinical platform for translational retinal research, combining disease modelling, therapeutic evaluation and comprehensive structural and functional analyses. Our key capabilities include: • **Preclinical animal models of retinal neurodegeneration**, including inherited retinal dystrophies (retinitis pigmentosa), ocular hypertension/glaucoma and diabetic retinopathy, for studying disease mechanisms and therapeutic interventions.

- **Comprehensive evaluation of neuroprotective therapies**, using complementary behavioral, functional, structural and molecular approaches to assess treatment efficacy. Our analytical platforms include **optomotor behavioral testing, electroretinography (ERG), optical coherence tomography (OCT), immunohistochemistry, confocal microscopy, Western blotting, PCR/qPCR, transcriptomic analysis and flow cytometry**.
- **Ex vivo and in vitro experimental platforms**, including **organotypic retinal explant cultures** and **retinal cell culture systems**, enabling mechanistic studies, drug screening and validation of novel therapeutic strategies.
- **Gene therapy and genome editing**, with expertise in **AAV-mediated gene delivery**, **CRISPR-based genome editing**, and the preclinical evaluation of gene therapies for inherited retinal diseases.
- **Integrated translational expertise**, combining animal models, human retinal pathology and multimodal functional and molecular analyses to accelerate the development of therapies for retinal and neurodegenerative diseases.

Capabilities or expertise you would especially like to access through the consortium

We are particularly interested in connecting with partners that provide complementary expertise and enabling technologies capable of expanding the translational scope of our research. In particular, we would welcome collaborations with teams offering:

- **Human retinal organoids and iPSC-derived retinal models** for disease modelling, target validation and therapeutic testing.
- **Single-cell and spatial omics technologies**, including single-cell RNA sequencing (scRNA-seq), spatial transcriptomics and multi-omics approaches to characterize retinal cell populations and therapeutic responses.
- **Advanced behavioral and functional vision assessment**, including innovative visual function tests and digital phenotyping platforms for both preclinical and clinical studies.
- **Artificial intelligence, machine learning and computational biology** for multimodal data integration, image analysis, biomarker discovery and predictive modelling.
- **Advanced retinal imaging technologies**, including adaptive optics, high-resolution OCT, OCT angiography and multimodal clinical imaging.
- **Clinical cohorts and patient-oriented research infrastructures**, facilitating validation of experimental findings in well-characterized patient populations and supporting precision medicine approaches.
- **Regenerative medicine and advanced cell therapies**, including retinal tissue engineering, cell replacement strategies and biofabrication technologies.
- **Nanoparticle-based gene delivery systems**, including non-viral vectors and targeted nanocarriers for retinal gene therapy and nucleic acid delivery.
- **Advanced gene therapy platforms**, particularly partners with expertise in the design, engineering and GMP production of **AAV vectors** and other viral vectors for preclinical and clinical gene therapy applications.
- **Biomaterials, nanotechnology and innovative drug delivery systems** for retinal therapeutics and sustained intraocular delivery.
- **Industrial partners and regulatory expertise**, providing knowledge on technology transfer, regulatory pathways, intellectual property, GMP manufacturing and clinical translation of advanced therapies and medical devices.
- **Large international clinical and translational networks**, promoting long-term collaborations, staff exchanges, joint grant applications and multidisciplinary research initiatives beyond the field of ophthalmology.

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	X Yes ☐ No
X We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
Examples: PIC pending approval, institutional approval, legal status, hosting rules, IP constraints, data access, ethics, clinical permissions, staffing continuity, resource availability.

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organization / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organization. Submission of this form confirms that the contributor agrees to this internal use. Organizations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

☐ **Understood**

Name: Nicolás Cuenca
Date completed: 10/7/2026

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)

Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	University College Dublin
Short name / acronym used in projects	NUID
Participant Identification Code (PIC), if available	999974359
Website(s)	www.ucd.ie https://people.ucd.ie/alison.reynolds
Country	Ireland
Organisation type	**X** Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	University College Dublin, Belfield, Dublin 4, D04 W6F6.
Department(s), institute(s), research group(s) or unit(s) that would be involved	UCD School of Veterinary Medicine (Veterinary Biosciences Section) UCD Conway Institute of Biomedical & Biomolecular Research
Additional institutional resources relevant to collaboration	e.g. Cat 3 lab, Cat 2 facilities, transcriptomics facilities, tissue culturing, veterinary hospital (if clinicians involved), organoid facility, genomics core, proteomics core, microscopy core, biomedical facility (rodent and fish).

Main contact person for this profile	Alison Reynolds
Main contact email	alison.reynolds@ucd.ie

Other contact person(s), if relevant	N/A

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
X Ocular surface	**X** Retina
X Inherited retinal dystrophies (IRDs)	**X** Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
☐ Advanced therapies / regenerative medicine	☐ Gene therapy / cell therapy / tissue engineering
X Inflammation / biomarkers / neuroprotection	**X** Drug delivery / formulations / nanomedicine
X Drug discovery / screening / repurposing	**X** Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	**X** Other (please specify): Public Patient Involvement, Surveys, Charles Bonney Syndrome

Free Keywords for internal searchable database (please provide up to 10)	Retinal (photoreceptor cell death, angiogenesis) and ocular surface (dry eye – inflammation, oxidation stress) disease models: zebrafish, organoid, mouse, *in vitro* (2D and 3D ocular surface cultures), drug discovery and development, space associated neuro-ocular syndrome (SANS),
Geographies of interest for exchanges (optional)	

Overview of your research / innovation focus

Please describe the main focus of the team / unit that could participate. Keep it concise but specific (recommended: 150-250 words).

Currently, the main focus is ocular surface: development of ocular surface disease models, ocular surface organoids, spheroids, stratified models, nanoformulated drug delivery (and discovery, development with “OcuHealth” in South East Technological University). We are also using zebrafish to see if we can develop a SANS model.

Why your team could be relevant to a translational eye-health and vision consortium

Please explain the specific added value your organisation / team could bring to a consortium centred on visual health, translational ophthalmology and related enabling technologies. ***Please think broadly: the contribution does not need to be directly ophthalmology-related. Interdisciplinarity is strongly encouraged, and expertise from apparently distant fields may open unexpected routes to innovation, new paradigms, and creative solutions to complex challenges that currently limit progress from research to patient benefit****.*

Expertise in both front of the eye and back of the eye disease modelling, understanding pathophysiology and drug discovery and development. This has covered inherited retail degeneration, angiogenesis, drug-induced models, together with a move to ocular surface disease modelling and understanding of disease progression, together with drug development and delivery technologies (with SETU). Commercialisation expertise (patenting, future spin out). PPI involvement (paper written to identify the gaps), survey design experience (Charles Bonnet Syndrome), also I am a board member of Fighting Blindness Ireland Charity.

Currently exploring how to model space-associated neuro-ocular syndrome.
I am also Associate Dean for Research, Innovation and Impact in the School of Veterinary Medicine, so keen to promote cross-discipline collaborations between our staff and graduate students.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

Please provide bullet points covering your strongest capabilities, methods, models, analytical approaches, translational assets or innovation strengths.

Very keen to collaborate to build better front of the eye models, in which we can learn more about cell communication, flare ups, and use these to design better drugs and drug-delivery methodologies.

Infrastructure, equipment, facilities or datasets relevant to consortium work

Examples: specialised laboratories, clean rooms, imaging platforms, -omics, biobanks, clinical cohorts, animal models, software tools, rehabilitation platforms, regulatory support or prototyping capacities.

Listed above: Cat 3 lab, Cat 2 facilities, transcriptomics facilities, tissue culturing, veterinary hospital (if clinicians involved), organoid facility, genomics core, proteomics core, microscopy core, biomedical facility (rodent and fish).
Datasets generated on novel early onset inherited retinal disease models.

Training capacity and knowledge-transfer potential

Please indicate what inbound staff could realistically learn in your organisation (scientific, technical, clinical, translational, regulatory, entrepreneurial or managerial skills).

Courses through the University (in multiple things- e.g. targeted graduate courses re specific research skills- proteomics, big data), technology transfer courses (entrepreneurial skills), innovation academy courses, research integrity courses. If wanted to pay, we have a professional academy with courses e.g. leadership & management.

What your team would hope to gain from participating

Examples: access to methods, models, technologies, patient-oriented validation, translational pathways, industrial insight, clinical exposure, data science, regulatory know-how, joint publications or longer-term strategic collaboration. ***Again, please think broadly and beyond ophthalmology.***

Building collaborative networks to learn more about how diseases progress.
Support in developing patient-relevant cures.
Helping facilitate more cross-EU contact between people with lived experience of blindness.
I co-chair the Irish National Animal Welfare Body (also we have just joined the European Network of National Animal Welfare Bodies ENAWB- I know Spain is also a member), so I would hope we could incorporate new ways to reduce, replace, refine animal use (and ex vivo animal part use) in the area.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Alison Reynolds	PI/Assistant Professor	School of Veterinary Medicine	Permanent/Academic	Mid	alison.reynolds@ucd.ie
Katarina Krsteva	MSCA Visiting Researcher	School of Veterinary Medicine	Visiting	Early	

Main disciplines represented in your team	Genetics, pharmacology, anatomy, physiology, embryology, commercialization, cell biology, molecular biology, animal expertise.
Experience in EU / international / multi-partner collaborative projects	I have been a PI on several EU RISE awards (3DNeoNet, Crystal3, Everest). I have participated as a work package manager and supervising incoming staff/students. I am currently involved with a cost Action (resubmission).
Experience with clinical, translational or patient-facing research (if relevant)	X Yes ☐ No ☐ Not applicable | Short note: Surveys of patients (including ethics writing), patient engagement days, board member on patient charity, published paper on PPI in vision research https://doi.org/10.12688/openreseurope.16057.1

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

Please prioritise achievements most relevant to your possible contribution. Please include DOI, URL or identifier.

I collaborated with drug delivery experts and obtained Commercialisation Funding (CF-2020-1511; €700,000) to develop novel nanoformulations (SETU) with novel drugs (UCD) for dry eye disease, pitching and winning UCD's VentureLaunch Accelerator competition (2025) and came 3rd in Retinnovate's pitch competition (2026). A High Potential Start-Up will be spun out in 2026 to develop and commercialise this work to first in human.
(DOI: 10.1039/d4na01086h)

I worked with collaborators on a H2020-funded Innovative Training Network to Explore Public/Patient Involvement and challenges in visual impairment research (doi:10.12688/openreseurope.16057.1)

I supervised PhD students who developed models of stargardt disease in zebrafish (DOI: 10.1007/978-3-031-76550-6_36) and early onset severe retinal degeneration in organoids (https://doi.org/10.3389/fnins.2021.784987)

With a colleague in the Biomedical facility, we gained funding (2025-26) and successfully ran *Mental Health First Aider* and *Mental Health in the Workplace* training for staff and students working with animals. This included a bespoke model on "Compassion fatigue" for those working with animals.

Up to 5 relevant previous projects, grants, networks or collaborations

Please provide detailed information (Title, Acronym, funding agency, GA-number, dates). Please indicate your role (coordinator / beneficiary / partner / service provider / collaborator). ***Indicate if international.***

1. OcuHealth: Enterprise Ireland Commercialisation Funding (CF-2020-1511; €700,000). Co-PI. National (2021 – 2026)
2. CRYSTAL3; MSCA/HE grant agreement, 101007931. Work Package leader, collaborator. International (2021 – 2024)
3. EVEREST MSCA/HE GA 101183034. Supervisor of visiting researchers. International (current)

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)
Please indicate only what can be shared at this stage. Previous patents awarded: USPTO and EPO (2 each), antiangiogenic compounds.

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	☐ Yes **X** Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	☐ Only Researchers **X** Technical Staff/Innovation **X** Managerial / project support **X** Admin Staff ☐ Other profiles **X** Maybe
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only **X** Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	If EU/ UK passport and citizen- this is fine. English language used in the lab- and staff seconding need to be competent to use on a daily basis. Health and Safety induction performed locally.

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
Disease modelling *Commercialisation interests* *Patient-centric support*

Capabilities or expertise you would especially like to access through the consortium
Please indicate the types of teams, methods, infrastructures or translational pathways you would like to connect with in the consortium. New methods for analysis of front of the eye disease models Development of new cures Bringing patients together

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	X Yes ☐ No
X We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
Examples: PIC pending approval, institutional approval, legal status, hosting rules, IP constraints, data access, ethics, clinical permissions, staffing continuity, resource availability. Ethics requirements can take some time (Human or animal ethics) – so if a specific project is required, planning may be required far in advance. DPIA agreements may be required. Limited availability to send out secondees. Would require internal approval to send PI on secondment (unlikely to happen prior to mid 2028, due to leadership role).

9. Additional collaborators you would like us to consider

If there are any additional organisations, teams or contacts - current collaborators, previous collaborators or desired future collaborators - that you think could be relevant for this MSCA-SE consortium, please list them below. The coordination team may contact them to explore their potential interest in joining the consortium-building process.

Please provide the following details (as many entries as needed).

Type of organisation: ☐ Academic ☐ Clinical ☐ SME/industry ☐ NGO/foundation ☐ Public body ☐ Other

Organisation / team name:

Country:

Website:

Existing link with your team: ☐ Current collaborator ☐ Previous collaborator ☐ Desired collaborator ☐ Other

Suggested contact person/email (if known):

Why could they be relevant for this MSCA-SE consortium?

Any additional comments or sensitivities before contacting them?

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

X Understood

Name: Alison Reynolds
Date completed: 24th September 2026

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)

Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	University College Dublin, National University of Ireland, Dublin
Short name / acronym used in projects	UCD
Participant Identification Code (PIC), if available	999986384
Website(s)	https://www.ucd.ie/
Country	Ireland
Organisation type	X Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public bodv ☐ Other
Main address	Belfield, Dublin 4, Ireland
Department(s), institute(s), research group(s) or unit(s) that would be involved	UCD School of Biomolecular and Biomedical Science; UCD Conway Institute of Biomolecular and Biomedical Research
Additional institutional resources relevant to collaboration	Zebrafish Research Facility, UCD Conway Imaging Core, Genomics Core, Proteomics Core, Mass Spectrometry Facility

Main contact person for this profile	Prof. Breandán Kennedy, Professor of Pharmacology
Main contact email	Brendan.kennedy@ucd.ie

Other contact person(s), if relevant	

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	☐ Retina
☐ Inherited retinal dystrophies (IRDs)	☐ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
☐ Advanced therapies / regenerative medicine	☐ Gene therapy / cell therapy / tissue engineering
☐ Inflammation / biomarkers / neuroprotection	☐ Drug delivery / formulations / nanomedicine
☐ Drug discovery / screening / repurposing	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	☐ Other (please specify):

Free Keywords for internal searchable database (please provide up to 10)	Zebrafish models; retinal dystrophies; visual function assays; drug discovery; small molecule screening; neuroprotection; ocular pharmacology; translational validation; angiogenesis; photoreceptors, circulating biomarkers
Geographies of interest for exchanges (optional)	

Overview of your research / innovation focus
The Kennedy laboratory at UCD focuses on ocular pharmacology and translational visual science, specifically targeting inherited retinal degenerations (IRDs) and acquired eye diseases. Utilizing high-throughput, in vivo zebrafish models of human visual disorders alongside cell-based platforms, the team investigates mechanisms of photoreceptor degeneration, vascular leakage, and visual cycle. A key strength lies in phenotype-based small molecule screening pipelines to identify novel therapeutic candidates, and omics pipelines to identify biomarkers.

Why your team could be relevant to a translational eye-health and vision consortium
Prof. Kennedy's group bridges fundamental disease mechanisms and early-stage drug discovery, providing a crucial in vivo link between in vitro target identification and mammalian/clinical evaluation. The team contributes validated zebrafish disease models for rapid target validation, functional visual testing (e.g., visual behavior/OKR), and pharmacodynamic assessment of novel formulations or delivery systems. This expertise complements IOBA's focus on advanced therapies, neuroprotection, and biomarkers, offering interdisciplinary synergy to accelerate translational ocular therapies

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute
High-throughput in vivo small molecule screening using zebrafish models of visual impairment.

Functional visual recovery assays (optokinetic response/OKR, visual behavior analytics). *Ocular angiogenesis, vascular permeability, and neuroprotection drug testing platforms.* *Target validation, histological analysis, bioimaging, and molecular profiling of retinal disease.*

Infrastructure, equipment, facilities or datasets relevant to consortium work
• *Fully equipped, high-capacity Zebrafish Research Facility with automated behavioral tracking systems.* • *UCD Conway Institute Core Facilities: Advanced Light Microscopy, Flow Cytometry, Mass Spectrometry, Genomics, and Proteomics platforms.* • *Curated chemical screening libraries and target validation datasets for ocular compounds.*

Training capacity and knowledge-transfer potential
• *Hands-on training for incoming staff in larval zebrafish handling, ocular phenotyping, visual behavioral assays, and high-content drug screening.* • *Transfer of skills in ocular pharmacology, experimental design for small molecule discovery, bioimaging techniques, and multi-omics data integration.*

What your team would hope to gain from participating
• *Access to advanced cell/tissue engineering models, human retinal organoid platforms, and novel biomaterials/drug delivery formulations.* • *Clinical exposure, translational validation pathways, and collaboration with non-academic/SME partners for downstream commercialization and regulatory navigation.*

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
tbc					

Main disciplines represented in your team	Ocular pharmacology, visual neuroscience, drug discovery, developmental biology, genetics, chemical biology
Experience in EU / international / multi-partner collaborative projects	Extensive experience coordinating and participating in EU FP7, Horizon 2020, Horizon Europe actions, and Marie Skłodowska-Curie Networks (ITN/DN, RISE/SE).
Experience with clinical, translational or patient-facing research (if relevant)	☑ Yes : Translational screening pipelines focused on identifying leads or biomarkers for clinical-stage development and commercial spin-outs.

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

1: Ruddin G, McCann T, Kaylor JJ, Fox MM, Fehilly JD, Ward R, Faulkner A, Moran AL, Wynne K, Radu RA, Monaghan MG, Thorpe SD, Travis GH, Kennedy BN. Dual Knockout Models of the Spatially and Functionally Conserved rgra and rgrb Zebrafish Genes Reveal the Requirement of RGR for the Integrity of Cone-Mediated Photopic Vision, the Photopic Visual Cycle and Bruch's Membrane Morphology. FASEB J. 2026 May 31;40(10):e71877. doi: 10.1096/fj.202504935R. PMID: 42160018.

2: Fehilly JD, McCann T, Ruddin G, Kaylor JJ, Grenville H, Ward R, Wynne K, Sánchez AG, Strachan E, Travis GH, Collery RF, Kennedy BN. Germline Disruption of Retinal Pigment Epithelium-Expressed Zebrafish rlbp1b^{-/-} Results in Selective Dim Light Visual Behavior Deficits and Provides a Screening Platform for Evaluating the Pathogenicity of Human RLBP1 Variants. FASEB J. 2025 Jun 30;39(12):e70754. doi: 10.1096/fj.202500600R. PMID: 40552921; PMCID: PMC12186719.

3: McCann T, Sundaramurthi H, Walsh C, Virdi S, Alvarez Y, Sapetto-Rebow B, Collery RF, Carter SP, Moran A, Mulholland R, O'Connor JJ, Taylor MR, Rauch N, Starostik MR, English MA, Swaroop A, Geisler R, Reynolds AL, Kennedy BN. Emc1 is essential for vision and zebrafish photoreceptor outer segment morphogenesis. FASEB J. 2024 Oct 15;38(19):e70086. doi: 10.1096/fj.202401977R. PMID: 39360639.

4: Tonelotto V, Costa-Garcia M, O'Reilly E, Smith KF, Slater K, Dillon ET, Pendino M, Higgins C, Sist P, Bosch R, Passamonti S, Piulats JM, Villanueva A, Tramer F, Vanella L, Carey M, Kennedy BN. 1,4-dihydroxy quininib activates ferroptosis pathways in metastatic uveal melanoma and reveals a novel prognostic biomarker signature. Cell Death Discov. 2024 Feb 10;10(1):70. doi: 10.1038/s41420-023-01773-8. PMID: 38341410; PMCID: PMC10858877.

5: Gómez Sánchez A, Colucci P, Moran A, Moya López A, Colligris B, Álvarez Y, Kennedy BN. Systemic treatment with cigarette smoke extract affects zebrafish visual behaviour, intraocular vasculature morphology and outer segment phagocytosis. Open Res Eur. 2023 Nov 23;3:48. doi: 10.12688/openreseurope.15491.2. PMID: 38283058; PMCID: PMC10822043.

Up to 5 relevant previous projects, grants, networks or collaborations

EVEREST (Horizon Europe MSCA Staff Exchanges)
Role: Project Coordinator (Led by UCD)
Focus: Harnessing extracellular vesicles (EVs) as non-invasive diagnostics and targeted, personalized therapies for complex conditions such as cancer and cardiovascular diseases.
Consortium Composition: 21 academic and industry institutions across 11 countries (including Trinity College Dublin, SiriusXT, Luxembourg Institute of Science and Technology, and BioReperia).

CRYSTAL3 (Horizon 2020 MSCA RISE)
Role: Consortium Coordinator
Focus: Exploring novel drug discovery and therapeutics focused on inflammation-related signaling and eye/cancer diseases through an industry-academia exchange framework.

3D-NEO-NET (Horizon 2020 MSCA RISE)
Role: Consortium Coordinator
Focus: Developing 3D disease models and identifying novel therapeutics for ocular diseases and ocular oncology (particularly uveal melanoma) by bridging academic research with biotechnology enterprise partners.

3D-NET (FP7 Industry-Academia Partnerships and Pathways - IAPP)
Role: Consortium Coordinator

Focus: Drug discovery and development targeting ocular diseases, integrating zebrafish functional genomics, targeted drug screening, and delivery solutions across academic and corporate partners.

CURAM – MedTrain+ (Horizon Europe MSCA Fellowship Program)
Role: Participating Principal Investigator / Supervisor
Focus: Identification, characterization, and optimization of novel immune-engineered peptides to treat metastatic uveal melanoma.

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)
☐ ***WO2014012889A1*** *— Anti-angiogenic compounds* ☐ ***US9388138B2 / US8916586B2*** *— Anti-angiogenic compounds* ☐ ***EP2875008A1*** *— Anti-angiogenic compounds* ☐ ***US20150191428A1*** *— Anti-angiogenic compounds*

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	☐ Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	☐ Only Researchers ☐ Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only ☐ Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
1. *Rapid, cost-effective in vivo small-molecule screening and target validation in zebrafish models of retinal degeneration and ocular vascular disease.* 2. *Automated visual function and behavioral assays (OKR) for quantitative functional recovery testing.* 3. *Comprehensive core infrastructure for high-resolution ocular imaging, proteomics, and multi-omics analysis.*

Capabilities or expertise you would especially like to access through the consortium
Human cell/tissue platforms, retinal organoid models, bio-formulation/nanomedicine delivery, and clinical validation pathways.

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	☐ Yes ☐ No
☐ We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
Examples: PIC pending approval, institutional approval, legal status, hosting rules, IP constraints, data access, ethics, clinical permissions, staffing continuity, resource availability.

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☐ **Understood**

Name: Breandan Kennedy
Date completed: 18/09/2026

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)

Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	University College Dublin
Short name / acronym used in projects	UCD
Participant Identification Code (PIC), if available	If no PIC is available yet, please indicate 'not yet available'.
Website(s)	https://www.ucd.ie/conway/
Country	Ireland
Organisation type	x Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	UCD Belfield, Dublin D04 V1W8, Ireland
Department(s), institute(s), research group(s) or unit(s) that would be involved	UCD School of Biomolecular & Biomedical Science UCD Conway Institute
Additional institutional resources relevant to collaboration	UCD Conway Genomics Core (https://www.ucd.ie/conway/technology/genomicscore/) UCD Conway Metabolomics Core (https://www.ucd.ie/conway/technology/metabolomics/) UCD Aquatics Facility

Main contact person for this profile	Derek Costello, Associate Professor, Neuropharmacology, UCD
Main contact email	Derek.costello@ucd.ie

Other contact person(s), if relevant	Name, role, email

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	☐ Retina
☐ Inherited retinal dystrophies (IRDs)	☐ Age-related ocular disease
x Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
☐ Advanced therapies / regenerative medicine	☐ Gene therapy / cell therapy / tissue engineering
x Inflammation / biomarkers / neuroprotection	☐ Drug delivery / formulations / nanomedicine
x Drug discovery / screening / repurposing	☐ Ocular vascular disease
x Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	x Other (please specify): Neuroinflammation, neurodegeneration and neurotoxicity

Free Keywords for internal searchable database (please provide up to 10)	Microglial culture, neuronal culture, zebrafish, metal dyshomeostasis, novel pharmacotherapeutics, microglial dysfunction, neurodegeneration models, Alzheimer's disease, Parkinson's disease, temporal lobe epilepsy, bacterial meningitis
Geographies of interest for exchanges (optional)	Spain; Sweden

Overview of your research / innovation focus
Our research is focused on understanding the cellular mechanisms underlying neuroinflammatory, neurodevelopmental and neurodegenerative disease, along with their primary risk factors. In particular, we investigate the immune dysregulation that drives the chronic neuroinflammation characteristic of neurodegenerative disease pathology. We use cultured mammalian and human microglia and macrophages to evaluate the inflammatory changes, metabolic dysfunction and reprogramming indicative of the dysfunctional microglia in the diseased brain. Using human and mammalian neuronal cultures we further explore the mechanisms underlying the neuronal injury and neurotoxicity characteristic of the neurodegenerating brain. We are particularly interested in the role of metal ion imbalance in promoting neuronal death. Accumulation of ions such as iron and copper in the brain is a hallmark of neurodegenerative disease, driving neuronal dyshomeostasis and cell death mechanisms including ferroptosis and cuproptosis. In parallel, we use model organisms, including zebrafish and c. elegans, to generate in vivo models of neuroinflammatory and neurodegenerative disease, including Alzheimer's disease, Parkinson's disease and epilepsy. We directly explore the toxicity and behavioural impairment that typify disease states, and further validate our in vitro analyses. Importantly, we use these in vitro and in vivo models to explore novel therapeutic targets and pharmacotherapeutic strategies for neurodegenerative disease. Currently there are limited/no effective treatments for neurodegenerative disease. Most attempts to date have focused on a single aspect of the pathology, and have resulted in minimal or no clinical efficacy. Our research explores strategies that target multiple facets of pathology, such as inflammation, oxidative injury, metal overload, metabolic imbalance. These strategies include naturally-occurring compounds and novel derivatives, specific inhibitors, Schiff base ligands, and novel small molecule agents.

Why your team could be relevant to a translational eye-health and vision consortium
Our team brings a neurobiological and neuropharmacological perspective to eye health, with particular expertise in mechanisms shared between neurodegenerative disease and retinal/visual-system dysfunction. We provide an experimental pipeline spanning mechanistic cellular studies, multi-omics discovery and therapeutic screening, through to functional in vivo validation. **Mechanistic neurobiology expertise:** We bring broad expertise spanning neuronal and glial physiology, neuroinflammation, neurodegeneration, cellular metabolism and pharmacological intervention, allowing us to interrogate mechanisms contributing to retinal and visual-system dysfunction. **Associating eye disease with neurodegeneration:** Our research addresses pathological processes relevant across retinal and CNS disorders, including neuroinflammation, neuronal dysfunction, metabolic stress and dysregulated cellular homeostasis. **Therapeutic investigation:** We investigate complex disease mechanisms with the aim of identifying actionable therapeutic targets and interventions, providing a bridge between fundamental disease biology and translational development. **Complementary to eye health:** Our experimental neurobiology capabilities can help translate clinical observations and disease-associated signatures into cellular and in vivo models for mechanistic investigation and therapeutic testing. **Therapeutic innovation**: Our experience in pharmacological intervention and development of novel small-molecule approaches provides opportunities to identify and validate multi-target therapeutic strategies appropriate for complex and multifactorial visual disorders.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute
Cellular and molecular neurobiology: In vitro models for investigating neuronal and glial function, neuroinflammation, neurotoxicity, metabolic dysfunction and mechanisms of neurodegeneration. **In vivo disease models and functional phenotypes**: Established models for translation of molecular and cellular findings to functional outcomes, including in vivo toxicity, behavioural phenotypes and visual assessment. **Multi-omic approaches:** Proteomic, transcriptomic and metabolomic approaches for pathway analysis and identification of disease biomarkers, molecular signatures and novel therapeutic targets. **Molecular and biochemical profiling:** Gene and protein expression analysis, cellular metabolic assessment, inflammatory signalling, oxidative-stress and cell-death pathways, and investigation of cellular homeostatic mechanisms. **Pharmacological screening and validation:** Experimental platforms for screening and characterising candidate therapeutic compounds across in vitro and in vivo models. **Small-molecule therapeutic development:** Expertise and access to novel and patent-protected compounds, including multi-target pharmacological approaches with potential for therapeutic development and repurposing. **Integrated translational pipeline:** Capacity to progress findings from pathway identification and cellular validation to pharmacological intervention and functional validation. This provides experimental support for clinically derived hypotheses and candidate therapeutics.

Infrastructure, equipment, facilities or datasets relevant to consortium work
Our team are based at the UCD Conway Institute of Biomolecular & Biomedical Research. (https://www.ucd.ie/conway) The UCD Conway Institute houses a state-of-the-art aquatics facility for zebrafish experimentation. This includes equipment for behavoiural assessment such as measurement of visual motor function and optokinetic response, microinjection equipment for embryonic and adult pharmacological interventions, and microscopy/imaging analysis. Conway core facilities include a genomics facility for gene expression and transcriptomic analysis, a proteomics core facility, metabolomics core for analysis of metabolic signatures, core technologies including *Seahorse* analysers and *Incucyte* imagers for live cell analysis of cell health, metabolism and respiration, and a dedicated electrophysiology suite for analysis of neuronal function in ex vivo brain tissue.

Newly-upgraded tissue culture suites allow dedicated facilities for culture and maintenance of cell lines, primary culture systems and human iPSC-derived cells and organoids.

Training capacity and knowledge-transfer potential

Skills and training opportunities for inbound staff:

Interdisciplinary training environment: Inbound staff will join the highly interdisciplinary and collaborative environment of the UCD Conway Institute, gaining exposure to expertise spanning neurobiology, neuroinflammation, neurodegeneration, visual dysfunction, pharmacology and therapeutic development.

Preclinical disease modelling: Hands-on training in the development, maintenance and application of in vitro neuronal and glial models and in vivo zebrafish models of neurological and visual disorders, including Alzheimer's and Parkinson's diseases, temporal lobe epilepsy, bacterial meningitis and inherited blindness.

Cellular and molecular analysis: Training in experimental approaches for investigating neuronal and glial toxicity, neuroinflammation, cellular metabolism, metal dyshomeostasis and disease-associated molecular signalling, together with interpretation and integration of molecular datasets.

Functional phenotyping: Practical experience linking molecular and cellular pathology with in vivo functional outcomes, including behavioural, neurological and visual/functional assessment in zebrafish disease models.

Experimental design and translational validation: Training in the design of complementary in vitro and in vivo studies, selection of appropriate experimental endpoints, and translation of mechanistic findings across model systems.

Therapeutic screening and pharmacological evaluation: Experience in screening and mechanistic characterisation of candidate therapeutics, including assessment of efficacy, toxicity and modulation of disease-associated pathways.

Novel therapeutic development: Opportunities to investigate multifunctional small molecules developed with pharmaceutical chemistry collaborators, alongside naturally occurring bioactive compounds, providing experience in progressing candidate interventions from mechanistic rationale through preclinical evaluation.

Translational research skills: Exposure to the wider pathway from disease-mechanism discovery and target identification through therapeutic validation, interdisciplinary collaboration and development of research findings with potential for therapeutic translation.

Transferable research skills: Development of competencies in experimental planning, critical data interpretation, scientific communication and collaborative working across disciplines. Our team routinely engages with scientific outreach and educational activities, along with patient/advocate representatives.

What your team would hope to gain from participating

What our team would hope to gain from participating:

Establish collaborations with consortium partners that offer advanced experimental models, in particular imaging technologies, computational approaches to complement our existing neurobiological and pharmacological capabilities.

Access to clinically derived samples and datasets, to further validate mechanisms identified in our experimental models and establish their relevance to human disease.

Opportunities to integrate findings from our cellular and in vivo models with patient-derived cells, iPSC-based systems, organoids etc, to strengthen the translational relevance of our mechanistic and therapeutic findings.

Access to specialist expertise in visual function, retinal biology, imaging and electrophysiology would enable us to extend our existing neurobiological models and develop more sophisticated approaches for evaluating visual-system dysfunction and therapeutic rescue.

We would particularly value partnerships that can help progress promising small molecules and other interventions beyond early preclinical discovery, including medicinal chemistry, pharmacokinetics, drug-delivery, toxicology and later-stage preclinical validation.

We hope to identify mechanisms shared between visual disorders and wider neurological, inflammatory, metabolic and age-related diseases, creating opportunities for therapeutic repurposing and development of interventions with relevance beyond ophthalmology.

Participation would provide a foundation for exchange expertise, joint publications and development of collaborative and cross-disciplinary funding applications.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Derek Costello	Associate Professor and Principal Investigator	UCD School of Biomolecular & Biomedical Science (SBBS)	Researcher	Experienced	Derek.costello@ucd.ie
Jodie Kearney	PhD Student	UCD SBBS	Researcher	Early-stage	Jodie.kearney@ucdconnect.ie
TBD	2x doctoral students	UCD SBBS	Researcher	Early-stage	

Main disciplines represented in your team	Neurobiology, pharmacology, immunology, neurodegeneration, neurotherapeutics
Experience in EU / international / multi-partner collaborative projects	Participation as a research partner in the *Crystal3* consortium. I supervised a 12-month secondment at UCD Conway, and a 3-month staff exchange from our team. To date this has yielded 2 publications, and 2 currently under review.
Experience with clinical, translational or patient-facing research (if relevant)	☐ Yes x No ☐ Not applicable | Short note: Primarily preclinical research, however routine engagement with PPI/patient advocates to inform research.

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

Clarke NC, McCabe E, Jensen LD, Creaven BS, Costello DA (2025) Salicylaldehyde benzoylhydrazone protects against ferroptosis in models of neurotoxicity and behavioural dysfunction, in vitro and in vivo. J Mol Neurosci 75:2. doi: 10.1007/s12031-025-02371-2

Galvin DM, Fernandez-Garcia S, Dawson E, Pryce C, Egan BP, Clarke NC, Reynolds AL, Costello DA (2025) Ergolide regulates microglial activation and inflammatory-mediated dysfunction: a role for the cysteinyl leukotriene pathway. Int Journal Mol Sci 26:11. doi: 10.3390/ijms26115050

Al-Thani NA, Stewart GS, Costello DA (2025) The role of the urea cycle in the Alzheimer's disease brain. J Neurochem 169:3. doi: 10.1111/jnc.70033

Al-Thani NA, Zinck D, Stewart GS, Costello DA (2024) Modulation of urea transport attenuates TLR2-mediated microglial activation and upregulates microglial metabolism in vitro. Metabolites 14:11. doi: 10.3390/metabo14110634

Howe AM, Burke S, O'Reilly ME, McGillicuddy FC, Costello DA (2022) Palmitic acid and oleic acid differently modulate TLR2-mediated inflammatory responses in microglia and macrophages. Molecular Neurobiology 59:2348–2362. doi: 10.1007/s12035-022-02756-z

Howe AM, Cosgrave A, Ó'Murchú M, Britchfield C, Mulvagh A, Fernandez-Perez I, Dykstra M, Jones AC, Costello DA (2020) Characterising lipoteichoic acid as an in vitro model of acute neuroinflammation. International Immunopharmacology 85:106619. doi: 10.1016/j.intimp.2020.106619

Up to 5 relevant previous projects, grants, networks or collaborations

Title: Exploring coumarin-derivative Shift base compounds as potential therapeutics for bacterial meningitis
Funder:Research Ireland (IRC)
Funder reference: GOIPG/2025/8836
01 Sep 2025 - 31 Aug 2029
Title: Investigation of novel neurotherapeutic strategies for drug-resistant epilepsy

Funder:Irish Research Council (IRC)
Funder reference: GOIPG/2024/3286
01 Sep 2024 - 31 Aug 2027
Title: Explore the therapeutic potential of seaweed extracts for Alzheimer's disease-related neurodegeneration
Funder: Institutional
01 Sep 2023 - 31 Oct 2025
Title: Commercial & Research Opportunity for Cysteinyl Leukotriene Signalling in Ocular & CNS Dysfunction Cancer and Cardiovascular Disease (CRYSTAL3)
Funder reference:101007931
Funder: Horizon 2020
01 Jan 2021 - 31 Dec 2024
Title: Investigating the therapeutic potential of phytocannabinoids in models of neuroinflammation
Funder reference: EPSPG/2020/73
Funder: Irish Research Council (IRC)
01 Apr 2020 - 31 Mar 2022
Title: Investigating the role of UT-B in the pathogenesis of Alzheimer's disease
Funder: Jack Pickard Research Trust (UK)
01 Mar 2019 - 28 Feb 2021

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)
Title: Novel coumarin-based neuroprotective agents Inventors: Creaven B.S., Costello D.A. Patent number: Application in preparation (Sept 2026)

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	x Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	x Only Researchers ☐ Technical Staff/Innovation ☐ Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only x Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	N/A.

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
Neurobiology and disease models: Integrated in vitro neuronal/glial and in vivo zebrafish models for investigating neurodegeneration, neuroinflammation and visual dysfunction, with molecular, behavioural and functional phenotyping. **Therapeutic discovery and preclinical validation:** Pharmacological screening and mechanistic evaluation of novel, repurposed and naturally derived therapeutic candidates, including access to patent-protected multifunctional small molecules and platforms for in vitro and in vivo efficacy testing. **Mechanistic and multi-omics discovery:** Integrated molecular, transcriptomic, proteomic and metabolomic approaches for identifying disease mechanisms, biomarkers and therapeutic pathways, with particular expertise in neuroinflammation, cellular metabolism and homeostatic dysfunction.

Capabilities or expertise you would especially like to access through the consortium

Clinical datasets and patient-derived resources: Access to clinical datasets and disease-associated biomarker data to validate mechanisms identified in our experimental models and establish their relevance to human disease. **Computational biology:** Expertise in bioinformatics, systems biology and advanced computational approaches for integration and interrogation of transcriptomic, proteomic, metabolomic and clinical datasets. **Medicinal and pharmaceutical chemistry**: Expertise in medicinal chemistry, structure–activity relationships and lead optimisation to support progression of promising therapeutic compounds from biological proof-of-concept towards candidates with drug-like properties. **Drug development and delivery:** Complementary expertise in pharmacokinetics, drug delivery etc to strengthen the translational pathway for novel and repurposed therapeutic candidates. **Human-relevant models and phenotypes:** Access to patient-derived cells, iPSC-based systems etc, and assessment of visual and neurological function to bridge our existing cellular and in vivo platforms with human disease.

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	x Yes ☐ No
x We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
N/A

9. Additional collaborators you would like us to consider

If there are any additional organisations, teams or contacts - current collaborators, previous collaborators or desired future collaborators - that you think could be relevant for this MSCA-SE consortium, please list them below. The coordination team may contact them to explore their potential interest in joining the consortium-building process.

Please provide the following details (as many entries as needed).

Type of organisation: ☐ Academic ☐ Clinical ☐ SME/industry ☐ NGO/foundation ☐ Public body ☐ Other

Organisation / team name:

Country:

Website:

Existing link with your team: ☐ Current collaborator ☐ Previous collaborator ☐ Desired collaborator ☐ Other

Suggested contact person/email (if known):

Why could they be relevant for this MSCA-SE consortium?

Any additional comments or sensitivities before contacting them?

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants.

The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

x Understood

Name: Derek Costello
Date completed: 31/08/26

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)